

Square-lattice FK interfaces and nested loops for $1 \leq q < 4$

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Abstract

For every fixed $1 \leq q < 4$, critical square-lattice random-cluster Dobrushin interfaces converge as ordered curves to chordal $\text{SLE}_{\kappa(q)}$, where $\kappa(q) = 4\pi / \arccos(-\sqrt{q}/2)$, confirming the Rohde–Schramm prediction in this parameter range. This holds in every bounded Jordan domain under uniform marked boundary approximation. The complete nested plane loop collections converge to whole-plane $\text{CLE}_{\kappa(q)}$ in a spherical matching topology retaining multiplicities and traversals. At $q = 1$ we obtain Cardy’s formula for square-lattice bond percolation with free boundary edges.

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1 Introduction and precise statements

Planar critical models are expected to have conformally invariant scaling limits. For random-cluster models this prediction identifies the law of an interface, the joint law of all surrounding loops, and crossing probabilities in arbitrary shapes. These are distinct conclusions: a limit for one interface does not by itself identify its coupling with other interfaces, and convergence of loop traces on compact sets does not by itself retain multiplicities, nesting, or complete traversals.

The random-cluster representation of Fortuin and Kasteleyn [19] places percolation, Ising, and Potts models in a common family with a real cluster weight. Its planar loop and transfer-matrix descriptions belong to the work of Temperley and Lieb [51] and Baxter, Kelland, and Wu [4]. Exact-solution and Coulomb-gas methods supplied much of the prediction of critical behavior [3, 35]. We use that finite algebraic lineage, and prove the analytic estimates needed to pass from the square-lattice law to its conformal curve limits.

Schramm’s characterization of conformally invariant interfaces led to Schramm–Loewner evolution [43]. Cardy’s crossing prediction [10] was proved by Smirnov for critical site percolation on the triangular lattice [48]; Camia and Newman subsequently constructed its full continuum loop limit [9]. For FK-Ising, Smirnov’s fermionic observable [49] and the isoradial analysis of Chelkak and Smirnov [11] led to convergence of the Dobrushin interface to $\text{SLE}_{16/3}$ [12]. Kemppainen and Smirnov proved convergence of the boundary-touching FK-Ising loops [25], and then joint convergence of the full loop ensemble and its exploration tree in bounded wired domains [23].

For general cluster weights, Duminil-Copin, Kozłowski, Krachun, Manolescu, and Oulamara proved asymptotic rotational invariance of critical square-lattice FK for $1 \leq q \leq 4$ [18, Theorem 1.2]. This establishes rotational invariance without identifying the limiting interfaces or loop collections as SLE or CLE. It is background for the present result, whose identification proceeds through the four-change observable.

There are also prior public claims for square-lattice bond percolation. The preliminary preprint of Tsai, Yam, and Zhou [53] claims SLE_6 convergence of its exploration path. Zhou’s later preprint [54] claims full-curve SLE_6 convergence and Cardy convergence for its stated

crossing event under Condition C, which imposes mesoscopic straight boundary segments and separation conditions. These are overlapping $q = 1$ claims. The statements here include every uniformly approximated marked Jordan domain in the specified lattice convention, every fixed $1 \leq q < 4$, and the complete nested whole-plane ensemble with multiplicities and traversal order.

For general cluster weights, the parafermionic program of Smirnov [49], the work of Riva and Cardy [41], and the criticality applications of Duminil-Copin [13] establish the importance of local winding identities. Riva and Cardy explicitly distinguish such identities from continuum holomorphicity, which also requires regularity. Chelkak and Smirnov's four-boundary-change construction [11, Section 6] supplies a further antecedent at $q = 2$. The main analytic task here is to close this regularity gap for the four-change observable throughout $1 \leq q < 4$, and then to retain the resulting tests after exploration.

The probabilistic inputs have similarly distinct roles. Positive association comes from the FKG inequality [20]. Beffara and Duminil-Copin identify the self-dual critical point [5]; Duminil-Copin, Sidoravicius, and Tassion prove continuity and buffered crossing estimates [15]. We use the isoradial crossing estimates of Duminil-Copin, Li, and Manolescu [16] for transfer diagrams, and the rough-quad and six-arm estimates of Duminil-Copin, Manolescu, and Tassion [17] for moving boundaries. Their exact parameter and boundary-condition ranges are stated below.

The continuum loop laws are built from branching SLE [29, 30, 46]; the simple-loop characterization [47] belongs to the complementary simple regime. For the nonsimple regime used here, the large-domain construction and sphere invariance are supplied by Miller et al. [32] and Gwynne et al. [21]. Our task is to identify the lattice tree and its complete loops with these laws. In doing so we distinguish convergence of drivers, ordered curves, loop sets, and loop collections retaining multiplicity. The results concern the nearest-neighbor square lattice in its ordinary Euclidean embedding and its original random-cluster probability law.

1.1 The model and its boundary conventions

Fix a real number $1 \leq q < 4$ and put

$$p_q = \frac{\sqrt{q}}{1 + \sqrt{q}}, \quad \kappa(q) = \frac{4\pi}{\arccos(-\sqrt{q}/2)}. \quad (1.1)$$

The inverse cosine has values in $[0, \pi]$. Thus $4 < \kappa(q) \leq 6$, with $\kappa(1) = 6$ and $\kappa(q) \downarrow 4$ as $q \uparrow 4$. On a finite graph $G = (V, E)$, let ξ be a partition of designated boundary vertices. For $A \subseteq E$, let $k_\xi(A)$ be the number of components of (V, A) after identification within each block of ξ , including isolated vertices. The random-cluster law is

$$\phi_{G,q}^\xi(A) = \frac{1}{Z_{G,q}^\xi} p_q^{|A|} (1 - p_q)^{|E| - |A|} q^{k_\xi(A)}. \quad (1.2)$$

An *actual open path* always consists of open edges of G . Identification in ξ is used to compute the probability weight and does not supply an additional edge in such a path.

Let D be a bounded Jordan domain. A lattice approximation at mesh $\delta_n \downarrow 0$ is the bounded interior D_n of a simple closed nearest-neighbor polygon on $\delta_n \mathbb{Z}^2$. Its graph G_n has every lattice vertex in $\overline{D_n}$ and every nearest-neighbor edge whose closed segment is contained in $\overline{D_n}$. In particular, boundary edges are part of G_n .

For two marked boundary points a, b , we require counterclockwise parametrizations $\beta_n, \beta : [0, 1] \rightarrow \mathbb{C}$, injective on $[0, 1]$, such that

$$\begin{aligned} \beta_n(0) = \beta_n(1) = a_n, \quad \beta_n(1/2) = b_n, \quad \beta(0) = \beta(1) = a, \quad \beta(1/2) = b, \quad (1.3) \\ \sup_{t \in [0, 1]} |\beta_n(t) - \beta(t)| \longrightarrow 0. \end{aligned}$$

The free arc F_n goes counterclockwise from a_n to b_n ; the other arc W_n goes counterclockwise from b_n to a_n . Wire together all vertices of W_n , including the two marked endpoints. All other boundary vertices remain singleton blocks. Remove from the sampled edge set the boundary edges of W_n and declare those edges open for the interface drawing. This is equivalent on all remaining edges to (1.2) with the same wired partition: the endpoints of each removed edge were already identified. Edges on F_n remain random.

The medial graph has vertices at the midpoints of primal edges, with medial edges joining the midpoints of consecutive sides of a lattice square. At an open primal edge the four incident medial half-edges are paired in quarter turns around the adjacent dual faces; at a closed primal edge they are paired around the primal endpoints. At the boundary one uses the turns between the primal wired arc and its complementary dual wired arc. There is a unique medial strand between the two changes of boundary condition. We orient it from a_n to b_n and denote it by η_n . This is the usual Dobrushin construction [14, Section 5.2.2]. Occurrence order at a twice-visited medial vertex is part of the curve. The tile drawings used later preserve this order and differ from the straight medial drawing by at most a constant times δ_n in uniform curve distance.

For continuous oriented curves $\gamma_1, \gamma_2 : [0, 1] \rightarrow \mathbb{C}$, write

$$d_{\text{curv}}(\gamma_1, \gamma_2) = \inf_{\varphi_1, \varphi_2} \sup_{0 \leq t \leq 1} |\gamma_1(\varphi_1(t)) - \gamma_2(\varphi_2(t))|, \quad (1.4)$$

where the infimum is over increasing homeomorphisms of $[0, 1]$. Zero-distance curves are identified. Chordal SLE_κ in $(D; a, b)$ is the conformal image of the upper-half-plane Loewner trace driven by $\sqrt{\kappa}B_t$ from 0 to ∞ , with B a standard Brownian motion. The remaining positive dilation of the conformal coordinate does not change this law.

1.2 Complete loop collections

Let ϕ_q^0 be the weak limit of (1.2) in increasing square boxes of $\mathbb{Z}^2$ with free boundary conditions. Form its complete medial interface collection, using the same local pairings as above, and then scale the embedding by δ . Denote the resulting collection by Γ_δ . It contains every loop at every nesting level. The elementary circuit argument in Section 2 shows that its discrete interfaces are almost surely finite.

Individual loops are unrooted and unoriented continuous maps of the circle into the Riemann sphere $\hat{\mathbb{C}}$. Their distance is the infimum of the uniform spherical distance over circular orientation-preserving homeomorphisms and reversal, with zero-distance representatives identified. We retain distinct loops even when their images coincide. For two collections, an ε -matching is a partial bijection that covers all loops of spherical diameter greater than ε in both collections, has matched-loop distance at most ε , and leaves only loops of diameter at most ε unmatched. Convergence of collections means the existence of such matchings with $\varepsilon \downarrow 0$. The continuum law in the next statement is the standard nested whole-plane CLE_κ in this topology.

Theorem 1.1 (Interfaces and all nested loops). *For every fixed real $1 \leq q < 4$, the following assertions hold.*

- (i) *For every bounded Jordan domain D with distinct boundary marks a, b , and every approximation satisfying (1.3), the Dobrushin strand η_n converges in law, in (1.4), to chordal $\text{SLE}_{\kappa(q)}$ from a to b in D .*
- (ii) *Under the specified free infinite-volume law, Γ_δ converges in law as $\delta \downarrow 0$ to the nested whole-plane $\text{CLE}_{\kappa(q)}$, in the spherical collection topology just defined. The limit is Möbius invariant.*

Both conclusions hold along every mesh sequence. No uniformity in q is asserted.

The word “critical” refers to the edge parameter p_q , whose criticality is established in Beffara and Duminil-Copin [5]. This gives a positive resolution, for $1 \leq q < 4$, of the FK-to-SLE prediction in Rohde and Schramm [42, Conjecture 9.7] and Schramm [44, Problem 2.6]. Those formulations also include $0 < q < 1$; the result proved here is the stated parameter-restricted part. The Jordan-domain theorem includes the square-domain formulation of Rohde and Schramm [42]. The parameter $q = 4$ is outside the scope of this article. The strict angular and six-arm estimates in the proof are used only at fixed $q < 4$.

1.3 Square-lattice bond crossings

For four distinct counterclockwise boundary marks a, b, c, d of D , require the same uniform convergence as in (1.3), with the respective four marks at parameters $0, 1/4, 1/2, 3/4$. Sample *all* edges of G_n independently with probability $1/2$, including every boundary edge, and make no identifications or external connections. Let C_n be the event of an actual open path from the closed counterclockwise arc $[a_n, b_n]$ to the closed counterclockwise arc $[c_n, d_n]$.

Theorem 1.2 (Cardy’s formula for square-lattice bond percolation). *Let $\psi : D \rightarrow \mathbb{H}$ send a, c, d to $0, 1, \infty$, and put $x = \psi(b) \in (0, 1)$. Under the boundary and approximation conventions above,*

$$\lim_{n \rightarrow \infty} \mathbb{P}(C_n) = \frac{\int_0^x [t(1-t)]^{-2/3} dt}{\int_0^1 [t(1-t)]^{-2/3} dt}. \quad (1.5)$$

Theorem 1.2 proves the square-lattice bond-percolation form of Cardy’s prediction with the free-boundary convention specified here. We obtain it directly from the alternating four-change calculation and a deterministic comparison on a common lattice graph. In particular, its proof supplies the boundary transfer for the actual crossing event; it does not infer continuity of that event from a curve limit.

The same Cardy function also governs limiting annealed closed-cell crossing probabilities in critical Poisson–Voronoi percolation in every fixed bounded Jordan quadrilateral [37, Theorem 1.1].

1.4 Square-lattice bond critical exponents

Write $\mathbb{P}_p^\square$ for independent bond percolation on $\mathbb{Z}^2$, and put $B_R = \{z \in \mathbb{C} : |z| < R\}$. Let $\pi_1(R) = \mathbb{P}_{1/2}^\square(0 \leftrightarrow \mathbb{Z}^2 \setminus B_R)$. For $k \geq 1$, let $\pi_{2k}(r, R)$ be the $\mathbb{P}_{1/2}^\square$ -probability that $2k$ disjoint actual primal-open and dual-open paths, cyclically alternating in color, join $\overline{B}_r$ to $\mathbb{C} \setminus B_R$. Here a dual-open edge crosses a closed primal edge. For the open cluster $C(0)$, set

$$\begin{aligned} \theta(p) &= \mathbb{P}_p^\square(|C(0)| = \infty), \\ \chi(p) &= \mathbb{E}_p^\square[|C(0)|; |C(0)| < \infty], \\ \xi(p)^2 &= \frac{1}{\chi(p)} \sum_{x \in \mathbb{Z}^2} |x|^2 \mathbb{P}_p^\square(0 \leftrightarrow x, |C(0)| < \infty) \quad (p \neq \tfrac{1}{2}). \end{aligned}$$

Thus χ is the truncated susceptibility and ξ is the second-moment correlation length.

Corollary 1.3 (Square-bond arm and thermodynamic exponents). *For every fixed $k \geq 1$ and every sufficiently large fixed r ,*

$$\pi_1(R) = R^{-5/48+o(1)}, \quad \pi_{2k}(r, R) = R^{-(4k^2-1)/12+o(1)} \quad (R \rightarrow \infty). \quad (1.6)$$

Moreover,

$$\begin{aligned}\theta(p) &= (p - \tfrac{1}{2})^{5/36+o(1)} & (p \downarrow \tfrac{1}{2}), \\ \chi(p) &= |p - \tfrac{1}{2}|^{-43/18+o(1)} & (p \rightarrow \tfrac{1}{2}, p \neq \tfrac{1}{2}), \\ \xi(p) &= |p - \tfrac{1}{2}|^{-4/3+o(1)} & (p \rightarrow \tfrac{1}{2}, p \neq \tfrac{1}{2}).\end{aligned}\tag{1.7}$$

Section 16 proves this consequence by buffered annular comparisons using the complete loop limit, followed by square-bond arm quasi-multiplicativity and the near-critical scaling relations. The arm assertion is for even, cyclically alternating words; it makes no assertion for odd or other color words. None of these exponent claims is made for $q \neq 1$.

1.5 Proof structure and technical contributions

For $1 \leq q < 4$, use the parameters

$$d = \sqrt{q} = 2 \cos \lambda, \quad \rho = \lambda/\pi, \quad 0 < \lambda \leq \pi/3.\tag{1.8}$$

The first part of the proof determines an alternating-boundary connection probability. Four marked points divide the boundary into arcs A, B, C, D , with the first and third assigned one color and the other two the opposite color. We study the event of an actual path of the first color from A to C . Section 6 defines the auxiliary finite law with four unmatched medial ends and no extra pairing weights, and computes its limiting probability $f(\chi)$ when the boundary marks have half-plane coordinates $0, \chi, 1, \infty$, $0 < \chi < 1$. Completing the two designated arcs by separate deterministic wires gives instead $f/(f + d(1 - f))$. We keep this normalization change explicit before returning to actual FK crossings.

Positive comparisons before complex observables. Section 2 identifies the capped tile law exactly with the random-cluster law and establishes the collar and passage-count comparisons used later. It keeps actual components distinct from boundary partition blocks. Bulk box crossing at $q \leq 4$ and the stronger up-to-boundary crossing estimates for $q < 4$ have different roles [16, 17]. Section 3 derives the local exponent inequalities needed to control corner, tip, and source factors. All complex currents are subsequently estimated in these positive laws.

Angular transfer and boundary holomorphicity. The four-change construction gives a discrete complex field depending on an interior probe; its boundary values can be computed exactly. To identify its limit with the conformal boundary map, we must prove that every subsequential limit h is holomorphic. The finite row and path calculations in Sections 4 and 5 supply an angular decay estimate. We apply it to two copies of the field placed on opposite sides of a wall, one probed inside and the other outside. Repeated reflections keep the external states in their norms, and the logarithmic estimates pay their complete costs.

The resulting mixed difference estimate tends to zero after division by the mesh squared. Closing the finitely many wall gaps and separating the distant exterior copy yields $\bar{\partial}h(z) \partial_j J(u) = 0$ in distributions, where J is the limiting exterior field and ∂_j is a coordinate derivative. The prescribed boundary values make J nonconstant, so a suitable exterior test forces $\bar{\partial}h = 0$. Sections 7–9 prove these steps, including control of the comparison factors through the successive limits. The boundary-value calculation of Section 6 then identifies the connection probability.

From fixed polygons to marked Jordan domains. The first calculation is made in rational orthogonal polygons with specified lattice corner parities. Proposition 10.1 in

Section 10 extends it to arbitrary uniform marked Jordan approximations. Easier and harder comparison quads are drawn on the same lattice and compared on their common edges. Monotonicity and a deterministic comparison of actual paths sandwich its crossing probability between their limits. This step also proves the $q = 1$ Cardy statement with all boundary edges sampled freely.

Moving domains and the chordal driver. The fixed-shape formula must remain useful after a random exploration. Section 11 proves this transfer using a finite chart approximation and an averaged six-arm estimate under the original switch law. It does not require a uniform six-arm theorem in every possible explored domain. Section 12 first normalizes a shrinking test into a bounded martingale and then transfers it to the true Dobrushin law. Two tests identify the driver and its bracket. Tightness and a separate traversal argument give the Euclidean uniform curve conclusion of Theorem 1.1(i) for $q < 4$.

Joint branches and loop identities. Sections 13–15 construct a target tree by uniform-walled peeling. They establish the reflected angular dynamics, exclude forcepoint variation supported on endpoint times, and identify independent continuation noises after separation, conditional on the common causal history at separation. The resulting branching SLE description is then matched to the original discrete loops: winding information determines loop labels, while additional estimates determine full traversal order. An escape bound upgrades convergence in bounded windows to the spherical matching topology for every nesting generation.

1.6 Conventions on constants and limits

Every theorem is for a fixed q . Constants may depend on q , macroscopic shapes, positive separations, and an explicitly fixed finite set of tests. Mesh limits are taken with those data fixed; auxiliary separations, gaps, cutoffs, and localization parameters are removed in the stated order. A claim uniform over random exploration histories is made only where the corresponding comparison has been proved. In particular, convergence in probability over histories is not replaced by a deterministic estimate for every history.

The proof uses established public results at their stated hypotheses. It uses no theorem about an annealed random planar map, no random embedding, and no transfer from a random-map metric limit to the square lattice. The finite Dobrushin experiments and the free infinite-volume experiment are different probability spaces; the theorems assert their separate convergence and require no coupling between their samples.

2 Tiles, positive laws, and localization

Fix $1 \leq q < 4$ and write

$$d = \sqrt{q} = 2 \cos \lambda, \quad \rho = \lambda/\pi, \quad 0 < \lambda \leq \pi/3. \quad (2.1)$$

All constants may depend on this fixed parameter and on the fixed geometric shapes under consideration. In particular, none of the estimates below is asserted uniformly as $q \uparrow 4$.

We first express the tile model in terms of positive FK laws, keeping actual paths distinct from boundary identifications. The resulting crossing and comparison estimates will control how local changes to walls and loop weights affect observations outside their neighborhoods.

2.1 Tiles and boundary caps

This positive loop description is the planar random-cluster representation [19], in the finite diagrammatic tradition of Temperley and Lieb [51] and Baxter et al. [4]. We give the exact counting with our boundary convention before using it in probability comparisons.

We use a square tile array whose corner sites alternate between primal (P) and dual (D) sites. The primal lattice consists of the primal diagonals, so its mesh is $\sqrt{2}$ times the tile-edge length. Rotating the tile coordinates gives the prescribed orientation of the primal square lattice. Each tile carries one of the two noncrossing pairings of its four side midpoints, called *ports*. Pairing around the dual corners declares the primal diagonal open; the other pairing declares the dual diagonal open. An *actual color path* uses these open diagonals and never uses boundary identifications as edges. A strand has an actual primal path on one side and an actual dual path on the other.

The two local pairings are shown in Figure 1.

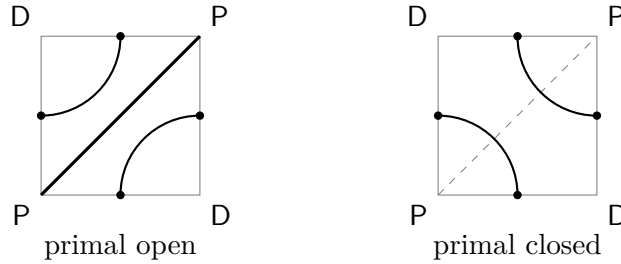


Figure 1: The two tile switches. The solid diagonal is an open primal edge; the dashed diagonal records the absent primal edge. The curved arcs specify the pairing of ports and hence the order of strand occurrences when a medial vertex is visited twice.

For topological arguments, draw the two arcs of a switch disjointly within its tile, normally to the sides at their ports. This drawing and the straight medial drawing can be parametrized piece by corresponding piece, preserving the order of occurrences at a double visit. Their uniform distance is at most twice the tile diameter. Thus these local changes of drawing disappear in the curve topology of the theorem.

A wall cuts tile connections at ports. Distinct sectors at a wall corner have distinct copies of that corner, as on a surface cut open. The end of a finite slit has one sector of angle 2π . All shapes in the tile arguments are unions of whole tiles; separating repeated boundary incidences by a small thickening leaves their combinatorics unchanged.

Partition each boundary component into intervals of alternating designated wire colors. Within an interval, pair the two intervening ports between successive occurrences of the designated color, including eligible end vertices. These are the *local caps*. A genuine change leaves one unmatched port, on the side whose color differs from the change vertex. Use the analogous cyclic caps on a boundary component without changes. Intervals will always be much longer than one mesh.

A *positive cap completion* joins the unmatched ports around the primal intervals in the added boundary disk. Equivalently, wire each primal interval separately, with a dual sea between these wires. This adds only boundedly many connections beyond the local caps when the number of changes is bounded. It adds no primal identification on a dual interval.

Lemma 2.1 (The positive loop law). *On a genus-zero tile surface with positive cap completions, the switch weight d^ℓ , where ℓ counts completed medial loops, is proportional to the critical FK weight with the corresponding boundary partition. Conditional laws on common subdomains are therefore FK laws with induced partitions. The same assertion holds after*

interchanging the colors. On a torus without caps the two weights differ by a factor bounded in terms of q alone.

Proof. Draw noncrossing exterior primal trees realizing the designated identifications and thicken the resulting open primal graph. Its perimeter is the medial loop collection, apart from a fixed number of exterior components. Euler's formula gives the configuration-dependent part of the perimeter count as

$$\ell = 2k_\xi(A) + |A| + \text{constant}.$$

Since $d^2 = q$ and $p_q/(1 - p_q) = d$, the weights agree up to a constant independent of A . The domain Markov property gives the conditional-law statement. On a torus the thickened graph has Euler characteristic $|V| - |A| = 2k - \ell - 2g$, with $g \leq 1$, so the additional factor is bounded. $\square$

2.2 Crossing estimates and finite plane loops

We use two distinct crossing inputs. The buffered bulk box-crossing theorem for canonical weights on doubly periodic isoradial graphs applies for $1 \leq q \leq 4$ [16, Theorem 1.1]. Square rhombi have canonical odds $\sqrt{q}$, so this includes our square lattice. For example, for fixed $a > 0$, horizontal crossing of $[-an, an] \times [-n, n]$ inside $[-(a+1)n, (a+1)n] \times [-2n, 2n]$ has probability bounded away from zero and one, uniformly in the boundary partition on the larger rectangle. The same theorem gives equality of the free and wired infinite-volume laws. Hence planar duality at these self-dual odds preserves the plane loop law, including when a tile translation exchanges the two colors.

When a crossing must reach a domain wall, we use the stronger theorem for critical square FK with $1 \leq q < 4$: a discrete simple-polygon quad of bounded continuum extremal distance has an actual open crossing with probability bounded away from zero, uniformly in its boundary partition [17, Theorem 1.2]. We do not apply this statement at $q = 4$. FKG and the domain Markov property apply because $q \geq 1$; duality supplies the corresponding statements for the other color.

Here is the concrete wall application needed below. A tile-coordinate rectangle contains a nearest-neighbor quad of either color graph with staircase sides within boundedly many mesh layers of its boundary. The continuum extremal distance required by the theorem is bounded for bounded aspect ratio. Indeed, straight segments parallel to the crossing direction join the staircase electrodes through a transverse band whose width is comparable to the rectangle's width. Their lengths are at most a constant times its length. Integrating their metric lengths across this band and applying Cauchy–Schwarz bounds the extremal distance by the same aspect-ratio constant. The discrete resistance is also bounded: distribute a unit flow among disjoint zigzag paths whose number is comparable to the width in mesh units and whose lengths are bounded by a constant times the length in mesh units. At an electrode on a wall, the last crossing vertex can be connected to a wall vertex of the same color using boundedly many fresh edges. Conditional finite energy bounds the extra cost below uniformly. Keep the electrode away from the ends of its designated wall interval.

Crossings in a buffered piecewise rectangular corridor follow by using strips of width a fixed fraction of its clearance and overlapping end squares. Transversal crossings in these squares join successive longitudinal crossings; FKG combines the finitely many requirements of one color. This constructs wall-to-wall paths in annular sectors with openings $\pi/2, \pi, 3\pi/2, 2\pi$, including between the two banks of a slit, as well as circuits in a full annulus. Different colors in tile-disjoint sectors can be imposed successively by their conditional lower bounds. All paths here are actual paths. Radii may use tile sup-norm rims, and all buffers have fixed relative width and sufficiently large fixed mesh width.

Lemma 2.2 (Finiteness in the plane). *For every fixed $1 \leq q < 4$, all medial strands of the specified free infinite-volume square FK law are almost surely finite loops. Finite deterministic surgeries admit positive normalized loop laws with the FK conditional laws of Lemma 2.1.*

Proof. Buffered bulk crossing gives a conditional probability bounded below for an actual primal circuit around any fixed box in each of successively separated annuli. The probability of missing the first m such screens is at most $(1 - c)^m$, regardless of the exterior partition. The same argument gives arbitrarily large actual dual circuits. These two colors of screens prevent a medial strand meeting the box from escaping to infinity. Taking the countable union over boxes proves finiteness.

A surgery at finitely many tile connections changes only finitely many loops at a fixed mesh. Relative to the unmodified plane law, its weight ratio is d^{Δ_ℓ} , bounded above and below by constants that may depend on this finite set of connections. Normalization defines its positive law. Equivalently, exhaust by planar domains; the same circuit argument makes the local completion independent of remote boundaries. Lemma 2.1 then gives the conditional law. $\square$

2.3 Comparison through a collar

Lemma 2.3 (Bounded densities across a collar). *Suppose two positive laws have the same graph in a collar between radii r and $16r$. In this collar the graph is a full annulus, or a bounded number of sectors separated by straight coordinate walls; coincident slit banks are allowed. Their deterministic wire portions agree there. Only boundedly many wire blocks traverse the collar, and no other identifications jump it. One radial side faces an arbitrary unknown partition, whereas the graph and deterministic wires toward the other side are fixed. The marginal densities on edges beyond that other side are mutually bounded by a constant independent of scale and of the far graph. Equivalently, their expectations of every nonnegative remote function are comparable with this same constant.*

The conclusion also holds for a fixed-shape rectangular bulk collar shielding a long box, with a constant depending on its shape.

Proof. First work on a finite graph and a finite remote edge set. Split each through wire transversally, preserving its connections on each side and retaining just one representative connection across the cut. The unknown data induce a partition ξ at their section. Include all common deterministic wires for every choice of ξ .

For two ordered partitions, the likelihood ratio of the coarser law to the finer law is increasing: the rank of the additional identifications decreases as open components merge. Write R for this full likelihood ratio. Its marginal ratio on a remote edge set E_0 is

$$\mathbb{E}_{\text{fine}}[R \mid \omega|_{E_0}].$$

Opening remote edges increases both R and the conditional FK law on the other edges, by partition monotonicity. Thus this marginal ratio is increasing as well. Consequently its maximum is attained at the all-open configuration and its minimum at the all-closed configuration. It is enough to compare those two events for the maximal and minimal partitions; every other partition lies between them.

An actual primal crosscut in every sector, or a primal circuit in the annulus, screens the remote graph with uniformly positive probability. Such screens can be discovered from the unknown side without examining edges strictly beyond them. In a sector, explore all actual dual paths from its unknown-facing arc and query edges incident to the reached dual vertices. If a primal wall-to-wall crossing exists, the search does not reach the remote arc. The elementary complementary crossing alternative then places a blocking primal path among

the queried edges: otherwise, setting the unqueried primal edges closed would produce a dual crossing. The alternative follows by tracing the perimeter of the actual component reachable from the starting side; corner gap vertices belong to the arcs when their color permits. The annular version is the alternative between a primal circuit and a radial dual connection. Perform the searches in a subcollar leaving space before both sections. Reversing colors gives the dual searches as well.

Condition on a successful primal search under the minimal law. Each screen is already connected. Joining all screens and the bounded number of through-wire representatives requires only boundedly many additional block identifications. Each has an unnormalized weight ratio in $[q^{-1}, 1]$, hence changes expectations by a bounded density factor. The resulting partition dominates every partition transmitted from the unknown side. In particular it dominates the maximal law restricted at this same screen geometry, without requiring that law to contain the screen. The positive probability of the search proves the desired comparison of all-open probabilities.

For all-closed probabilities, make the actual dual searches under the maximal law. No actual primal path crosses these screens. Only the boundedly many through-wire representatives can still transmit ties; remove these ties at bounded cost. The resulting free cut is stochastically smaller on the remote graph than any other imposed cut partition, including the restriction of the minimal law. This proves the decreasing-event comparison. The extremal likelihood-ratio argument now gives the assertion for every remote event and every nonnegative remote function. Exhaustion proves the infinite-exterior version. For a rectangular bulk collar use the corresponding enclosing-circuit searches. $\square$

Lemma 2.4 (Relative comparison). *Under the hypotheses of Lemma 2.3, suppose successive collars have one of the following geometries:*

- (i) *a full annulus, or one sector whose through wires are precisely continuous primal wire portions along its side rays;*
- (ii) *two sectors, with primal wires along both side rays of one sector and no through wires in the other, and no further fixed contacts between the sectors.*

After m further collars of a sufficiently large fixed radius ratio, the remote expectations of every fixed nonnegative function under two possible unknown-side laws have ratio $1 + O((1 - c)^m)$, or both expectations vanish. The conclusion holds in either radial direction, also after positive modifications or tilts supported on the unknown side.

Proof. Use a radius ratio such as 256 to leave two subcollars of the preceding type. Lemma 2.3, applied nearer the unknown side, gives a common marginal part of uniformly positive mass on the other subcollar. Couple those configurations identically on a fixed fraction of a reference marginal. In case (i), additionally impose a primal screen meeting and joining every through wire. In case (ii), impose a primal screen on the wired side and a dual screen on the other side. These requirements have uniformly positive conditional probability. On their occurrence the partition transmitted to the next section depends only on the shared configuration: every route through the unknown region must enter and leave through a primal screen already connected to its through wires, and the other sector transmits no primal connection.

For a nonnegative remote function, take the largest and smallest expectations over partitions at each successive section. Conditioning nests these intervals of possible expectations. The common part just constructed contracts their difference by a factor $1 - c$. Their ratio is bounded initially by Lemma 2.3; iterating therefore gives the stated relative estimate. A positive unknown-side tilt is a mixture over the same partitions. In particular, when propagating away from a gap, wire connections made or lost inside that gap belong to the unknown partition, which explains the need for both colors of screen in case (ii). $\square$

2.4 Passages and positive local tilts

Raw strands use only tile switches and stop when they reach a wall port. A passage count in a buffered annulus counts disjoint strand passages through its prescribed inner and outer rims; the outer buffer allows cropping at both rims.

Lemma 2.5 (Exponential passage moments). *In any fixed buffered annulus or sector of the preceding types, a raw passage count X satisfies*

$$\mathbb{E}[\exp(BX) \mid \text{exterior}] \leq C_B \quad (0 \leq B < \infty), \quad (2.2)$$

uniformly in scale and in exterior states inducing the specified graph and arbitrary boundary partition. The same holds for a fixed finite family of macroscopic passage counts in a fixed geometry.

Proof. First obtain one positive exponential rate. Disjoint simple radial strand subarcs split a slightly cropped annulus into ordered strips. Along each passage is an actual primal crossing path. Paths in distinct strips cannot join in the cropped graph, because open primal edges cannot cross the intervening medial separators. Choose the endpoint rim bands strictly inside the region traversed by the separators. Thus X passages force at least $X/2 - O(1)$ different crossing components in the actual cropped graph.

Explore complete actual components successively from one rim, ignoring all external identifications when deciding membership in a component. After a component is removed, its incident unexplored boundary edges are closed. The surviving FK law is dominated by full wiring at the original rims and walls; removing vertices can only decrease actual crossings. Even with this full wiring an actual dual blocker across the cropped annulus has probability at least $c > 0$. Conditional on every successful discovery, the chance of another crossing is therefore at most $1 - c$. The number of components has a geometric tail, proving (2.2) for one $B > 0$.

To obtain any fixed rate, divide the radial distance into J separated slices of thickness comparable to r/J . Cover a middle rim in each slice by $O(J)$ small boxes whose buffered annuli remain within the slice. Use wall-centered boxes near a straight wall and smaller bulk boxes elsewhere, counting the two sides of a wall separately. The overlap graph has a bounded coloring independent of J . Every original passage crosses one small annulus in each slice, so

$$JX \leq \sum_i X_i,$$

with $O(J^2)$ counts having the first exponential bound. Conditional iteration within each color class, followed by Hölder across the bounded number of colors, gives a rate $c'J$. Here iteration uses the same bound conditional on every configuration outside each buffer; it does not require independence between buffers. Choose J arbitrarily large but fixed. Before these boxes fit in the mesh, a deterministic finite-size bound suffices. A finite covering by bulk, straight-wall, and corner buffers gives the macroscopic version: a strand segment of the designated diameter must leave one box of an inner cover whose outer buffer is much smaller than that diameter. $\square$

Let N_j count raw nests wholly inside radius r_j , where the radii are dyadic and the local sector geometry extends beyond the largest radius by a buffer. A nest is a closed loop surrounding a bulk site, or an arc joining the opposite side rays around a boundary gap, corner, change, or slit tip. Then

$$0 \leq N_j - N_{j-1} \leq X_j, \quad (2.3)$$

where X_j is a finite family of passage counts in a band near r_j . Its buffer is disjoint from $B_{r_{j-k}}$ for a fixed k . Indeed a newly included nest either travels a fixed fraction of the radius or goes around the point within the band. At a slit tip, reaching the other bank without approaching the tip requires traversing the full-angle sector. Small-box coverings away from the center give the asserted X_j . Start the levels at a fixed mesh radius so the initial count is bounded.

Lemma 2.6 (Local tilt bounds). *For fixed $b > 0$, put*

$$F_j = \mathbb{E}b^{N_j}, \quad d\nu_j = F_j^{-1}b^{N_j} d\mathbb{P}.$$

The ratios F_j/F_{j-1} are bounded above and below uniformly. Under ν_j , the shell counts X_i at and beyond the outer tilt level, $i \geq j$, have all fixed exponential moments uniformly bounded. The assertions remain conditional on exterior data of the specified geometry and allow adjoining a fixed family of passage counts with buffers at or beyond that level.

Proof. The case $b = 1$ is Lemma 2.5. For $0 < b < 1$, set $D_i = \log(F_{i-1}/F_i) \geq 0$. Jensen's inequality and the exponential bound give, for any sufficiently small fixed $\varepsilon > 0$,

$$\begin{aligned} D_i &\leq |\log b| \mathbb{E}_{\nu_{i-1}} X_i \\ &\leq \varepsilon \log \mathbb{E}_{\nu_{i-1}} \exp(|\log b| X_i / \varepsilon) \\ &\leq C_\varepsilon + \varepsilon \sum_{i-k < l < i} D_l. \end{aligned}$$

In the last inequality, drop decreasing factors back to level $i - k$. The remaining tilt is supported outside the buffer of X_i , so the conditional passage bound applies. Choose $\varepsilon(k-1) < 1$; induction bounds every D_i . Dropping the same bounded number of levels then proves every asserted exponential moment.

For $b > 1$, choose $B > k \log b$ as large as desired and induct on j that $\mathbb{E}_{\nu_j} e^{BX_i} \leq C_*$ for $i \geq j$. When $i \geq j + k$, the tilt and the buffer are disjoint. Otherwise let $j_0 = i - k < j$. The normalizing increment from j_0 to j is at least one, so

$$\mathbb{E}_{\nu_j} e^{BX_i} \leq \mathbb{E}_{\nu_{j_0}} \exp \left(BX_i + (\log b) \sum_{j_0 < l \leq j} X_l \right).$$

Give each summand in the sum Hölder weight $\log b/B$ and use the inductive bound at the earlier tilt j_0 . The remaining Hölder factor is an exponential moment of X_i at a larger fixed rate, whose buffer is disjoint from the tilt at j_0 . The right side is at most $CC_*^{k \log b/B}$, which closes the induction for sufficiently large C_* . Negative levels mean no tilt. The moment bounds give the ratio bounds using (2.3). The argument is unchanged after adjoining the indicated passage counts. $\square$

Consequently the expectations of these local weights and their fixed powers have logarithms of size $O(\log n)$ up to radius $O(n^C)$ in mesh units. Changing a cutoff radius by a fixed factor costs a bounded ratio. Spatially separated local tilts can be treated successively by conditioning on their disjoint patches, or by Lemma 2.3. Finally, log convexity between a given power and twice that power shows that increasing the power by ε costs only $n^{O(\varepsilon)}$.

Lemma 2.7 (Absorbing macroscopic errors). *Suppose $Y_n \geq 0$, $\mathbb{E}Y_n \geq n^{-A}/C$, and $\mathbb{E}Y_n^2 \leq Cn^A$, whereas $X_n \geq 0$ has uniformly bounded exponential moments at every fixed rate. For fixed $C_0 \geq 1$ and every $\varepsilon > 0$,*

$$C_\varepsilon^{-1} n^{-\varepsilon} \leq \frac{\mathbb{E}[Y_n C_0^{\pm X_n}]}{\mathbb{E}Y_n} \leq C_\varepsilon n^\varepsilon.$$

Proof. Choose $\eta > 0$ so that $\eta \log C_0 < \varepsilon$ and split at $X_n = \eta \log n$. On the first event the multiplicative factor lies between $n^{-\varepsilon}$ and n^ε . Cauchy–Schwarz and the exponential moment bound at an arbitrarily large fixed rate make both $\mathbb{E}[Y_n; X_n > \eta \log n]$ and $\mathbb{E}[Y_n C_0^{X_n}; X_n > \eta \log n]$ smaller than any prescribed inverse power after accounting for the polynomial second moment. Choose that power larger than $A + 1$. Division by the stated lower bound for $\mathbb{E}Y_n$ proves both estimates, enlarging constants for bounded n . $\square$

2.5 Comparing slit partition functions

Lemma 2.8 (Surface partition comparison). *Let finitely many straight finite wall segments have fixed macroscopic shapes and positive mutual separation. Complete each as a positive isolated boundary slit. Let Z_s be its partition function relative to the unmodified plane law, and Z_{all} the corresponding partition function with all slits. Uniformly in the mesh,*

$$Z_{\text{all}} \asymp \prod_s Z_s. \quad (2.4)$$

Reversing every cap-wire polarity on both banks of a straight isolated segment with boundedly many changes alters Z_s by a bounded factor. The same comparison holds for a segment and its image, with transported cap data, under a symmetry of the homogeneous tile lattice, with the two color names exchanged when the symmetry exchanges the color lattices.

Proof. Enclose each segment in its own patch with a vacant bulk tile collar and room for a further disjoint buffer. Let X_s be a passage count through that collar. Compare the incremental loop change $\Delta \ell_s$ of the slit surgery with $\Delta \ell_s^\circ$ computed after closing the patch ports by the same arbitrary pairing before and after the surgery. Their difference is $O(X_s)$. To see this, cut the segment ports. The background pairing is determined within the patch except for ends whose strand first traverses the collar. There are $O(X_s)$ such ends, and changing their pairing changes either completed cycle count by at most their number, also when other surgeries have been made outside.

Put $G_s = d^{\Delta \ell_s^\circ} \mathbf{1}_{\{X_s \leq K\}}$. For fixed sufficiently large K , simultaneous truncation has probability bounded below under the normalized law with all completed slits, by Lemma 2.5. On the truncated event the surgery weights and their local versions are comparable. Therefore

$$Z_{\text{all}} \asymp \mathbb{E}_0 \prod_s G_s, \quad Z_s \asymp \mathbb{E}_0 G_s,$$

where $\mathbb{E}_0$ is the unmodified plane expectation. Apply Lemma 2.3 successively to the separated patch factors to obtain (2.4).

Translate an isolated segment by one tile step in its tangent direction. This is a symmetry of the homogeneous plane loop law, with duality interchanging the colors when needed. Its shifted local caps agree with the reversed caps except at boundedly many ports near tips and changes. Their remaining pairings differ on only boundedly many ends, which has bounded cycle-count cost. Applying such a lattice symmetry to the isolated slit and its cap data gives the last assertion. This is a comparison of partition functions; it does not assert bounded pointwise density for changing all pairings along a wall. $\square$

3 Logarithmic short-distance comparisons

Only the strict range $1 \leq q < 4$ is considered in this section. We require inequalities between logarithmic costs, not the existence of limiting critical exponents. Put

$$\Delta = \frac{\pi}{2\lambda} - \frac{3}{2}, \quad s = \Delta + 1, \quad t = -\Delta, \quad f(g) = \frac{\sin((2-g)\lambda) \sin((1+g)\lambda)}{\sin(2\lambda) \sin \lambda}. \quad (3.1)$$

In the local experiments below, the cutoff radius is comparable to n in mesh units, the specified local geometry extends beyond it by a fixed buffer, and the exterior has any fixed compatible positive completion.

3.1 Local costs and their conventions

At a flat change of wire color, let $P(n)$ be the logarithm of the probability of an actual arm of the gap vertex's color reaching the cutoff radius. Let N count raw arcs straddling the change, from one side ray to the other, wholly inside that radius, and set

$$D_*(n) = \log \mathbb{E}[d^N].$$

At a corner without a wire flip, write $i = 0$ if the gap color agrees with the designated wire color and $i = 1$ otherwise. For a convex right-angle corner and a reentrant right-angle turn respectively, define

$$C_i(n) = \log \mathbb{E}[f(1/2)^{N/2}], \quad L_i(n) = \log \mathbb{E}[f(-1/2)^{N/2}],$$

where again N counts local straddling raw arcs. Finally, at a bulk vertex, let N count closed raw nests surrounding it and set

$$F_{\pm}(a; n) = \mathbb{E}[u_{\pm}(a)^N], \quad u_{\pm}(a) = \frac{\cos((1 \pm a)\lambda)}{\cos \lambda}, \quad (3.2)$$

for fixed

$$0 < a < 1, \quad (1 + a)\lambda < \pi/2. \quad (3.3)$$

We usually suppress n . In particular, $D_* \geq 0$, $C_i \geq 0$, and $L_i \leq 0$: comparing the products of sines at their common mean gives $f(1/2) > 1$ and $0 < f(-1/2) < 1$.

Lemma 3.1 (Local conventions and strict arm decay). *Changing a cutoff radius by a fixed factor, changing the fixed compatible exterior completion, applying a tile-lattice symmetry to the experiment, or interchanging both color names and wire designations changes each of these logarithms by $O(1)$. The expectations of the local weights, their fixed powers, and the same arm indicator have logarithms of size $O(\log n)$. For some $c > 0$,*

$$P(n) \leq -c \log n + O(1). \quad (3.4)$$

Proof. For nest weights these claims, apart from the signs already noted, are Lemmas 2.3 and 2.6. Symmetries of the local loop weights and caps give the geometric and color symmetries. At a flat change reflection exchanges the two assignments relative to the gap color. Bending or erasing distant cap connections changes only the possible exterior partition.

We give the arm argument because it concerns actual connectivity. In a tile half-box, boundary regions separated by raw boundary-to-boundary arcs are precisely the actual color components touching boundary gaps. Closed interior loops cannot split these boundary classes. If the region at the marked gap does not reach the outer rim, some raw arc with both ends on the flat wall straddles the gap. One way to see this is to form the tree of regions separated by the disjoint arcs. The wall and rim walks both traverse the tree path between their common end regions. A wall visit to a region absent from the rim excursion is entered and left across the same separating edge within the wall portion. Conversely, a straddling arc cuts off the gap and its actual color paths. Thus absence of all straddling arcs inside radius r forces an arm to a comparable radius, and an arm to a larger fixed multiple excludes all such arcs.

Arm probabilities at constant multiples of a radius are comparable. Given the shorter arm, impose a same-color wall-to-wall screen in an intermediate annulus and a crossing from its inner side to the desired outer radius. The arm meets the screen and the added crossing meets it; FKG and the wall crossing estimates give a uniform extension cost. The construction is valid under arbitrary distant boundary data. Starting from a bounded-mesh event and iterating gives a polynomial lower bound. Conversely, actual opposite-color screens in separated annuli obstruct the arm and have uniformly positive conditional probability. Their iteration gives (3.4). Dual-color requirements use FKG for decreasing primal events. No wire identification has been used as an edge of an arm. $\square$

For clarity, a one-sided error denoted $o(\log n)$ in an upper bound means an allowance

$$\varepsilon \log n + O_\varepsilon(1) \quad \text{for every fixed } \varepsilon > 0. \quad (3.5)$$

Its constants may depend on all fixed charges and geometric choices.

Proposition 3.2 (Three logarithmic comparisons). *With the definitions above,*

$$2P + 2D_* + 4(C_0 + C_1) \leq o(\log n), \quad (3.6)$$

$$C_0 + C_1 + L_0 + L_1 \leq o(\log n), \quad (3.7)$$

$$P + \sum_j \log \frac{F_-(a_j)}{F_+(a_j)} \leq o(\log n), \quad (3.8)$$

where the finite list of fixed charges in the last inequality satisfies (3.3) and $\sum_j a_j = 1$.

We prove the proposition in three finite auxiliary geometries. Their common algebra and cap bookkeeping are recorded first. The rectangle compares the flat-change and convex-corner costs; the long cylinder pairs convex with concave corners; the charged cylinder compares the bulk endpoint weights. Reversing all wire designations in each geometry cancels the positive partition-function ratios between its two winding expansions.

3.2 The two winding expansions

Orient every switch strand and, for either global sign, assign the turn weight $\exp(\pm i\rho W/2)$, where W is its signed turn. Give both switches the common coefficient $\sin(\lambda/2)/\sin \lambda$ and sum consistent orientations. The resulting real arrow tensor is identical for the two signs. Indeed it requires two incoming arrows. For adjacent incoming positions only one pairing works and its two turns cancel; for opposite positions the two pairing contributions sum to $2\cos(\lambda/2)$. Thus every finite contraction, with any prescribed boundary spin factors and inserted edge matrices, has the same value in these two expansions. Common tile coefficients will always cancel.

Traverse a boundary with the domain on its left. Let θ_j be a lift of its inward normal angle at port j , continued by the boundary turn at a corner. For spin $c = +1$ or -1 according as the arrow points inward or outward, use

$$b_j(c) = (2\sin \lambda)^{-1/2} \exp\{ic(\rho\theta_j/2 + \lambda h_j)\}. \quad (3.9)$$

A raw arc from port i to port j bounding a disk against the intervening forward boundary interval has turn $\theta_j - \theta_i - \pi$: close it against that interval, using the normal endpoint tangents. Summing its two orientations cancels curvature in the positive-sign expansion and gives

$$w(h_i - h_j), \quad w(m) = \frac{\cos((m - 1/2)\lambda)}{\sin \lambda}. \quad (3.10)$$

For the negative-sign expansion set $h'_j = -h_j - \theta_j/\pi$; its weight is $w(h'_i - h'_j)$.

Take $h_j = H_j + \Delta \mathbf{1}_{\{j \text{ high}\}}$, where H is piecewise constant. A port is high if its immediately preceding gap vertex in the interval has the designated wire color. Across a wire flip the adjacent ports are both high or both low. Since the ports strictly inside a disk arc's boundary interval pair among themselves, its endpoint high statuses are opposite for an even number of flips and equal for an odd number. Thus $h_i - h_j$ is the sum of the drops of H plus $r\Delta$, where $r = \pm 1$ for an even number of flips (positive when the first port is high) and $r = 0$ otherwise. In the negative-sign expansion, high ports refer to the complementary wires; base drops are negated and augmented by the boundary turn divided by π . The identities

$$w(g + \Delta) = \frac{\sin((2 - g)\lambda)}{\sin \lambda}, \quad w(g - \Delta) = \frac{\sin((1 + g)\lambda)}{\sin \lambda} \quad (3.11)$$

give weights $d, 1$ at zero drop.

Lemma 3.3 (Local cap elimination). *For disk arcs whose boundary intervals contain at most one wire flip, their positive cap-loop factors are d for a zero-flip high starting port and 1 for a zero-flip low starting port or a one-flip arc, up to a bounded overall loop-count error and an error $d^{O(X)}$ for remaining macroscopic arcs. A nest of N straddling arcs at a corner of turn $g\pi$ without a wire flip has negative-expansion weight relative to its caps comparable to $f(g)^{N/2}$.*

Proof. Take a family of disk arcs closed under the arcs inside their disks, with no holes, and remove them from the inside outward. Removing two adjacent remaining endpoints creates a loop exactly when their exterior partners already pair them; otherwise reconnect those partners. The caps still pair successive high-low ports on each remaining interval, with arbitrary pairings of its unmatched ends. A high-low arc within an interval creates one loop. A low-high arc merely transports the pattern. Across a flip, equal high statuses transport an unmatched end without creating a loop. If two transported unmatched ends were already tied, the extra loop erases an entire remaining interval. There are boundedly many such exceptions; also allow the final bounded error for a no-flip circle. The unremoved boundary loops are at most the number of remaining arcs.

Include all sufficiently small arcs, together with arcs inside their disks. Any others contribute a macroscopic passage count X in boundary bands. Along a local straddling nest the starting parities alternate, because ports between consecutive starts pair internally on that side. The two relative weights in (3.11) multiply to $f(g)$ after division by the corresponding cap weights d and 1 . One unpaired factor is bounded. This proves the claimed $f(g)^{N/2}$ comparison. Cutoff patches are disjoint with wider buffers, and a straddling arc omitted because it exits its patch is charged to X . $\square$

3.3 A rectangle: the flat-change and convex-corner costs

Proof of (3.6). Take a convex tile rectangle with both side lengths comparable to n . Put a change in the middle of the bottom side and another in the middle of the top side. Give H drops s, t at these changes and no others. Since $s + t = 1$ and the total boundary turn is 2π , the factors (3.9) are periodic around the contour.

In the positive-sign expansion, an arc not separating the changes uses an interval containing neither change and has its ordinary cap weight. An arc enclosing one change has $w(s) = w(t) = 1$. Closed interior loops have weight d . Lemma 3.3 and Lemma 2.5 therefore bound the spin partition function above by a constant times the positive cap partition Z_+^{cap} . The latter includes the common tile coefficients.

In the negative-sign expansion the change drops are $-s, -t$ and each convex corner contributes $1/2$. An arc containing neither change has an interval containing at most two

corners and has positive weight for both high statuses. An arc separating the changes can use the interval containing the s mark and k corners; its weight is $\sin(k\lambda/2)/\sin\lambda$. These weights are nonnegative and vanish exactly for $k = 0$, namely a straddling arc on the bottom side. Let E be the event that the s -gap vertex has an actual path of its color to another side of the rectangle. Every weight is strictly positive on E , and the local straddles at t have weight d . Relative to the cap law with the complementary wires, the spin partition is consequently at least

$$\mathbb{E}_- \left[\mathbf{1}_E d^{N_t} \prod_{v \text{ corner}} f(1/2)^{N_v/2} C^{-(1+X)} \right]. \quad (3.12)$$

Here X counts macroscopic boundary passages. For instance, choose all small arcs below a fixed diameter fraction so their disks contain at most one feature. The cap elimination accounts for those arcs, and remaining factors on E lie in a fixed finite set of positive values.

Without $C^{-(1+X)}$, the expectation in (3.12) is bounded below by a constant times

$$\exp \left(P + D_* + \sum_v C_{i(v)} \right).$$

Indeed the separated positive local factors compare conditionally by Lemma 2.3. After conditioning on their small patches, extend the local s arm through a fixed corridor avoiding those patches to another side, by the screens and crossing extensions in Lemma 3.1. The remaining law is FK with an induced partition. Lemma 2.7 loses only n^ε upon reinstating the passage factor, because all local first and second moments have polynomial cost by Lemma 2.6.

Choose odd horizontal tile length and even vertical tile length. The four corners then have the same relative gap-versus-wire type, since the left and right sides belong to opposite wire intervals. Compare the two tensor expansions and repeat on this very same rectangle with every wire polarity reversed. The ratios $Z_+^{\text{cap}}/Z_-^{\text{cap}}$ cancel upon multiplication. One run has four type-0 corners and the other four type-1 corners; both have the same P, D_* up to bounded changes of local convention. Taking logarithms proves (3.6) with the error convention (3.5). $\square$

3.4 A long cylinder: balancing the two corner turns

Proof of (3.7). Take a tile cylinder whose spatial period has two positive coordinate components, each comparable to n , with even sum. In a lift, its lower boundary repeats all the east steps followed by all the north steps. The upper boundary is a parallel translated copy at transverse distance of order $Kn \log n$, where K will be a sufficiently large fixed constant. Each boundary has one convex and one concave corner. There are no wire flips and H is constant. Use positive caps for the reference laws and free periodic spin contraction, with no flux projection, for the tensor. The normal-angle lift is periodic.

Every same-boundary raw arc bounds a disk against the boundary interval it cuts off. Its lifted endpoint interval spans less than one period: otherwise its endpoints and those of a translate would interleave, forcing intersecting arcs. Its positive-expansion weights are ordinary cap weights, and its negative-expansion drops are 0 or $\pm 1/2$, so those weights are positive. A contractible interior loop has weight d . An essential loop has total turn zero and hence weight 2; the zero-turn assertion follows by isotopy of a simple generator in the flat cylinder.

Reserve bottom and top bands of width $O(n)$ containing all boundary features and their enlarged buffers. In the empty middle strip let k_0 count essential closed medial loops wholly within that strip, and put

$$M = (2/d)^{k_0}. \quad (3.13)$$

It depends only on middle-strip tiles and is at least one. Under either positive cap law,

$$\mathbb{E}M^p \leq n^{O_p(K)} \quad (p < \infty). \quad (3.14)$$

To prove this, cover the middle strip by $O(K \log n)$ buffered bulk boxes of size a small fraction of the period. Each essential loop supplies a passage, and conditional iteration of Lemma 2.5 with a bounded coloring proves the bound. Essential loops omitted from M , macroscopic boundary arcs, and other macroscopic loops approaching features are charged to a count X in the enlarged boundary bands. This X has all fixed exponential moments uniformly, independently of K .

Let T be the event of a raw arc joining the two boundaries. Then $M = 1$ on T , and

$$\mathbb{P}_\pm(T) \leq Cn^{-cK}. \quad (3.15)$$

Indeed, in each successive transverse band of width comparable to the period, actual crossings can form a noncontractible cycle of one color. Use a necklace of overlapping strip crossings with transversal joins; its buffers are injective on the cylinder. In the lift the connected crossings continue through every translate, and their projection contains a noncontractible cycle. FKG and crossing estimates give a uniform conditional success probability. Such a cycle blocks T , so iteration through order $K \log n$ bands proves (3.15).

The positive expansion has absolute partition value at most

$$Z_+^{\text{cap}} \mathbb{E}_+[MC^{1+X}].$$

Off T , the negative expansion is positive and its integrand relative to Z_-^{cap} is at least

$$M \prod_v f(g_v)^{N_v/2} C^{-(1+X)}, \quad g_v \in \{1/2, -1/2\}.$$

On T , its absolute integrand is bounded by the same local factors times C^{1+X} , without M . These statements are Lemma 3.3; through arcs are macroscopic, so their bounded weights are included in X . Hölder, the polynomial local tilt bounds, and (3.15) bound the possible signed through contribution by $n^{-c'K+O(1)}$ relative to its caps.

Condition on the middle-strip tiles and use the separated feature patches to compare the positive main expectation, without its X factor, to

$$(\mathbb{E}_- M) \exp \left(\sum_{v: g_v=1/2} C_{i(v)} + \sum_{v: g_v=-1/2} L_{i(v)} \right).$$

It is bounded below by an inverse power of n whose exponent is independent of K , because $M \geq 1$. Choose K large enough that the through error is negligible. Now keep K fixed and apply Lemma 2.7, using (3.14), to absorb all $C^{\pm X}$ factors at an arbitrarily small power loss.

Choose the two boundary wire colors and run the comparison, then repeat with both colors reversed on exactly the same cylinder, cutoffs and middle strip. Tensor equality cancels the two ratios of $Z_\pm^{\text{cap}} \mathbb{E}_\pm M$ when the inequalities are multiplied. The two runs together count twice each of C_0, C_1, L_0, L_1 . Taking logarithms and dividing by two proves (3.7). $\square$

3.5 A charged cylinder: the bulk costs

Proof of (3.8). Use a cylinder with straight horizontal boundaries, with even horizontal period comparable to n , and lower boundary traversed eastward. Put the two flat changes on the bottom, separated by order n , and no change on top. Place the finitely many bulk points z_j at mutually separated distances of order n in the bottom band. Their buffered

cutoff patches are disjoint; make the band wide enough to leave clear corridors nearer the bottom. The height is again of order $Kn \log n$.

Draw tile-edge strings from the s -gap to each z_j and multiply the spin crossing its string by $\exp(-i\lambda a_j c)$, where $c = +1$ denotes an arrow along the left normal of the oriented string step. Use raw drops $s-1, t$ for H on the lower boundary, with total zero, and constant H on top. Oriented intersections give the following sign convention: in the positive expansion, a string of charge $-a_j$ shifts a boundary disk arc's cosine argument, in units of λ , by $+a_j$ for a launch in its interval and by $-a_j$ for an endpoint in its disk. The boundary child lies to the arc's right when the arc is followed from the earlier to the later port. Thus the local effective bottom drops are s, t ; in the negative expansion they are $-s, -t$, with $a = \sum_{z_j \text{ in disk}} a_j$ added for the endpoints inside.

For a boundary disk arc the negative-expansion numerators, over the common denominator $\sin \lambda$, are

number of enclosed flips	numerators	
0	$\sin((2-a)\lambda), \quad \sin((1+a)\lambda)$	
1	$\sin(a\lambda) \quad [s], \quad \sin((2-a)\lambda) \quad [t]$	
2	$\sin((3-a)\lambda), \quad \sin(a\lambda).$	(3.16)

The interval spans less than one period, so each change is counted at most once. Top boundary disks use the zero-flip row. Since $0 \leq a \leq 1$ and $0 < \lambda \leq \pi/3$, every entry is nonnegative. A vanishing entry always belongs to a child interval containing s ; this includes the additional zero $\sin(3\lambda) = 0$ at $q = 1$ in the two-flip row. Contractible closed loops in the negative expansion have relative weight $\cos((1-a)\lambda)/\cos \lambda > 0$; their interior is on the left for counterclockwise orientation. The two expansions therefore give the prescribed local weights $u_{\pm}(a_j)$. Essential loops have positive weight $2 \cos(a\lambda)$, where a is the charge separated from the launches, and loops in the empty middle strip have weight 2. Keep M as in (3.13).

Let E be the event that the s -gap vertex connects by actual paths of its color to a noncontractible open cycle. Construct that cycle near the bottom, before the conditioned endpoint patches and middle strip. Conditionally on those patches and the middle,

$$\mathbb{P}_-(E \mid \text{patches and middle}) \gtrsim e^P.$$

This follows from the local arm comparison, same-color FKG and crossing extension, and the periodic necklace construction used for (3.15). On E no child disk arc can cut off s , since the noncontractible cycle cannot fit behind it, and there is no through arc.

Set $U_{\pm} = \prod_j u_{\pm}(a_j)^{N_j}$ and let $Y_{\pm}$ be the spin integrands relative to their respective positive cap laws. Local cap and loop counting gives

$$\begin{aligned} |Y_+| &\leq MU_+ C^{1+X}, \\ Y_- &\geq MU_- C^{-(1+X)} \quad \text{on } E, \\ Y_- &\geq 0 \quad \text{off } T. \end{aligned}$$

The local t -straddles in the negative expansion contribute d each, which may be discarded in this lower bound. A loop whose endpoint enclosure differs from a local nest has a macroscopic passage near a feature: a contractible disk containing z_j must separate it from the bottom. Thus it is charged to X , with all cutoffs small compared with the feature separations. On the through event T one still has

$$|Y_-| \leq U_- d^{N_t} C^{1+X}, \quad M = 1.$$

All these local costs are polynomial and the through probability is $O(n^{-cK})$, so choose K fixed and large to dominate the signed through error.

Conditioning and Lemma 2.3 compare the separated local factors to $\prod_j F_{\pm}(a_j)$, the middle factor to $\mathbb{E}_{\pm}M$, and the event E to its lower arm cost e^P . Lemma 2.7 removes the passage errors at n^ε cost. Repeat with every wire designation reversed, using the same domain, strings, drops, cutoffs and M . Equality of the two tensor expansions cancels both cap and middle-factor ratios upon multiplying the two comparisons. The resulting inequality is

$$2P + 2 \sum_j \log \frac{F_-(a_j)}{F_+(a_j)} \leq o(\log n),$$

which gives (3.8). $\square$

3.6 The polygon budget

We now combine the rectangle and corner comparisons for a polygon with balanced corner types, retaining a strictly negative contribution from the arm logarithm P . The final assertion records the separate charge estimate in the form needed below.

Corollary 3.4 (Balanced corner parities). *Let a fixed simple rectilinear polygon have m concave and $m + 4$ convex true corners, with wire changes in side interiors. It has tile approximants differing by $O(1)$ mesh units such that at least $\lfloor m/2 \rfloor$ corners of each relative wire type occur in each of the convex and concave sets, also after global wire reversal. For these approximants,*

$$3P + D_* + \sum_{v \text{ convex}} C_{i(v)} + \sum_{v \text{ concave}} L_{i(v)} \leq \frac{1}{2}P - \frac{3}{2}D_* + o(\log n). \quad (3.17)$$

Each fixed family of negative endpoint charges with magnitudes summing to one has total wrong-to-correct local logarithmic cost at most $-P + o(\log n)$.

Proof. First prescribe the desired relative corner types for one designation. There remain at least four spare convex corners, one of which adjusts the parity of the sum of the absolute corner parities. If b_i denotes the integer coordinate parity of side i , a corner between sides $i - 1$ and i has parity $p_i = b_{i-1} + b_i$ modulo two. This cyclic system is solvable exactly when $\sum_i p_i = 0$ modulo two. Choose the side coordinates with those parities within boundedly many mesh units of the prescribed ones. For the fixed simple polygon, sufficiently small independent shifts preserve simplicity. Global wire reversal just exchanges the relative types.

Pair the balanced convex and concave sets and apply (3.7). If m is even, four convex terms remain; if m is odd, five convex terms and one nonpositive concave term remain. Since $C_i \geq 0$, Equation (3.6) implies

$$P + D_* \leq o(\log n), \quad C_i \leq -\frac{1}{2}(P + D_*) + o(\log n).$$

The first inequality permits paying for five convex terms even when only four remain, at an additional one-sided $o(\log n)$ error. This yields (3.17). The same first inequality is the corresponding flat budget when there are no corner factors. The endpoint-charge assertion is exactly (3.8). $\square$

4 Angular transfer and its two leading modes

Throughout this section, $1 \leq q < 4$, $d = \sqrt{q} = 2 \cos \lambda$, $\rho = \lambda/\pi \in (0, 1/3]$, $v = \pi/2$, and $Q = -e^{i\lambda}$. The argument has two parts. We first construct a Hilbert space from infinite-plane row correlations and prove an angular estimate there. We then identify its scalar products with reflection across diagonal cuts of the physical square array. Boundary conditions enter the resulting estimate through the norms of the exterior states.

4.1 Rows, twists, and the estimate

The local row algebra is the six-vertex form of the planar loop transfer formalism [4, 51]. Its finite identities are verified below; the infinite-plane reflection space and the angular decay estimate require the additional analytic arguments of this section.

In the spin basis $+, -$, set

$$J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad L_- = |- \rangle \langle +|, \quad L_+ = | + \rangle \langle -|, \quad K(\zeta) = \text{diag}(\zeta, \zeta^{-1}).$$

Here and below twists satisfy $|\zeta| = 1$. In the ordered pair basis $++, +-, -+, --$, let

$$R(u) = \begin{pmatrix} a(u) & 0 & 0 & 0 \\ 0 & b(u) & 1 & 0 \\ 0 & 1 & b(u) & 0 \\ 0 & 0 & 0 & a(u) \end{pmatrix}, \quad a(u) = \frac{\sin(\lambda - u)}{\sin \lambda}, \quad b(u) = \frac{\sin u}{\sin \lambda}. \quad (4.1)$$

Time and space steps have unit length and angles α_t and ω_j . Their successive sums give a rhombic tile array; initially $0 < \omega_j - \alpha_t < \pi$. A spin is positive when its arrow points along the left normal of the corresponding step. If h_0, h_1 are the two time sides before and after advancing in space, and s_0, s_1 the two space sides before and after advancing in time, the tile tensor has output indices (h_0, s_1) , input indices (h_1, s_0) , and value $R(\rho(\omega_j - \alpha_t))$.

On a spatial circle of even size N , write

$$T_N^D(\alpha) = \text{Tr}_c D_c R_{c0}(\rho(\omega_0 - \alpha)) \cdots R_{c, N-1}(\rho(\omega_{N-1} - \alpha)). \quad (4.2)$$

The auxiliary index is c . At the seam, $D_{x,y}$ uses the spin x from tile -1 and y from tile 0 . Let Ω_N be the positive unit Perron vector of the untwisted rows on the zero-total-spin sector, and $\Lambda_N(\alpha)$ their eigenvalue. Normalized rows are $A_N^D(\alpha) = T_N^D(\alpha)/\Lambda_N(\alpha)$. Their common Perron data, and the limiting operators $A^D(\alpha)$, are constructed below. In the homogeneous transverse array $\omega_j = v$, put

$$\begin{aligned} P_\zeta(z) &= A^{K(\zeta)}(z), & M(z) &= e^{i\tau\rho z} A^D(z), \\ N(z) &= M(z)P_\zeta(z) + P_\xi(z)M(z), \end{aligned} \quad (4.3)$$

where

$$\zeta/\xi = Q^\epsilon, \quad \epsilon \in \{-1, 1\}, \quad (D, \tau) = (L_-, \epsilon) \text{ or } (L_+, -\epsilon). \quad (4.4)$$

The two summands of N describe the two possible positions of a source in two consecutive rows of the same angle. Products act from right to left: rows before the source carry twist ζ , and rows after it carry twist ξ . Thus a source in the second row gives MP_ζ , whereas one in the first gives $P_\xi M$. The phase in M records the direction of the source row. Section 5 will identify this sum with a two-step path increment; here we prove the row estimate independently of the path's exterior geometry.

Theorem 4.1 (Angular transfer estimate). *Fix a compact interval $I \subset (-v, v)$. Let B_{in} be a product of P_ζ rows and B_{out} a product of P_ξ rows, at real angles in $[-v, v]$. Suppose each product contains at least $n \geq 1$ angles in I . Uniformly for $x, y \in [-v, v]$,*

$$\|B_{\text{out}}N(x)B_{\text{in}}\| \leq Cn^{-1+\rho}, \quad (4.5)$$

$$\|B_{\text{out}}\{N(x) - e^{i\tau(x-y)}N(y)\}B_{\text{in}}\| \leq Cn^{-1-\rho}. \quad (4.6)$$

The constant depends on q and I . Boundedly many additional pairs of before-twist rows may be inserted immediately to the right of N , and additional pairs of after-twist rows immediately

to its left. The two terms in (4.6) may use different such insertions; on each side their counts must be even. The constant may depend on a bound for these counts. Multiplication of D by a common scalar multiplies the bounds by its absolute value.

The same estimates hold in the physical square array sliced in a diagonal time direction: its time angles are $\pm\gamma_0$, where $\gamma_0 = \pi/4$, and its space angles alternate between $\pi - \gamma_0$ and γ_0 . The exterior states may contain any finite number of physical row and spin insertions outside the sandwich. A scalar contraction is bounded by the displayed rate times the two reflection norms, with the squares centered on a cut included in its reflection kernel.

The improvement in the second estimate comes from removing one angular mode, not from assuming that the exterior states have bounded norms. We first construct the plane reflection space and continue its rows analytically in the angle. Boundary unitarity then controls the scalar spectral functions near modulus one. A product of n strict rows selects spectral energies of order n^{-1} . On those low bands, endpoint reversal singles out the mode $e^{i\tau x}$ with bound $O(n^{-1+\rho})$; the next odd mode and the remaining errors have bound $O(n^{-1-\rho})$. The combination in (4.6) cancels the former. The final subsection transfers this estimate, with its exterior norms, to diagonal cuts of the physical square array.

4.2 Row algebra and the infinite-plane state

The finite-dimensional identities used throughout are

$$R_{12}(x-y)R_{13}(x)R_{23}(y) = R_{23}(y)R_{13}(x)R_{12}(x-y), \quad (4.7)$$

$$R(u)R(-u) = (1 - b(u)^2)I, \quad (4.8)$$

$$J_c R(u)^{t_c} J_c = R(\lambda - u). \quad (4.9)$$

For completeness, the nontrivial one-minus block of (4.7) is symmetric precisely by the three sine identities

$$b(x-y) + a(x)b(y) = b(x)a(y), \quad b(y) + a(x)b(x-y) = b(x)a(x-y),$$

$$a(x) + b(x-y)b(y) = a(x-y)a(y).$$

Spin reversal gives the one-plus block; the remaining blocks are scalar. The other identities follow directly from (4.1). Sliding the intertwiner in (4.7) through two complete rows shows that transfers with a common diagonal twist commute: the intertwiner preserves total spin and hence commutes with the tensor square of that twist. For arbitrary insertions D, E , the same operation conjugates $D \otimes E$ by a two-spin intertwiner. Expanding the conjugated matrix in matrix units expresses the exchanged rows as a finite sum with new insertions. Its coefficients are analytic wherever that intertwiner is invertible. Division by the scalar Perron eigenvalues preserves this exchange identity.

For strict angles, the untwisted transfer is primitive on each total-spin sector. It has positive diagonal entries; moreover an auxiliary spin can swap with any chosen opposite pair of site spins, without swapping at the other sites. Such exchanges connect the sector. Commutation therefore gives common positive left and right Perron vectors. In the homogeneous array, auxiliary transposition reverses the product in (4.2), since the distinct site operators commute. Equation (4.9) then gives

$$A^D(\alpha)^* = A^{J\bar{D}J}(-\alpha) \quad (\alpha \in \mathbb{R}). \quad (4.10)$$

The corresponding finite-volume identity makes the left and right vacua equal.

More explicitly, a formal row word h specifies a finite list $(D_1, \alpha_1), \dots, (D_m, \alpha_m)$ of matrices and strict angles. Its finite-circle vector and vacuum coefficient are

$$h_N = A_N^{D_m}(\alpha_m) \cdots A_N^{D_1}(\alpha_1) \Omega_N, \quad \langle \Omega_N, h_N \rangle.$$

For two such words h, g , the proposed Gram form is

$$\langle h, g \rangle = \lim_{N \rightarrow \infty} \langle h_N, g_N \rangle.$$

The following lemma establishes these limits. The empty word becomes Ω , and prefixing a row to the operator product defines its left action. A bare vacuum coefficient means the same limit with the empty word in the first slot; the adjoint identity expresses every Gram coefficient as such a coefficient of a longer word.

Lemma 4.2 (Plane kernels). *Finite words of homogeneous normalized strict-angle rows have limiting Gram matrices as $N \rightarrow \infty$. Their quotient and completion give a separable Hilbert space $\mathcal{H}$ with unit vector Ω and bounded left row actions satisfying*

$$\|A^D(\alpha)\| \leq 2\|D\|. \quad (4.11)$$

Bare vacuum coefficients are continuous in strict angles and are unchanged by a common translation of all time angles, as long as the translated angles remain in $(-v, v)$.

Proof. We first identify the plane state, including the effect of the zero-spin projection. Take periodic lists of time and space angles in a strict range, with the two direction ranges ordered and transversely separated in a half-circle. Use tori whose period sizes are even multiples of the list periods. The resulting rhombic square array is embedded and has bounded angles. Indeed the cyclic order at each vertex gives local injectivity, while averages of runs of either family have uniformly positive length and uniformly transverse directions. Thus index distances control Euclidean distances in both directions, giving properness and global embedding.

Expand each R into its two adjacent-side pairings with oriented turn weight $e^{i\rho W/2}$. The pairing at the corner between the two positive steps, and at its opposite corner, has coefficient $a(u)$; the other pairing has coefficient $b(u)$. For example, the mixed entry follows from $a(u)e^{iu} + b(u)e^{-i(\lambda-u)} = 1$ and its conjugate. In a single-pairing entry the two turns cancel. A contractible loop therefore has weight d . An essential simple oriented torus loop has total turn zero. All essential loops are parallel up to orientation. Their number k is even, by the even intersection parities with both period cycles and primitivity of their common homology class. The allowed orientation count after projection onto zero spin on either chosen quantum section is

$$\chi = 2^k \quad \text{if the intersection is zero,} \quad \chi = \binom{k}{k/2} \quad \text{otherwise.} \quad (4.12)$$

Since $d < 2$, there is $c(d) > 0$ such that $c(d)d^k \leq \chi \leq 2^k$ for every even k .

Replacing χ by d^k gives the canonical isoradial FK law, up to the bounded torus genus factor of Lemma 2.1. To check the parameter, the relative open weight at a primal corner between the positive steps is $\sqrt{q}b/a$, and at the other primal corner it is $\sqrt{q}a/b$. If γ is the angle at the primal corner, the odds divided by $\sqrt{q}$ are

$$\frac{\sin(\rho\gamma)}{\sin(\rho(\pi - \gamma))}.$$

This is the canonical weight, with the complementary-angle convention of [16, Equation (1.1), Theorem 1.1, and Equation (1.3)]. That theorem applies to these fixed doubly periodic

bounded-angle isoradial grids. It gives uniqueness and buffered crossing estimates, uniformly over the boundary partition on the outer buffer. The bulk passage-count argument of Lemma 2.5 consequently applies on each such grid. No estimate as an angle degenerates is used.

Here is why the essential-orientation tilt does not change the local limit, even for unbounded aspect ratios. Let $r \leq R$ denote the shorter and longer period sizes. Cover the torus by $O(R/r)$ buffered boxes of size a small fraction of r , with a bounded number of disjoint-buffer colors, so that every essential loop crosses one of them. Include $O(R/r)$ translates, by medium periods, of a target annulus about the query. Let Y be the sum of the passage counts in this cover. For each fixed $c > 0$, conditional passage moments, iteration within each color, and Hölder's inequality give

$$\mathbb{E}e^{cY} \leq e^{C_c R/r}, \quad k \leq Y. \quad (4.13)$$

The density of the tilted law is bounded above by $Ce^{c_0 Y}$ before normalization and its normalizing denominator is bounded below. Consequently $\mathbb{E}_{\text{tilt}} e^Y \leq Ce^{CR/r}$, and Jensen's inequality gives $\mathbb{E}_{\text{tilt}} Y \leq CR/r$. Translation invariance then bounds the expected count at the target annulus uniformly.

Condition on the exterior of its inner box. The number of exterior excursions from that boundary to distances comparable to r before returning is tight. If it is at most K , changing the filling can alter only $O(K)$ essential loops: every affected loop must leave the injective chart along such an excursion. The common homology class can change only when no essential loop remains unaffected. Thus the ratios of the genus and orientation tilt over all fillings are bounded in terms of K, q . For the binomial case this also follows directly by comparing central binomial coefficients with indices differing by $O(K)$; a switch between the two cases in (4.12) can occur only when there are $O(K)$ essential loops. The conditional inner law is therefore comparable with FK under the induced partition. Apply this density comparison once to the whole event that an arm or medial strand from a fixed window reaches distance comparable to r . On the event of at most K exterior excursions, its conditional probability is at most $C(K, q)$ times the corresponding FK bound. Successive buffered circuits make that FK bound tend to zero as $r \rightarrow \infty$. The excursion count is tight, so the complementary probability becomes arbitrarily small by then letting $K \rightarrow \infty$. No density factor is accumulated separately at each circuit scale.

Now condition outside a large fixed test box containing the query, and let z_0 be its center. Choose a fixed $\eta > 0$ so small that the torus metric ball $B_T(z_0, 2\eta r)$ has an injective planar lift; the transverse period bounds above give this uniform choice. For large r the fixed box lies in $B_T(z_0, \eta r/2)$. The preceding escape estimate shows that, with probability tending to one, no exterior actual component or medial arc incident to the box reaches the rim of $B_T(z_0, \eta r)$. Every such object is connected and meets the box, so their entire union, together with every possible filling of the box, lies in the same injective disk $B_T(z_0, 2\eta r)$. Only exterior components and arcs incident to the box can be changed by a new filling. Every changed primal component is therefore planar, so its genus contribution is unchanged; every changed medial loop is contractible. The essential loop count, homology class, and orientation factor therefore remain fixed. The conditional law of the filling is ordinary FK. Letting the test box grow, comparison of increasing cylinder events with free and wired laws and uniqueness identifies the local plane law.

The same conclusion holds with finitely many marked matrices D_i . Given the unoriented pairing, the absolute marked orientation ratio is at most $\prod_i 2\|D_i\|$: only marked loops are broken, each newly cut arc has at most two orientations, unaffected contractible loops retain weight d , and any residual flux constraint has no more orientations than the central count in (4.12). When all marked strands stay in a fixed window, they do not encounter the distant flux projection. The absence of escaping strands therefore proves marked local convergence too.

Sending the time length to infinity first gives the Perron-normalized coefficient at fixed spatial circumference. Sending the circumference to infinity, and diagonalizing the two limits, gives the stated plane coefficient. Any finite word can be included in a periodic list; unmarked filler rows disappear from its normalized longitudinal vacuum coefficient.

There is also a transverse quantization. Take a circle in the time index, transfer in decreasing space index, and let the auxiliary space spin flow in increasing time. The same difference relation makes the untwisted transverse transfers commute and gives common Perron data. The seam marks of the original word become one tensor-product quantum insertion. If the time circle has length L , write this insertion as $\mathcal{D}_L = \bigotimes_{t=0}^{L-1} D_t$, where $D_t = I$ at unmarked times, using the seam-index convention of (4.2). Let ℓ_L, r_L be the common left and right zero-spin Perron vectors of the untwisted transverse transfers. Sending the transverse cylinder length to infinity gives the normalized coefficient

$$\frac{\langle \ell_L, \mathcal{D}_L r_L \rangle}{\langle \ell_L, r_L \rangle}.$$

The transverse Perron eigenvalues cancel between the marked and unmarked partitions. Their common eigenvectors, and hence this coefficient, do not depend on the strict transverse angles. Sending $L \rightarrow \infty$ gives the same plane coefficient as the longitudinal limit already identified; this is why changing transverse angles does not change it. Rotating both families and then returning the transverse angles to v proves common time-angle translation invariance. Continuity follows from the finite-box FKG sandwich and uniqueness as a common finite periodic medium varies; sufficiently large fixed circuits at the limiting medium control long marked strands, and all remaining local orientation factors are continuous.

Finite-volume positivity now gives the limiting Gram form. To prove boundedness before completing, let h be a formal word vector and set $T = (A^D)^* A^D$, with the adjoint understood first as the reflected formal row in (4.10). The marked expansion bounds $\|T^j h\|^2 \leq C_h (2\|D\|)^{4j}$. Iterated Cauchy–Schwarz applied to the positive form $\langle h, Th \rangle$ gives

$$\|A^D h\|^2 \leq \|h\|^{2-2^{1-m}} \|T^{2^{m-1}} h\|^{2^{1-m}}.$$

Letting $m \rightarrow \infty$ gives (4.11); the same inequality shows that null vectors remain null. Continuity of kernels, and words with rational angles and matrix entries, give separability. $\square$

We may initially enlarge this construction by finite-position quantum spin insertions and shifts. Moving a shift through a word only translates finitely many markings, so the same proof gives the enlarged Gram space and bounds. In that space the endpoint rows are already defined by

$$A^D(v) = S D_0, \quad A^D(-v) = (J D^t J)_0 S^{-1}, \quad (4.14)$$

where $(S\sigma)_j = \sigma_{j+1}$. These formulas follow from $R(0)$ being the swap and (4.9); their vacuum eigenvalue is one. We next show that these endpoints preserve the original cyclic space $\mathcal{H}$.

4.3 A self-adjoint angle generator

For a formal word h , translate all its angles by t and call the result $V_t h$, for t in a neighborhood of zero where the word is strict. Reflection in its Gram formula negates the angles in the reflected word. Common translation invariance therefore gives

$$\langle V_t h, V_s g \rangle = \langle h, V_{s+t} g \rangle \quad (4.15)$$

whenever both sides are defined. These curves are continuous. If h is null, then $\|V_t h\|^2 = \langle h, V_{2t} h \rangle = 0$ for small t , so the translation is locally well defined on actual vectors.

Lemma 4.3 (Angle generator). *There is a self-adjoint operator B on $\mathcal{H}$ such that $V_t h = e^{tB} h$ locally for each word h . Ordinary strict-angle words form a graph core for B . Each family A^D extends to a bounded holomorphic operator function on $|\operatorname{Re} z| < v$, with*

$$A^D(x + ib) = U_b A^D(x) U_{-b}, \quad U_b = e^{ibB}. \quad (4.16)$$

Proof. For each word h , the function $f_h(t) = \langle h, V_t h \rangle$ is continuous and exponentially convex on a sufficiently small interval: every permitted Hankel matrix is a Gram matrix by (4.15). The interval Bernstein–Widder theorem, in the form of [7, Theorem 3.4 and Corollary 3.5, p. 735], represents f_h as a Laplace transform of a positive measure with finite integrals throughout that interval. In particular it is analytic near zero. Inner products of finite differences of $V_t h$ are the corresponding differences of $f_h(s + t)$. Their limits show that all Hilbert-space derivatives $h^{(k)}$ exist and satisfy

$$\|h^{(k)}\|^2 = f_h^{(2k)}(0) \leq C_h (2k)! r_h^{-2k}. \quad (4.17)$$

The Taylor series recovers the vector curve: this follows by comparing its inner products and its squared norm using (4.15).

On the span of all these derivative vectors, define $B_0 h^{(k)} = h^{(k+1)}$. The cross versions of (4.15) make this well defined and symmetric. More explicitly, if a linear combination of derivatives vanishes, its proposed derivative has zero inner product with every word by differentiating the cross identity; density makes that derivative zero. The same identity gives symmetry between derivative vectors. The domain is dense and invariant. Bound (4.17), also with k replaced by $k + j$, implies that every vector in this domain is analytic for this same operator: for some $t > 0$ its series $\sum_{j \geq 0} t^j \|B_0^j h^{(k)}\|/j!$ converges. Nelson’s analytic-vector theorem [34, Lemma 5.1], in the form stated in Neeb [33, Definition 4.7 and Theorem 4.8], makes B_0 essentially self-adjoint. Denote its closure by B .

Finite differences of ordinary words approximate $h^{(k)}$ both in norm and after applying B , by the Taylor series just obtained. Ordinary words are thus a graph core. The vector series gives $V_t h = e^{tB} h$ for small real t on each word.

For fixed D , the row family is strongly continuous, by the Gram continuity and uniform bound. The word identity $A^D(x + t) V_t h = V_t (A^D(x) h)$ yields, weakly on the graph core,

$$\frac{d}{dx} \langle g, A^D(x) h \rangle = \langle Bg, A^D(x) h \rangle - \langle g, A^D(x) Bh \rangle.$$

Integrate this equality and approximate g, h in graph norm to extend it to $\operatorname{Dom} B$. The extension in Equation (4.16) then satisfies the Cauchy–Riemann equations between domain vectors, with continuous partial derivatives. Its uniform bound $2\|D\|$ extends weak holomorphicity from that dense set to all vectors, hence gives bounded operator holomorphicity throughout the strip. $\square$

Words whose angles all lie in the middle third $(-v/3, v/3)$ already span densely: test orthogonality to them and apply analyticity successively in every argument of a word. For a fixed twist, analytic continuation of the commuting-row identity and (4.10) gives

$$[P_\zeta(z), P_\zeta(w)] = 0, \quad P_\zeta(z)^* = P_\zeta(-\bar{z}). \quad (4.18)$$

Thus this family lies in an abelian von Neumann algebra. Decompose its representation into cyclic summands, or equivalently into a direct integral with multiplicity spaces: each algebra element acts on a fiber by a scalar times the identity, while Hilbert-space vectors remain vector-valued on that fiber. The cyclic representation and spectral calculus are recalled in Peterson [38, Sections 3.8 and 4.3]. Its simultaneous spectral variables can therefore be

chosen as scalar functions $f(z)$, analytic almost everywhere on the strip, bounded by 2, and satisfying $f(-\bar{z}) = \overline{f(z)}$. To obtain one common exceptional set, start with a multiplication representation of the algebra generated by countably many arguments in a dense set, and use the norm-bounded Taylor series on countably many discs. Covariance (4.16) says that $f(z) \mapsto f(z + ib)$ preserves the spectral measure class.

4.4 Continuation through the two edges

To estimate long products of rows, we next need uniform control of their scalar spectral functions up to the two boundary lines of the strip.

Lemma 4.4 (Scalar continuation in a fixed band). *Almost every spectral function f extends analytically and without zeros to bands of one fixed positive width about the lines $\operatorname{Re} z = \pm v$. It has modulus one on those lines and satisfies $|f(z)| \leq 1$ in the intervening strip. At any specified point of the strip, the set of spectral functions vanishing there has measure zero.*

Proof. We first obtain a uniform finite-volume vacuum estimate. Temporarily write $T_D(u) = T_N^D(v - u/\rho)$. For u in a fixed sufficiently small complex disc about zero, both

$$T_D(u)T_I(u)^{-1}, \quad T_I(u)^{-1}T_D(u) \quad (4.19)$$

are holomorphic with norms at most $C\|D\|$, uniformly for all sufficiently large N .

To prove the first assertion, multiply by the transfer $\tilde{T}_I(-u)$ having reversed auxiliary order. Fold its auxiliary index c' by transposition. The product is the two-auxiliary partial trace of $(D \otimes I) \prod_j G_j$, in increasing site order, where

$$G_j = R_{cj}(u)R_{c'j}(-u)^{t_{c'}}.$$

The line spanned by $|++\rangle + |--\rangle$ is invariant in the auxiliary space, and on this line G_j is $r(u)I$, where $r(u) = 1 - b(u)^2$, by inversion. At $u = 0$ the range of G_j is entirely in that line tensored with the site, by the swap identity. In the common triangular decomposition, $G_j/r(u)$ has identity on this line, a complementary diagonal block of norm at most $C|u|$, and a bounded off-diagonal block. A product of these matrices has uniformly bounded norm, since the off-diagonal block is bounded by a geometric series. Its partial trace without D differs from the identity by at most $C(C|u|)^N$: the off-diagonal block has zero partial trace, the line block gives the identity, and the remaining diagonal product has the stated norm. This partial trace is invertible on one fixed disc, proving the first bound in (4.19). Reversing the product in G_j gives the same argument with the invariant covector, proving the second. Crossing gives the corresponding bounds near $u = \lambda$.

The common vacuum is an eigenvector of $T_I(u)$ by analyticity, and its eigenvalue is nonzero in these discs. Hence $A_N^D(z)\Omega_N$ has a uniform vector bound in discs of a fixed radius about $z = \pm v$. Pairing with each fixed limiting word, normal-family compactness gives analytic subsequential limits; their values on the interior fix them uniquely. The vector norm bound identifies these limits with a bounded holomorphic $\mathcal{H}$ -valued function. To identify its endpoint values, Cauchy's derivative estimate on a smaller fixed disc gives, uniformly in N ,

$$\|(A_N^D(x) - A_N^D(e))\Omega_N\| \leq C_D|x - e|, \quad e \in \{-v, v\},$$

for real interior x sufficiently close to e . The enlarged Gram construction includes the endpoint swap vectors from (4.14) and their mixed scalar products with strict row words. Passing to those Gram limits preserves the squared norm inequality. Letting $x \rightarrow e$ therefore identifies the continued vacuum vector with the swap vector strongly in the enlarged space. This is

convergence in the angle, not a claim of strong convergence between different finite-circle spaces.

Now commute this endpoint row through a word with all angles in $(-v/3, v/3)$. Each exchange uses an intertwiner whose inverse is analytic on one fixed disc about the endpoint, independent of the word: the middle-third angle restriction keeps its parameter uniformly away from inversion zeros. The exchange expresses the result as a finite sum of bounded interior rows applied to the continued vacuum vectors. It follows that the endpoint preserves $\mathcal{H}$ and that $A^D(z)h$ continues holomorphically on this same disc for every middle-third word h . Its bound may depend on the word, because the number of exchanges does. This does not alter the radius. Strong convergence of interior rows to the endpoint holds first on this dense set and then on all of $\mathcal{H}$, by the uniform interior operator bound.

Choose a countable dense set of middle-third words h_j , and a finite equivalent measure in the spectral representation of P_ζ . If r is the common continuation radius, write the continued vector as $F_j(z) = \sum_{k \geq 0} a_{jk}(z - v)^k$ at the right endpoint. For $r' < R < r$, Cauchy's estimate gives

$$\|a_{jk}\|_{L^2} \leq M_{j,R} R^{-k}, \quad \sum_{k \geq 0} (r')^k \|a_{jk}\|_{L^2} < \infty.$$

Since the measure is finite, the corresponding pointwise series of fiber norms is summable almost everywhere. Intersect these full-measure sets over the countable core and a countable family of smaller radii. We obtain pointwise holomorphic vector series on the same disc for every j . The left endpoint is identical. On its interior portion the vector equals $f(z)h_j$. Analyticity consequently makes the continued fiber proportional to h_j everywhere. At almost every spectral point at least one vector in the countable dense set is nonzero; ratios using any such vector agree on the interior and hence throughout the disc. This constructs the scalar continuation with one common radius.

The swap formulas show that $P_\zeta(v)$ and $P_\zeta(-v)$ are unitary inverses. Strong boundary convergence and (4.16) give unitary horizontal limits at $\pm v + ib$. For countably many b within the continuation discs, the scalar value there has modulus one almost everywhere. Continuity gives modulus one on each full vertical diameter. Schwarz reflection across that diameter now implies $f(z)\overline{f(2v - \bar{z})} = 1$, and the analogous identity at $-v$, throughout the symmetric disc. Thus there are no zeros there. Rational imaginary translations preserve null sets and cover both boundary lines by discs of the same radius. This gives the asserted fixed bands.

The bounded-function maximum principle on the strip gives $|f| \leq 1$. Finally each nonzero analytic f has isolated zeros, so Fubini's theorem shows that on a fixed vertical orbit almost every argument has no zero for almost every spectral function. The covariance of the spectral measure class transfers this null-set assertion to any prescribed point of that orbit. $\square$

4.5 Inner factorization and low spectral bands

Use $w = e^{iz}$ to map $|\operatorname{Re} z| < v$ to the right half-plane. We keep the notation f for the resulting function. Lemma 4.4 says that f is inner, continues through every finite nonzero point of the imaginary axis, and that all its zeros satisfy

$$\operatorname{Re} w_j \geq c|w_j| \tag{4.20}$$

for one $c > 0$ independent of the spectral point. The disk factorization and its boundary-spectrum description [1, pp. 1367–1368, Equations (1)–(2)], transported by $(w - 1)/(w + 1)$, therefore give

$$f(w) = \eta e^{-aw - b/w} \prod_j \beta_j(w), \quad a, b \geq 0, \quad |\eta| = 1. \tag{4.21}$$

Here β_j is $(w - w_j)/(w + \overline{w_j})$ with the canonical constant phase, chosen positive at $w = 1$ when $w_j \neq 1$. The zeros satisfy the half-plane Blaschke condition $\sum_j \operatorname{Re} w_j / (1 + |w_j|^2) < \infty$. There is no other singular factor: analytic continuation across the rest of the boundary excludes singular measure there, leaving only atoms at 0 and ∞ . Under the Cayley map these two atoms give exactly b/w and aw .

Suppose $-\log|f(1)| \leq s$, where $s > 0$ is sufficiently small. There is then no zero at 1, and evaluation of (4.21) gives

$$a + b + \frac{1}{2} \sum_j \log \left(1 + \frac{4 \operatorname{Re} w_j}{|1 - w_j|^2} \right) \leq s. \quad (4.22)$$

The cone condition implies, with a fixed constant,

$$a + b + \sum_j \min\{|w_j|, |w_j|^{-1}\} \leq Cs. \quad (4.23)$$

Indeed, for $r = |w_j|$, the fraction inside the logarithm is at least $4cr/(1+r)^2$, and the resulting logarithm is bounded below by a positive constant times $\min(r, r^{-1})$.

Choose $c_1 > 0$ sufficiently small. On the half-annulus

$$c_1^{-1}s < |w| < c_1/s, \quad |\arg w| \leq \pi/2, \quad (4.24)$$

f and f^{-1} are bounded by fixed constants. On the unit semicircle,

$$f(w) = f(1) + O(s). \quad (4.25)$$

To see both assertions directly, (4.23) separates every zero into $|w_j| \leq Cs$ or $|w_j|^{-1} \leq Cs$. For a small zero the ratio $\beta_j(w)/\beta_j(1)$ differs from one by at most $C(1+|w|^{-1})|w_j|$; for a large zero its error is at most $C(1+|w|)/|w_j|$. Their logarithms are summable and uniformly bounded on (4.24), after decreasing c_1 . The exponential factor is controlled by $a|w-1|+b|w^{-1}-1|$. On $|w| = 1$ the same estimates are $O(s)$. This proves the claims, including bounded invertibility.

For a twist ζ , define the nonnegative spectral operator

$$H_\zeta = -\log|P_\zeta(0)|, \quad E_\zeta(s) = \mathbf{1}_{[0,s]}(H_\zeta).$$

The no-zero assertion of Lemma 4.4 makes this definition unambiguous almost everywhere. Since $P_\zeta(0)$ is self-adjoint, split $E_\zeta(s)$ further according to its sign, denoting these projections by $E_\zeta^\sigma(s)$, $\sigma = \pm 1$. Equations (4.24)–(4.25) say that the compressed P_ζ is boundedly invertible on the half-annulus and

$$P_\zeta(w)E_\zeta^\sigma(s) = \sigma E_\zeta^\sigma(s) + O(s) \quad (|w| = 1, \operatorname{Re} w \geq 0). \quad (4.26)$$

4.6 Reversal, Laurent modes, and the sandwich

The low-band estimates now control the surrounding twisted rows. It remains to identify the leading angular mode of N and the improved decay obtained by canceling it in (4.6).

The endpoint formulas give the exact reversal identity

$$M(v) + P_\xi(v)M(-v)P_\zeta(v) = 0. \quad (4.27)$$

For example, if $D = L_-$, then $K(\xi)DK(\zeta) = (\zeta/\xi)D$ and

$$e^{-i\tau\rho v}(\zeta/\xi) = -e^{i\tau\rho v}.$$

The raising case is the analogous computation, with $\tau = -\epsilon$. The identity at $-v$ follows using the inverse endpoint twists. Imaginary covariance extends these relations after every common imaginary shift: both M terms acquire the same scalar multiplier. Boundary convergence is strong.

Compress M between the low bands of the two twists:

$$M_c(w) = E_\xi(s)M(w)E_\zeta(s).$$

For $\operatorname{Re} w < 0$ define it from the right half-plane by

$$M_c(-w) = -P_\xi(w)^{-1}M_c(w)P_\zeta(w)^{-1}, \quad \operatorname{Re} w > 0, \quad (4.28)$$

where the inverses are only on the stated low bands. The two versions of (4.27) give equality on both imaginary rays. Uniform bounds and strong boundary continuity allow weak holomorphic gluing, for instance by Morera's theorem on small rectangles crossing the rays. Thus M_c is holomorphic on the full annulus (4.24). Throughout it,

$$\|M_c(w)\| \leq C|w|^{\tau\rho}. \quad (4.29)$$

Write its Laurent expansion as $\sum_{j \in \mathbb{Z}} w^j M_{c,j}$. Cauchy estimates on circles near the inner or outer edge give

$$\|M_{c,j}\| \leq C(C's)^{|j-\tau\rho|}. \quad (4.30)$$

On $|w| = 1$, summing these estimates yields

$$\|M_c(w)\| = O(s^\rho), \quad (4.31)$$

$$\frac{M_c(w) - M_c(-w)}{2} = w^\tau M_{c,\tau} + O(s^{1+\rho}), \quad \|M_{c,\tau}\| = O(s^{1-\rho}). \quad (4.32)$$

Here the nearest odd integer to $\tau\rho$ is τ ; the next possible distance is $1 + \rho$. This is where $0 < \rho \leq 1/3$ is used in the ordering of the relevant exponents.

Refine the left and right low bands by signs σ_ξ, σ_ζ . If the signs agree, (4.28) and (4.26) imply $M_c(-w) = -M_c(w) + O(s)\|M_c(w)\|$ on the unit semicircle. By (4.31), its even part is $O(s^{1+\rho})$. On this block,

$$E_\xi^{\sigma_\xi}(s)N(w)E_\zeta^{\sigma_\zeta}(s) = 2\sigma_\xi w^\tau M_{c,\tau} + O(s^{1+\rho}).$$

If the signs disagree, the two leading signs in $M_c P_\zeta + P_\xi M_c$ cancel, giving $O(s^{1+\rho})$ directly. Consequently, on the full low bands,

$$\|E_\xi(s)N(x)E_\zeta(s)\| \leq Cs^{1-\rho}, \quad (4.33)$$

$$\|E_\xi(s)\{N(x) - e^{i\tau(x-y)}N(y)\}E_\zeta(s)\| \leq Cs^{1+\rho}. \quad (4.34)$$

All real $x, y \in [-v, v]$ correspond to the closed unit semicircle. A product of an even bounded number of extra rows is $I + O(s)$ on each sign block, by (4.26). Its error multiplied by (4.31) is $O(s^{1+\rho})$. This proves exactly the same conclusions with the extra pairs stated in Theorem 4.1, even if the two terms use different pairs.

To finish the sandwich estimate, decompose each H_ζ, H_ξ into $[0, n^{-1}]$, dyadic intervals above n^{-1} up to a small fixed threshold, and the remaining interval. Schwarz–Pick gives, uniformly for x in the fixed compact interval I ,

$$1 - |f(x)| \geq c_I(1 - |f(0)|).$$

One can obtain this directly from the pseudohyperbolic contraction between 0 and x ; their hyperbolic distance stays bounded on I . A band with lower endpoint u below the fixed

threshold is therefore damped by the n strict steps by e^{-cnu} ; the remainder is damped exponentially in n . All other rows are contractions, including the endpoint rows. Apply (4.33) or (4.34) with s the larger of the two band upper endpoints. The double dyadic sum of

$$C \max(u, u', n^{-1})^\beta e^{-cn(u+u')}, \quad \beta = 1 - \rho \text{ or } 1 + \rho,$$

is at most $C_\beta n^{-\beta}$. Terms involving a remainder band use the uniform unrestricted row bound and are exponentially small. This proves (4.5) and (4.6) in the homogeneous representation.

4.7 Diagonal cuts and the physical reflection norm

It remains to prove the final assertion of Theorem 4.1. Work in the diagonal time frame and write

$$\gamma_0 = \pi/4, \quad X = e^{-i\gamma_0}, \quad Y = e^{i\gamma_0}, \quad H = -X, \quad L = Y.$$

Space steps alternate H, L , starting with H ; time steps are X or Y . Put $R_* = R(\lambda/2)$ and let

$$G = \prod_j (\text{swap } R_*)_{2j, 2j+1}. \quad (4.35)$$

The factors act on disjoint pairs. Each is positive definite: its middle block has eigenvalues $1 \pm b(\lambda/2) > 0$, and the two outer entries are $a(\lambda/2) > 0$. Use G as the finite-circle metric. Directly from the collapsed tiles,

$$T_N^D(\gamma_0) = D_{N-1} S G, \quad T_N^D(-\gamma_0) = (J D^t J)_0 S^{-1} G. \quad (4.36)$$

Here is the geometry behind both the metric and these formulas. The lower slice has vertices $p, p - X, p + Y - X, \dots$. Reflection in the cut through p , perpendicular to time, gives upper order L, H . For lower spins (c_0, c_1) and upper spins (d_0, d_1) , the intervening square kernel is

$$(R_*)_{-c_0, d_0; -d_1, c_1} = (\text{swap } R_*)_{d_0, d_1; c_0, c_1}.$$

Shifting the slice origin by Y or X puts the seam at site -1 or 0 respectively, with reversed conventions in the latter case. Equivalently, every second indexed tile has collapsed. A coincident-step tile gives $R(0)$, the swap; an opposite-step tile gives $R(\lambda)$, which imposes $h_0 = -s_1, h_1 = -s_0$, with weight one. Its duplicate sides are therefore spin identifications. The adjoint for the G metric exchanges the two time steps and sends D to $J \bar{D} J$.

Both untwisted operators $S^{\pm 1} G$ are primitive on the zero-spin sector. Over a complete shift cycle, paired bonds of both parities are available. One can hold every pair or swap a chosen adjacent opposite pair, and such swaps connect the sector. The transfers commute, have a common positive right Perron vector, and have the same eigenvalue: their relation by S^2 and uniqueness make the vector S^2 -invariant. Their left Perron data are its G -dual. Normalize the vacuum to unit G norm.

These degenerate indexed arrays are exactly physical square tori; they need no limiting estimates for degenerate rhombi. To check this in both quantizations, let p_t follow a periodic sequence of X, Y steps with even period size, and set

$$q_{2j} = j(Y - X), \quad q_{2j+1} = q_{2j} - X.$$

The uncollapsed tiles of the array $p_t + q_j$ biject with the physical squares centered at each time level. The collapsed identifications join duplicate physical edges as just described. There are no free cycles consisting solely of duplicates: each collapsed pair uses a time edge, which by the alternating space pattern is also incident to an uncollapsed tile. Hence zero-spin projection on either an unmarked zigzag slice or a periodic time path is precisely the corresponding

physical flux projection. Marks at the time seam mark that physical path. The periods preserve parity and are uniformly transverse, so the torus-to-plane argument of Lemma 4.2 applies directly to these square kernels.

For clarity, the transverse period product is primitive too. Prescribe arbitrary zero-sum spin sequences on the time paths p_t and $p_t + m(Y - X)$. Interpret them as increments of integer heights with consistent sublattice parity, periodically bounded on the two lifted paths. Each path is geodesic in the square graph with steps X, Y , so its own heights are 1-Lipschitz. Between the two paths the graph distance is at least $|m|$: if the time-level difference is ℓ , the two lower bounds are $|\ell|$ and $2|m| - |\ell|$. Thus for all sufficiently large $|m|$ the prescribed heights together are 1-Lipschitz. Extend them by the minimum of the integer distance cones $h(a) + \text{dist}(a, \cdot)$ from prescribed vertices a . This extension is time-periodic; its parity is fixed, so every edge increment is exactly ± 1 . It gives an allowed spin configuration of positive weight across the transfer strip. Therefore a sufficiently large power of either oriented two-space-step transfer has all entries positive. The difference relation makes this period product commute with the untwisted transfer at strict transverse angle v on the same time chain; the intertwiners involved are invertible. They consequently share Perron data. Seam-marked transverse vacuum coefficients agree with those at that strict angle.

Construct the limiting Gram space with the G metric, allowing the two endpoint rows, local site matrices, and S^2 . The adjoint of a site matrix, conjugated by G , is still local and bounded. The preceding physical cylinder limits and the marked moment argument give the Gram limits and bounded row actions. Let $\mathcal{H}_*$ be the closed subspace generated by seam words in $A^D(\pm\gamma_0)$ alone. Their Gram kernels are the homogeneous transverse kernels, by the adjoint formula, the transverse Perron comparison, and then the plane limit. Thus $\mathcal{H}_*$ identifies isometrically with the subspace of $\mathcal{H}$ generated by those words.

This subspace also contains the physical states required in the theorem. It is reducing for $P_1(\pm\gamma_0)$. Those operators have dense range there, because their adjoints have no kernel by the no-zero assertion at the specified interior angles. The finite-row identities give

$$S^2 P_1(-\gamma_0) = P_1(\gamma_0).$$

Both shifts by two therefore preserve $\mathcal{H}_*$, first on a dense range and then by continuity. Equation (4.36) with arbitrary D similarly shows that site matrices at -1 and 0 preserve it; shifting by two gives every finite site. Every physical edge between tiles lies on a time-level zigzag slice, so arbitrary finite physical modifications are such site insertions interleaved with layer transfers. They belong to $\mathcal{H}_*$. For ordinary homogeneous slicing the same assertion follows from the endpoint shifts; several time-edge insertions on a strict row can also be moved to one seam by commuting through finite strings of invertible R matrices and expanding their matrix elements.

Finally, reflection in a time-level cut, followed by complex conjugation and arrow reversal, matches lower spins to upper spins through the intervening kernel G . Normalize by the unmarked vacuum, or equivalently by the unit G vacuum and one common Perron factor per layer. The squared norm of a lower state is then exactly the vacuum expectation ratio of its reflected configuration. Between two cuts, transfer from the lower slice of the first to the lower slice of the second, and glue to the reflected future there using G . Strings immediately below and above the last cut act on the respective states and leave this kernel unchanged. The homogeneous estimate restricted to $\mathcal{H}_*$ is therefore the asserted diagonal estimate. This completes the proof of Theorem 4.1.

5 Local path calculus for the current

The transfer estimate applies to a lattice current because its path dependence can be removed locally. This section proves the needed tensor identities and records the phase bookkeeping when a current is placed between two diagonal cuts.

5.1 A path primitive and its deformation rule

Direct a simple path $\mathcal{C}$ of tile edges from its initial point. Measure each crossing spin using the left normal of the path step. At a crossing, a matrix has row index on the right tile and column index on the left tile. Lift the tangent angle continuously along a rounded version of the path, starting with a specified lift. Define its primitive $\mathcal{J}(\mathcal{C})$ by summing over the choice of one marked step:

- at that step insert $e^{i\rho\alpha}L_-$, where α is the lifted tangent angle;
- on all preceding steps insert $K(Q)$;
- on all later steps insert the identity.

Thus both strand arrows at the marked crossing point away from it. All insertions are additional to the tile tensors. Other primitives and other marked insertions are kept outside the deformation corridor, unless their order is explicitly included.

Lemma 5.1 (Local deformation). *In the positive winding expansion, the primitive is invariant, pairing by pairing, under simple-path isotopies that continue its initial lift, keep its endpoints in their regions, and avoid the other insertions. The statement is a tensor identity for arbitrary outside spins. A pure diagonal string obeys the corresponding deformation rule without a source.*

Proof. Draw the strands smoothly and disjointly, with transverse path crossings. At a marked crossing use the tangent-angle lift rotated by less than $\pi/2$ to the orthogonal crossing direction; this agrees with the lattice convention at a tile edge. Sliding the source on its strand while its tangent changes by b changes the sum of the turns of the two departing pieces by $-2b$. The winding factor is consequently multiplied by $e^{-i\rho b}$, which cancels the change of the marker gauge. All other spin constraints and prefix factors correspond directly.

A generic isotopy also creates or removes a pair of crossings at a tangency. Terms marked elsewhere are unchanged because the two new diagonal factors cancel. The two terms marked at the new crossings cancel one another. To check the sign, orient a test tangent β of the strand approximately in the path direction near tangency, so crossings 1, 2 occur in that order both along the path and along this test direction. With the source at crossing 2, let c_1 be the signed spin at crossing 1; the strand there travels opposite to β , away from the source. The orthogonalized lifts satisfy

$$(\alpha_2 - \beta_2) - (\alpha_1 - \beta_1) = -c_1\pi. \quad (5.1)$$

For example, when the first path tangent is clockwise from β , the offsets are $-\pi/2, +\pi/2$ and $c_1 = -1$; the other case reverses the signs. The ratio of the marker gauge times the source-piece winding factors is therefore $e^{-ic_1\lambda}$. The extra prefix factor is Q^{c_1} , and

$$e^{-ic_1\lambda}Q^{c_1} = -1.$$

The outside constraints and factors of the two terms are identical. This proves the cancellation.

A generic isotopy has only isolated tangencies of this kind, besides transverse slides, and can be chosen to keep all fixed insertions disjoint. This proves the primitive identity. For a pure diagonal string, only the first cancellation is needed; equivalently it is the tile ice-conservation identity. $\square$

5.2 Boundary-tree coefficients

We use the boundary factors from Section 3. For reference, if θ_j is the lifted inward normal at port j , h_j its real height parameter, and $c = \pm 1$ is its inward/outward arrow sign, they are

$$b_j(c) = (2 \sin \lambda)^{-1/2} \exp\{ic(\rho\theta_j/2 + \lambda h_j)\}.$$

Consider a disk, order its boundary from a flat gap p , and assume there are no obstructing extra strings. Start a path from p toward an interior vertex z with initial tangent lift strictly between the positive boundary tangent and that tangent plus π . Every boundary arc bounds a child interval not containing p ; these arcs form the separator tree of the complementary regions.

Lemma 5.2 (Tree evaluation). *For an arc $e : i \rightarrow j$ bounding its child interval, put $m = h_i - h_j$ and*

$$\begin{aligned} w_e &= \frac{\cos((m - \frac{1}{2})\lambda)}{\sin \lambda}, & k_e &= -\frac{\cos((m + \frac{1}{2})\lambda)}{\sin \lambda}, \\ a_e &= e^{-i\lambda(h_i + h_j)}, & c_* &= \frac{e^{-i\lambda}}{2 \sin \lambda}. \end{aligned} \tag{5.2}$$

For each fixed pairing, the primitive to z is the sum over a marked edge of the separator-tree path from p to the region of z : use k_e on earlier edges, $c_ a_e$ on the marked edge, and w_e on every other boundary arc. Unmarked internal loops have weight d , and a marked closed loop contributes zero.*

For a pure $K(Q)$ string to z , use k_e on the full separator-tree path and w_e off it. An internal loop enclosing z then has weight -2 instead of d . The parity of the number of such loops records the color change from the boundary-tree region to the lattice color of z .

Proof. By Lemma 5.1, for each pairing we may choose a simple path crossing minimally the boundary arcs separating p from z , followed, if necessary, by the nested closed loops. Such paths exist in the disk regions cut out by the disjoint arcs and loops. A sole outward source on a closed loop is inconsistent with the remaining arrow constraints, so it gives zero.

For a parent-to-child crossing, the boundary-ordered arc travels along the positive normal of the path. Let β be its tangent lift, continued from θ_i . The correct lift relation is the exact identity

$$\beta - \alpha = \pi/2, \tag{5.3}$$

without a multiple of 2π . To verify this, close a simple curve by taking the boundary from p to i , the arc to the crossing, and the reversed path back to p . Its positive full turn, with the left right-angle turns at the port and crossing and the short turn at p , is $\beta - \alpha + 3\pi/2$. It equals 2π , proving (5.3).

For an outward source the sum of strand turns is $\theta_i + \theta_j - \pi - 2\beta$. Multiplying its winding weight by the two outward boundary factors and by $e^{i\rho\alpha}$ gives

$$\frac{1}{2 \sin \lambda} e^{-i\lambda(h_i + h_j)} e^{i\rho(\alpha - \beta - \pi/2)} = c_* a_e.$$

Summing the two orientations of an unmarked arc gives w_e . If a prefix string crosses it, multiplication by $K(Q)$ changes that sum to k_e . These are precisely the tree weights in (5.2).

For a pure string, the same calculation holds on every separating boundary arc. A counterclockwise enclosing loop has intersection factor Q^{-1} , so the sum of its two orientations is $e^{i\lambda}Q^{-1} + e^{-i\lambda}Q = -2$. Crossing each such medial loop switches the adjacent color, which proves the final parity assertion. $\square$

5.3 Reflection and two-step increments

Reflect in a line of angle ϕ , and combine this operation with complex conjugation and arrow reversal. Relative to the mirrored step direction, spins are unchanged, while the left and right tiles are exchanged. Consequently the reflected primitive has the form

$$L_+, K(Q^{-1}), \quad e^{i\rho\alpha'}, \quad \alpha' = 2\phi - \alpha,$$

up to a common scalar. It obeys the reflected local deformation identities, using the opposite winding expansion. In the notation of (4.4), it has $\epsilon = -1$ and still $\tau = 1$. Further reflections alternate between these two types. Moreover, a high difference with coefficient $e^{i(\arg d_1 - \arg d_2)}$ becomes the same expression with the reflected directions d'_1, d'_2 .

A *low stencil* is a primitive increment on a straight two-step extension from a vertex z . A *high stencil* is the difference of two such increments in distinct tile-axis directions d_1, d_2 , with coefficient $e^{i(\arg d_1 - \arg d_2)}$ on the subtracted term. Both extensions have one common arriving prefix with fixed lift. The local path choices are simple and extensible.

Proposition 5.3 (Extraction in a clear slab). *Suppose a diagonal-time slab of macroscopic thickness about a low or high stencil contains no other non-diagonal tensors or endpoints. Diagonal chains may cross the slab. The stencil has an intact neighborhood disjoint from all other chains except its own prefix, and the slab has spare margins for routing and cuts. Bounded lattice errors are permitted inside these margins. Then the stencil contraction can be written as the corresponding low or high expression in Theorem 4.1, between exterior states independent of the choice of stencil term. Only boundedly many compensating pairs of rows and a common bounded scalar are introduced.*

In particular, if the two clear parts of the slab contain at least n physical steps, its scalar contraction is bounded by $Cn^{-1+\rho}$ for a low stencil, and by $Cn^{-1-\rho}$ for a high stencil, times the two exterior reflection norms. Reflected stencils satisfy the same statement.

Proof. There are two matters to check: the compared paths must have common endpoints, and rerouting the other charges must contribute the same phase to every possible source position.

Let a, b be the two positive-time unit steps, namely X, Y in the diagonal slicing. An increment in a negative direction is exactly the negative of the corresponding forward increment ending at z . This rewriting is local, even if previous cuts have been made. Choose a point on the common prefix in the intact neighborhood, express each increment as a difference of two complete primitives from that point, cancel their common initial marked terms, and apply Lemma 5.1 inside the neighborhood. Thus no earlier path or cut is altered.

Use the following common endpoints for the high stencil:

- For directions a, b , start at z and append two after-twist steps in the other direction to end at $z + 2a + 2b$.
- For directions $-a, -b$, start at $z - 2a - 2b$, prepend two before-twist steps in the other direction, and end at z .
- For directions $a, -b$, use the trajectory starting at $z - 2b$ with steps b, b, a, a .

The case $-a, b$ is the same with a, b interchanged. Backward differencing changes the relative sign exactly as required to obtain the coefficient for the two positive directions.

Approach the common start by the same prefix, with a final positive-direction segment. A simple replacement can first bend around the stencil and approach from the required side, even when the original prefix arrived from later times. In a box with spare room, choose lattice corridors preserving the entrance and reserve a clear rectangle for the last positive steps. All marker lifts now differ by the short angle between a and b in their common time

frame; no extra 2π turn is introduced. The low case is the same construction with one increment. The explicit prefix charge is Q^ϵ , with $\epsilon = 1$ or -1 according to the reflected type.

We next consolidate diagonal chains in the clear band. Keep as common background every chain except the explicit prefix steps between the common start and the marker. In particular, the charge arriving at the common start is included in this background. Using real charges modulo 2π , route it to carry a total phase ζ on a common positive-time seam before the start, and a phase ξ from the start through the comparison block and to the future seam. These are the net oriented charges through ordinary sections, and

$$\zeta/\xi = Q^\epsilon.$$

All endpoints are preserved. Joining and funneling the chains can be done outside the sandwich cuts but inside the larger vacant margins, including transverse lattice displacements. Choose the cuts through level vertices of the positive-time seam with many positive steps on each side. This construction does not require a monotone path between the original remote cut positions.

Inside the comparison rectangle, first route the ξ background along one common skirt on the higher-space-coordinate side of both positive-step paths, avoiding every possible marker. In X, Y coordinates it runs around the northwest side of the small positive rectangle between the common endpoints. Tile ice conservation then moves this skirt onto either compared path, always on the same, left spin side of a broken source edge. At a marker it therefore multiplies L_- or L_+ on a fixed side by the diagonal twist. This multiplication is one common scalar.

It remains to justify that the change from the original background routing to this common skirt has a common phase. Their difference is a closed tile-edge chain in the larger slab, hence a weighted sum of face boundaries. Tile ice conservation cancels every face contribution except at a source edge, whose signed spin jump is fixed and equal to two relative to its positive step. All prospective source edges lie in the same winding region of this common closed-chain difference. Indeed other original chains avoid the intact neighborhood; the chosen prefix approaches the common start from the past without entering the positive rectangle; the replacement before the start follows that same final incoming line with net charge ζ ; and the common skirt and outgoing seam stay outside the rectangle. No edge of the routing difference separates the interiors of two possible marker edges. Extend the coincident clear incoming and outgoing parts by many lattice edges if needed. The face coefficient seen by the source is consequently identical in every stencil term. Summing conservation gives one common phase. There are no contributions from other defects, since all changes remain inside the clear band and its margins. The same reasoning can be phrased as a sequence of path slides consistently kept on one side of the entire marker rectangle. Every phase remains unitary.

We have now obtained exactly the before- and after-twist operators in (4.3), with their required relative phase, the compensating even pairs described above, and one common bounded scalar. The two exterior states are independent of which term is used. Theorem 4.1 gives the claimed bound.

For later repeated cuts, note also how these states glue. A consolidated chain at a diagonal cut ends at a shared level vertex. Reflection conjugates its charge to the inverse in the mirrored direction, preserving the spin convention. Joining after reversing path direction therefore continues with the same charge and creates no endpoint at the cut. At preliminary tile-axis cuts, route the chains transversely over the line through its level vertices, with no segment lying on the line inside the spare margin; the same argument applies. Reflections preserve the high or low type at every other intact stencil point. This proves the reflected assertion and the compatibility needed for successive extractions. $\square$

The proposition is a local norm estimate. In later applications the number of available clear slabs and the sizes of their exterior norms are estimated separately by the positive partition comparisons; the local extraction itself makes no uniform bound on those norms.

6 The four-change observable and its boundary values

Throughout Sections 6–9, let $1 \leq q < 4$, $d = \sqrt{q} = 2 \cos \lambda$, and $\rho = \lambda/\pi$. The construction has two distinct parts. We first determine the boundary values of a disk observable by an exact calculation on its boundary-arc tree. We then prove its holomorphicity by placing two copies on opposite banks of a wall. The boundary identification in Proposition 6.5 is explicitly conditional until Theorem 9.3 supplies this last premise.

6.1 Boundary data and the open-cap law

The winding-field construction follows the parafermionic approach of Smirnov [49] and Riva and Cardy [41]. Alternating four-boundary-change observables were used for FK-Ising in Chelkak and Smirnov [11, Section 6]. Here we compute the finite boundary tree for every fixed $1 \leq q < 4$; the later reflection argument supplies its continuum holomorphicity.

Let Ω be a fixed simple orthogonal polygon in tile coordinates, with four distinct boundary points p, b, c, d_0 in counterclockwise order, each in the relative interior of a side. Write A, B, C, D for the successive intervals beginning at p . Approximate Ω by whole-tile disks of mesh n^{-1} , with side positions and marks changed by $O(n^{-1})$, using the balanced corner parities of Corollary 3.4. Give A, C one designated wire color $\mathfrak{A}$, and B, D the other, $\mathfrak{B}$. At each mesh choose the global color assignment for which the deterministic-cap partition, completed around A, C , is at least that for the reversed assignment. Changing just the pairing of the four unmatched ports changes a positive partition by a bounded factor.

Under the *open-cap law*, the four unmatched ports remain open; every closed loop, including those closed by local caps, receives weight d , in addition to the common tile coefficients. This law and either deterministic completion have uniformly bounded density ratios. Let $E_{\mathfrak{A}}$ be the event of an actual $\mathfrak{A}$ -path in the disk joining A to C , and define $E_{\mathfrak{B}}$ similarly for B, D . Eligible endpoints of intervals count as boundary vertices. Set $P_{\mathfrak{A}} = \mathbb{P}_{\text{open}}(E_{\mathfrak{A}})$ and $P_{\mathfrak{B}} = \mathbb{P}_{\text{open}}(E_{\mathfrak{B}})$.

Use the boundary factors and inward-normal lift of Section 3, with bases

$$(H_A, H_B, H_C, H_D) = (s + 1, 1, s, 0), \quad h_i = H_R + \Delta \mathbf{1}_{\{i \text{ is high}\}}, \quad i \in R. \quad (6.1)$$

The boundary order starts at p . On continuing to a second lap the base decreases by 1, to account for the curvature lift, so the locally forward raw drop at p is $-s$. Introduce

$$x = e^{-i\lambda}, \quad y = e^{i\lambda}, \quad P_0 = x^{s+1}, \quad G = 1 - y^2, \quad A_R = x^{2H_R + \Delta}. \quad (6.2)$$

In particular, $d = y + y^{-1}$ and $x^{2\Delta} = -y^3$.

6.2 The boundary-arc tree

Cut the disk along its raw boundary arcs, leaving its closed internal loops in place only for their ordinary weights. The adjacency graph of the resulting regions is a tree, rooted at the region incident to p . Its vertices have the common color of their boundary-gap visits; colors alternate across every edge. For a lattice vertex z , let $V(z)$ be the region containing z when the internal loops are ignored. Call a tree vertex *wired* if it visits a gap of its own color on an interval wiring that color.

Lemma 6.1 (Tree normalization and connection alternatives). *The number of loops closed by the local boundary caps is*

$$\mathcal{I} = \#\{\text{unwired tree vertices}\}. \quad (6.3)$$

Exactly one of $E_{\mathfrak{A}}, E_{\mathfrak{B}}$ occurs. Completing the four open ports around A, C adds respectively one and two loops on these events.

For an edge e represented by an arc $i \rightarrow j$, $i < j$, put

$$m_e = h_i - h_j, \quad w_e = \frac{\cos((m_e - \frac{1}{2})\lambda)}{\sin \lambda}, \quad k_e = -\frac{\cos((m_e + \frac{1}{2})\lambda)}{\sin \lambda}, \quad a_e = x^{h_i + h_j}. \quad (6.4)$$

Let $\mathcal{P}(V)$ be the root-to- V edge path. Define

$$D_V = d^{-\mathcal{I}} \prod_{e \in \mathcal{P}(V)} k_e \prod_{e \notin \mathcal{P}(V)} w_e, \quad (6.5)$$

$$H_V = d^{-\mathcal{I}} \sum_{e \in \mathcal{P}(V)} a_e \prod_{f \text{ before } e \text{ in } \mathcal{P}(V)} k_f \prod_{f \text{ not before } e, f \neq e} w_f. \quad (6.6)$$

Thus $H_{\text{root}} = 0$. If e joins its parent u to its child v , the common product D_e° , with the factor on e omitted, satisfies

$$H_v - H_u = a_e D_e^\circ, \quad D_u = w_e D_e^\circ, \quad D_v = k_e D_e^\circ. \quad (6.7)$$

Proof. Attach a thin exterior annulus containing the local caps, with the four unmatched ports continued to its outer boundary. Each cap cuts off a sliver at the intervening gap. The remaining annular regions connect the wire-color visits of each interval and lead to the four outer intervals. Consequently precisely the wired classes remain accessible from that boundary. Two noncrossing open strands cut a disk into three boundary regions, with one further region for each boundary loop. The other tree regions give exactly the loops in (6.3). This also gives the opposite-pair alternative.

These alternatives concern actual connections. Indeed the switch arcs are perimeters separating the two thickened diagonal graphs. Visits of one color in a single boundary region belong to the same actual component; reinstating internal loops cannot divide that boundary class. There are only two designated intervals of each color. The two possible pairings of four ports (Figure 2) agree or disagree with the deterministic exterior pairing, adding two or one loops, respectively, as asserted. Finally, the path rule of Lemma 5.2 gives (6.5)–(6.6); all ordinary internal-loop weights cancel. Separating the term marked on e gives (6.7). $\square$

The normalized observable is the expectation of this tree variable:

$$h_n(z) = \frac{\mathbb{E}_{\text{open}} H_{V(z)}}{P_0 G}. \quad (6.8)$$

To relate it to the current, take the primitive of Section 5 from p , initially directed inward, to z . Lemma 5.2 identifies its diagram, divided by the open-cap weight, with $c_* H_{V(z)}$, where $c_* = e^{-i\lambda}/(2 \sin \lambda)$. Thus h_n is equivalently its spin contraction divided by the open-cap partition and by $c_* P_0 G$. The finite tree expressions will determine its boundary values; the current representation will later control its bulk differences.

For later use we make the finite tree computation explicit. Let $L(e)$ be the number of marks among b, c, d_0 in the child boundary interval of e . Those marks form a consecutive

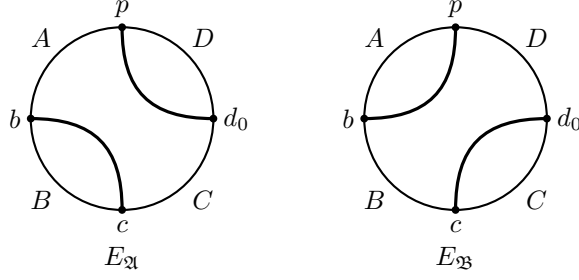


Figure 2: The two four-port pairings after local caps are attached. The boundary is drawn as a circle only to show cyclic order; A, C wire $\mathfrak{A}$ and B, D wire $\mathfrak{B}$. Completing around A, C adds one loop on the left and two on the right. The thick curves are open medial strands, not cluster paths.

subset. Their drops are s, t, s ; the endpoint high statuses agree when L is odd and differ when it is even. Substitution in (6.4) gives

L	w_e	k_e
0	$d, 1$	$-1, -d$
1 (b or d_0)	1	0
1 (c)	1	$-d$
2	$1, d$	$0, -(d^2 - 1)$
3	0	1

(6.9)

Two entries refer, in order, to a high and a low first port.

The $L > 0$ subtree has four terminal legs, possibly of zero length. The p -leg has $L = 3$, and the other legs have $L = 1$. Between the branches is either one shared node or a chain of $m \geq 1$ edges with $L = 2$. A chain of type X encloses c, d_0 and has ports on B, D ; write $\mathfrak{C} = \mathfrak{B}$ in this case. A chain of type Y encloses b, c , has ports on A, C , and has $\mathfrak{C} = \mathfrak{A}$. These are the only possibilities because descendant marked sets are nested. Number the chain nodes $0, \dots, m$ away from the p -leg, and write c_j for their colors. For a shared branch use $m = 0$.

Leg and branch nodes are wired: each meets intervals of both designated colors, also when a leg collapses at a mark. An internal chain node meets only intervals wiring $\mathfrak{C}$, and cannot meet a mark because its descendant marked set is unchanged along the chain. The child of an $L = 0$ edge is unwired exactly when its first port is high. Hence

$$\mathcal{I} = \mathcal{I}_0 + \mathcal{I}_{\text{mid}}, \quad \mathcal{I}_{\text{mid}} = \#\{0 < j < m : c_j \neq \mathfrak{C}\}, \quad (6.10)$$

where $\mathcal{I}_0$ counts high-starting $L = 0$ edges. If $m > 0$, the endpoint regions of each chain edge both visit the two $\mathfrak{C}$ -intervals, so $E_{\mathfrak{C}}$ holds. If $m = 0$, the shared node visits all four intervals, giving the connection of color c_0 . Since a chain edge has $w_e = d$ precisely when its child has color $\mathfrak{C}$,

$$F := d^{-\mathcal{I}_{\text{mid}}} \prod_{L(e)=2} w_e = \begin{cases} d^{\mathbf{1}_{\{c_0 \neq \mathfrak{C}\}}}, & m > 0, \\ 1, & m = 0. \end{cases} \quad (6.11)$$

Every off-path $L = 0$ weight cancels its contribution to $d^{\mathcal{I}}$. Inside an $L = 0$ branch contained in R , define

$$g_R(\text{wire color}) = \frac{1}{2}, \quad g_R(\text{other color}) = -\frac{1}{2}. \quad (6.12)$$

Then

$$\frac{k_e}{w_e} = -d^{g_R(\text{col}(v)) - g_R(\text{col}(u))}, \quad a_e = A_R. \quad (6.13)$$

In particular, two successive increments of H in such a branch cancel.

Lemma 6.2 (Marked values and corrected observable). *Call the tree typical when all four terminal legs have positive length. On a typical tree,*

$$D_p = D_b = D_{d_0} = 0, \quad \frac{H_b}{P_0 G} = \begin{cases} 1 & E_{\mathfrak{A}}, \\ d & E_{\mathfrak{B}}, \end{cases} \quad \frac{H_{d_0}}{P_0 G} = y \begin{cases} d & E_{\mathfrak{A}}, \\ 1 & E_{\mathfrak{B}}. \end{cases} \quad (6.14)$$

On the gap visits of any interval R ,

$$\Re(H_V/A_R) - c_R(\text{col}(V))D_V \text{ is constant,} \quad (6.15)$$

$$c_R(\text{wire color}) = -\cos \lambda, \quad c_R(\text{other color}) = \cos(2\lambda).$$

Set

$$\Gamma(\mathfrak{A}) = P_0 y^2, \quad \Gamma(\mathfrak{B}) = -P_0 y, \quad H_V^\circ = H_V - \Gamma(\text{col}(V))D_V. \quad (6.16)$$

The variables H_V° are bounded uniformly over all trees and vertices. On a typical tree they have value $P_0 GT$ along the c -leg and all its descendant branches, where

$$T = \mathbf{1}_{E_{\mathfrak{A}}} + y \mathbf{1}_{E_{\mathfrak{B}}}. \quad (6.17)$$

On gap visits along B , their only possible values are $H_b, P_0 GT$; along C , they are $H_{d_0}, P_0 GT$.

Proof. An unpassed $L = 3$ edge kills every term of H , so only its last edge can mark. Once the first b - or d_0 -leg edge has been passed, $k_e = 0$ kills all later marked terms. At node 0,

$$D = F, \quad H/P_0 = \begin{cases} F, & c_0 = \mathfrak{A}, \\ -y^3 F, & c_0 = \mathfrak{B}. \end{cases} \quad (6.18)$$

The first b -leg coefficient a_e/P_0 is $-y^2$ or y^{-1} , according as the parent color is $\mathfrak{A}$ or $\mathfrak{B}$; the corresponding d_0 -leg coefficients are $-y^4$ or y . This gives (6.14) at the near branch or shared node.

On the middle chain, $a_e/P_0 = y$ in type X and $-y^2$ in type Y . If the first chain edge has $k_e = 0$, its weight is $w_e = 1$ and only that chain mark contributes. Otherwise $F = d$; the chain terms and the target mark contribute $(2 - d^2)$ times the chain amplitude. Indeed the second chain edge already has $k = 0, w = 1$, or, when $m = 1$, the target edge has the same amplitude. The identities

$$d + (2 - d^2)y = yG, \quad -dy^3 - (2 - d^2)y^2 = G \quad (6.19)$$

give the far-branch values; the first- $k = 0$ cases give the same values.

For (6.15), consider first the earlier port of an arc, and let $r = 1$ for a high port and $r = -1$ otherwise. With “before” and “after” referring to boundary-gap visits,

$$\Re(a_e/A_R) = \cos((m_e - r\Delta)\lambda) = c_{\text{after}}k_e - c_{\text{before}}w_e. \quad (6.20)$$

At the later port, using its own high status,

$$-\Re(a_e/A_R) = -\cos((m_e + r\Delta)\lambda) = c_{\text{after}}w_e - c_{\text{before}}k_e. \quad (6.21)$$

These product-to-sum identities and (6.7) prove the invariant, including across a corner in the same interval.

For $R = B, C$, direct substitution gives

$$\Re(\Gamma/A_R) = c_R, \quad d\Gamma(\text{wire color}) + \Gamma(\text{other color}) = -A_R. \quad (6.22)$$

Thus H° is unchanged on $L = 0$ edges in B, C . It is unchanged on c -leg edges as well: those have $w = 1, k = -d$, and $a = A_C$ for a high first port, whose parent color is $\mathfrak{B}$, and $a = A_B$ otherwise. The table bounds all other parts of the tree: an unpassed p -leg kills terms; the middle chain kills a complete prefix in at most two steps; and an $L = 0$ tail has the two-step rule (6.13). The c -leg is the only place where the uncorrected variables can grow, and the correction is constant there. This proves the uniform bound for H° , including atypical trees.

At a node incident to a c -leg edge, the B, C line invariants, anchored by (6.14), intersect at P_0GT when $d > 1$. The four line directions after division by P_0G are

$$1, \quad -y^{-1}, \quad y^2, \quad -y \quad \text{on } A, B, C, D, \quad (6.23)$$

as follows from $iA_R/(P_0G)$ up to nonzero real factors. At $d = 1$ the same identity follows by continuity for each fixed finite tree. The descendants beyond a c -leg edge lie in its B, C boundary disk; its $L = 0$ side branches therefore preserve this value.

The gap walk along B goes from the end of the b -leg to the end of the c -leg. Its $L = 0$ excursions preserve H° , so only type- X middle nodes need checking. Similarly only type- Y nodes need checking for C . At node 0,

$$H^\circ/(P_0G) = F \quad (c_0 = \mathfrak{A}), \quad H^\circ/(P_0G) = yF \quad (c_0 = \mathfrak{B}). \quad (6.24)$$

After the first zero k , the value stays at the far-branch value. If the first k is nonzero, the first jump is $-y^{-1}$ in type X with $c_0 = \mathfrak{A}$, and $-y^2$ in type Y with $c_0 = \mathfrak{B}$. Thus node 1 already has value T in normalized units. These are precisely the alternatives in the statement. $\square$

6.3 Moments, shielding, and disappearance of the diagonal term

The tree calculation gives the marked values and the corrected values along each boundary interval. To pass these statements to subsequential limits, we need moment bounds and boundary shielding, followed by cancellation of the diagonal residual.

Lemma 6.3 (Compactness and boundary shielding). *For every fixed positive power, the moments of $|H_{V(z)}| + |D_{V(z)}|$ are bounded by a power of $\min(1, |z - c|)^{-1}$, uniformly when z is farther than a mesh-scale distance from c . The expectations in (6.8) are asymptotically equicontinuous on interior compact sets. Every lattice parity therefore has the same locally continuous subsequential limits.*

Typicality has probability tending to one. As a macroscopic interior point approaches an ordinary boundary point, including a polygon corner, its tree vertex visits that point's interval with probability tending to one after the mesh limit. Near b, c, d_0 it lies beyond an edge of the corresponding terminal leg with high probability; near p an unpassed $L = 3$ edge remains with high probability.

Proof. The table and the two-step cancellation bound $|H_V| + |D_V|$ by $C \exp(C\ell)$, where ℓ is the number of c -leg edges passed on the root-to- V path. A passed c -leg arc has a child disk containing z . It cannot be wholly inside a box at c of radius a small fixed multiple of $\min(1, |z - c|)$, because the flat wall makes such a child disk local. Fixed buffers and the dyadic bands between that radius and macroscopic scale count these passages. The conditional passage estimates of Lemmas 2.5 and 2.6, grouped into disjoint buffer colors, give the asserted moments. Bounded open-cap reweighting allows their use under a deterministic-cap FK law. If two nearby interior vertices have different boundary-tree regions, a raw boundary arc travels from their vicinity to the wall. Bulk RSW circuits make this probability uniformly small; Hölder's inequality and the moment bound give equicontinuity, including between lattice parities.

For the boundary claims, work on separated scales in the relevant wall sector. In disjoint buffers impose two actual wall-to-wall crosscuts, of opposite colors, the first inside the second. Up-to-wall RSW gives a uniform conditional lower bound. A raw medial separator lies between the two thickened paths and joins the two banks. One direct verification is to trace the perimeters: on either bank the gaps at the two crosscut ends have opposite colors, so an odd number of ports lies between them; the paths prohibit radial escape, and at least one of these ports must be paired to the other bank. Such an arc separates the deeper box from the exterior. The same argument works in a corner sector. Iteration gives probability tending to one at diverging scale ratio. It produces the claimed local raw arch at an ordinary point and the appropriate terminal-leg separator at each mark. $\square$

Lemma 6.4 (Diagonal residual cancellation). *For every bounded function γ of the two colors and every boundary point u distinct from the four marks,*

$$\lim_{z \rightarrow u} \limsup_{n \rightarrow \infty} \left| \mathbb{E}_{\text{open}}[\gamma(\text{col}(V(z))) D_{V(z)}] \right| = 0. \quad (6.25)$$

Here z in the outer limit is a macroscopic interior location, and any lattice vertices tending to it may be used in the inner limit.

Proof. We first cancel a positively weighted residual at a fixed interior z . Replace the marked primitive by the full $K(Q)$ string from p to z . If N is the number of internal closed loops surrounding z , their weight ratio is $-2/d$. Let σ take opposite signs on the two colors. Nesting parity gives $(-1)^N = \sigma(\text{col}(z))\sigma(\text{col}(V))$. Equality of the two winding expansions implies

$$\mathbb{E}_{\text{open}}[\sigma(\text{col}(V)) D_V (2/d)^N] = o(1). \quad (6.26)$$

Here is the required absolute estimate in the wrong expansion. Relative to the complementary cap partition it is at most

$$K_*^{1+X_*} \prod_{\nu=p,b,d_0} \mathbf{1}_{\{N_\nu=0\}} d^{N_c} \prod_{v \text{ a corner}} f(g_v)^{N_v/2}, \quad g_v \in \{-\frac{1}{2}, \frac{1}{2}\}. \quad (6.27)$$

The counts use disjoint small macroscopic cutoffs. The variable X_* counts macroscopic passages and has every fixed exponential moment; its constants may depend on z . Away from singularities, ordinary small arcs have the complementary zero-drop cap weights. At b, d_0, c the effective wrong drops are $-s, -s, -t$, with weights $0, 0, d$. At p , use a locally forward boundary interval. Its arc meets the exiting string with net intersection -1 ; after changing the winding convention the effective drop is $s + 1$, again giving weight zero. This depends on the net intersection, not the placement of the string. Corner alternation and cap elimination give the remaining factors in (6.27). Internal loops around z have absolute weight $2|\cos(2\lambda)| \leq d$, and each nonlocal arc costs at most a bounded factor per passage.

Without the $K_*^{1+X_*}$ factor, the logarithm of the expectation is bounded above by

$$3P + D_* + \sum_{v \text{ convex}} C_{i(v)} + \sum_{v \text{ concave}} L_{i(v)} + O(1). \quad (6.28)$$

Let Y_n be the nonnegative product in (6.27) without $K_*^{1+X_*}$. Its second moment is polynomial in n . For $0 < \eta < 1$, Hölder and interpolation between its first two moments give

$$\mathbb{E}[Y_n K_*^{1+X_*}] \leq C_\eta (\mathbb{E} Y_n)^{(1-\eta)/(1+\eta)} (\mathbb{E} Y_n^2)^{\eta/(1+\eta)}.$$

The constant uses the fixed exponential moment of X_* at rate $(1 + \eta) \log K_*/\eta$. The budget in (6.28) has size $O(\log n)$, so this costs only $O(\eta \log n)$ in that upper bound; no lower bound on $\mathbb{E} Y_n$ is needed. The balanced-corner budget (3.17) is strictly negative by (3.4). Our global

color choice makes the complementary cap partition at most a constant times the correct open-cap partition. This proves (6.26).

We next remove the positive weight, retaining relative error estimates. Let μ be the correct deterministic-cap FK law and $R_* = d^{\ell_{\text{open}} - \ell_{\text{det}}}$; open expectation is $\mathbb{E}_\mu[R_* \cdot] / \mathbb{E}_\mu R_*$. For the interval R containing u , put

$$B_* = R_* \sigma(\text{col}(V)) d^{-g_R(\text{col}(V))} D_V, \quad T_* = d^{g_R(\text{col}(V))} (2/d)^N. \quad (6.29)$$

Thus $\mathbb{E}_\mu[B_* T_*] = o(1)$ at every fixed z . Choose scales in the regular sector at u with

$$|z - u| \ll r_1 \ll r_2 \ll r_3 \ll 1, \quad (6.30)$$

all ratios diverging as $z \rightarrow u$, and take the mesh limit first.

Except on an event of vanishing probability, B_* is determined outside r_2 . To see this, use actual screens to ensure that no raw strand from the r_2 rim reaches r_3 . Every changing boundary arc then lies in one small part of R , so has $L = 0$. All $L > 0$ data, the marked subtree and R_* are fixed. Regard arcs touching the connected patch as affected and arcs wholly outside it as stable. Stable separators either separate the entire patch from p or do not; noncrossing puts the affected on-path arcs after them. Off-path factors cancel, while the on-path k/w suffix telescopes by (6.13). The color factor in B_* cancels that suffix exactly. Set $B_{\text{out}} = B_*$ on this exterior good event and zero otherwise.

Truncate T_* to T_{in} by counting only enclosing loops wholly inside r_1 and reconstructing the tree color from their parity and the fixed lattice color of z . This is exact unless a raw path runs from $O(|z - u|)$ to the r_1 scale: every enclosing loop meets the segment from z to its nearest wall point. To measure both truncation errors, let N_0 count the enclosing loops wholly inside the box of radius $\text{dist}(z, \partial\Omega)/8$ at z , and put $U_z = (2/d)^{N_0}$. Work under the positive tilt $\nu_{U_z} \propto U_z \mu$. The rare events remain rare under this tilt by conditional RSW outside its support. The variables B_* , T_*/U_z , and T_{in}/U_z have uniformly bounded fixed moments as $z \rightarrow u$. For B_* , every relevant c -leg arc gives a macroscopic passage away from u ; an arc wholly local at u would have $L = 0$. Every extra loop in T_* gives a passage in a band at distance comparable to $\text{dist}(z, \partial\Omega)$, by the segment observation and its escape from the cutoff. Finitely many bulk and sector buffers cover this band uniformly. Near a corner, use a corner buffer and small straight-bank buffers, or only the latter if the corner is farther away than this scale. The tilt-edge passage estimates apply in each buffer. Hölder therefore makes the product truncation errors $o(\mathbb{E}_\mu U_z)$, and

$$\mathbb{E}_\mu T_{\text{in}} \asymp \mathbb{E}_\mu U_z \geq 1. \quad (6.31)$$

The inward relative comparison of Lemma 2.4 between r_1 and r_2 , applied to this positive inner factor and the absolute parts of B_{out} , gives

$$\frac{\mathbb{E}_\mu[B_{\text{out}} T_{\text{in}}] - \mathbb{E}_\mu B_{\text{out}} \mathbb{E}_\mu T_{\text{in}}}{\mathbb{E}_\mu U_z} = o(1). \quad (6.32)$$

This is the one-sector case of that lemma: the two side rays carry the common primal wire when R is primal, and there are no through primal wires when R is dual. First use (6.26) at fixed z , then let $z \rightarrow u$. Equations (6.31)–(6.32) show that $\mathbb{E}_\mu B_*$ vanishes in this order of limits. Finally apply the same separation without a tilt to B_* times any bounded function of the tree color, reconstructing that color from the inner-loop parity. Its expectation factors up to $o(1)$ into $\mathbb{E}_\mu B_{\text{out}}$ and a bounded inner expectation. Taking the bounded color factor that reverses the extra factors in (6.29), and dividing by $\mathbb{E}_\mu R_*$, proves (6.25). $\square$

6.4 The boundary problem, conditional on holomorphicity

Proposition 6.5 (Identification of a holomorphic subsequential limit). *Suppose an interior subsequential limit h of h_n is holomorphic in the ordinary complex coordinate, and pass simultaneously to a limit r of $P_{\mathfrak{A}}$. Then $0 < r < 1$. Put*

$$U_0 = r + d(1 - r), \quad V_0 = dr + 1 - r, \quad T_0 = r + y(1 - r). \quad (6.33)$$

The function h maps Ω conformally onto the interior of the simple polygon with successive vertices $0, U_0, T_0, yV_0$, taking p, b, c, d_0 to those vertices. If a conformal half-plane coordinate takes the marks to $0, \chi, 1, \infty$, respectively, then

$$r = \frac{\int_1^\infty u^{\rho-1} |u - \chi|^{\rho-1} |u - 1|^{1-3\rho} du}{\int_\chi^\infty u^{\rho-1} |u - \chi|^{\rho-1} |u - 1|^{1-3\rho} du}. \quad (6.34)$$

Under deterministic separate $\mathfrak{A}$ caps the corresponding crossing probability is $r/(r + d(1 - r))$.

Proof. Fixed up-to-wall RSW corridors give both actual crossings positive probability under deterministic caps, and hence under open caps; thus $0 < r < 1$. Shielding and the moment bounds give the interior limits $0, U_0, yV_0$ at p, b, d_0 : near p an unpassed $L = 3$ edge kills H , and near b, d_0 the first $k = 0$ edge fixes it at its marked value. At every other boundary point the line invariant, shielding, and Lemma 6.4 put h on the corresponding line of direction (6.23), with offsets fixed by these values. In a conformal half-plane coordinate, whose boundary extension is homeomorphic, the line-normal harmonic component vanishes continuously on each unmarked interval. Harmonic Schwarz reflection, followed by extension of its harmonic conjugate, extends h holomorphically across that interval.

On B, C the traces lie on the actual segments $[U_0, T_0]$ and $[T_0, yV_0]$, respectively. Indeed H° has, for each connection event, only the two values in Lemma 6.2; the segment direction is fixed, so averaging keeps it between the averaged endpoints. Its correction has zero limiting expectation at every fixed boundary point. As that point approaches c , shielding gives a c -leg separator enclosing it and nearby interior points. The uniform bound for H° therefore makes the boundary trace tend to T_0 .

There is also continuity at c from the interior. In a half-plane coordinate ζ with $c = 0$, the flat wall and the moment bounds give polynomial growth of $h - T_0$. Its two boundary directions are $-y^{-1}$ on the negative and y^2 on the positive real axis. Consequently

$$e^{-2i\lambda} \zeta^{3\rho-1} (h(\zeta) - T_0) \quad (6.35)$$

has real boundary values on both sides, with the branch in the upper half-plane. Schwarz reflection makes it holomorphic on a punctured disk; polynomial growth makes its singularity at worst a pole. Its leading Laurent power, after undoing the multiplier, must have positive exponent because the two boundary traces tend to zero. Thus $h \rightarrow T_0$ throughout the interior approach as well.

The target polygon is counterclockwise and simple. For $d > 1$ its vertex T_0 is reflex: in the basis $1, y$ it lies strictly inside the triangle with vertices $0, U_0, yV_0$. When $d = 1$, it lies on the third side of that triangle. For any point off the four boundary lines, straighten each continuous boundary trace in its own line. Its winding number is that of the target polygon. The argument principle therefore gives one preimage of every generic interior point and none of every generic exterior point. The open mapping theorem puts the open image inside the polygon, and then the traces on A, D also lie on their actual segments. The segment control on B, C already established is needed at the reflex corner. No interior point lies on the trace, and the preimage count proves conformal bijectivity.

The four target angles are $\rho\pi, \rho\pi, (2-3\rho)\pi, \rho\pi$. The Schwarz–Christoffel formula therefore has derivative proportional to $\zeta^{\rho-1}(\zeta-\chi)^{\rho-1}(\zeta-1)^{1-3\rho}$; see, for example, [39]. The B and C side vectors are $-(1-r)y^{-1}$ and ry^2 , so their lengths are $1-r$ and r . Their ratio gives (6.34). At $d=1$ the straight marked point has exponent zero, as required. Lemma 6.1 reweights the two open-cap alternatives by d and d^2 , giving the last assertion. $\square$

For $q=1$, one has $d=1$, $\rho=1/3$, and the substitution $v=\chi/u$ turns (6.34) into Cardy’s integral [10]. No boundary-law reweighting of actual paths is then needed. Holomorphicity and the boundary approximation transfers are still required before this conditional computation becomes a scaling limit statement.

7 Paired walls and distant charges

We now put a source primitive on each bank of the polygon wall. The purpose of the exterior source is to leave a nonconstant second factor after applying a mixed bulk estimate; no conformal invariance of the exterior probability law is used.

7.1 The paired-wall tensor and its normalization

Let $O = \mathbb{C} \setminus \overline{\Omega}$. At every used wall port, replace the ordinary spin contraction by the product of two boundary factors. Use the same factor b_i on the interior and exterior banks, each evaluated on the arrow sign relative to that bank’s inward normal. The exterior designated color is opposite to the interior one. Starting at p , the exterior order is D, C, B, A , and the high statuses on the two banks agree. Its physical normal lift is $\theta_O = \theta_\Omega + \pi$, has total turn -2π , and has raw bases $H_R - \frac{1}{2}$.

Take the two source primitives from p to $z \in \Omega$ and from p to u in an exterior collar strip along D . The strip is connected and simply connected and approaches both p and d_0 ; initially we restrict u to its interior compact sets. Use the respective inward initial lifts. A *pure string of charge a* along a directed tile path contributes $e^{i\lambda ac}$ at a crossed spin, where $c=1$ denotes an arrow in the path’s left-normal direction. Add the following strings:

- (i) In exterior boundary order put

$$P_D = 0, \quad P_C = -2s, \quad P_B = -2(s+t), \quad P_A = -2(2s+t). \quad (7.1)$$

For each nonzero P_R , run a charge- P_R string forward, parallel to its interval in boundary homotopy, with its endpoints at the interval ends and its interior off the wall. Sliding this path out through a complete wall adds P_R to that interval’s raw base.

- (ii) Choose two finite sets of positive numbers a_j , each with sum 1, satisfying $a_j < 1$ and $(1+a_j)\lambda < \pi/2$. Run charge- $(-a_j)$ strings from p to exterior bulk vertices w_j , at successively separated distances

$$r_j \asymp R_0 L^{j-1} \quad (7.2)$$

from a fixed pole $z_* \in \Omega$. Their initial portions follow a common simple cut from p out to distant scales before splitting; they never return near the polygon. Continue this cut notionally to infinity, and back through Ω to z_* .

First fix a sufficiently large L , then let R_0 become large after the mesh limit. Exterior probe paths stay on the starting D side of the cut and avoid the short strings and their sliding homotopies. To arrange this, choose thin topological boundary strips separated at the marks by sector curves, reserve the D strip for probes, and send the cut out at p . Each fixed geometry admits piecewise smooth paths with clear corridors and then tile-edge

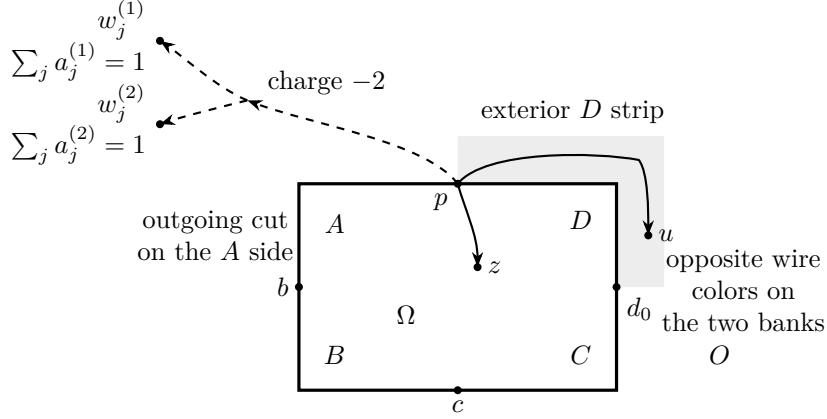


Figure 3: The paired-wall geometry in the original plane. The solid arrows are the two source paths from p ; the exterior probe stays in the shaded strip along D . The dashed pure strings leave on the other side of p . Each of their two endpoint lists has total charge -1 , so their common initial cut has charge -2 . Only representative distant endpoints are shown, not their separated metric scales. The polygon is schematic; the short interval strings and the fine parallel lanes at p are omitted. Inversion is used only to describe the ideal exterior turn factors, not to change its probability law.

approximations. Figure 3 shows the two probe regions and the common outgoing cut before its distant branches separate.

At shared initial tile edges use ordered parallel lanes, with pure strings on the non-probe side when they separate. This specifies the matrix order: the lanes cross the same smooth raw strand near a port midpoint, and diagonal phases act on the specified side of L_- . Near tile vertices they can be joined without extra strand crossings. Pure lanes need no mutual ordering. Source paths may be chosen in fixed homotopy with continued lift throughout their domains, also when small wall gaps are present away from the marks and these corridors.

For a positive reference law, cap each designated primal interval separately on each wall component. At a fragment tip the designated color changes between banks. Write μ for this deterministic-cap FK law, and Z^{cap} for its all-loop partition relative to the uncut plane vacuum, including common tile coefficients. If $\Xi(z, u)$ denotes the vacuum spin expectation of the boundary factors, sources, and pure strings, the winding expansion gives

$$\frac{\Xi(z, u)}{Z^{\text{cap}}} = \mathbb{E}_\mu \mathcal{O}(z, u), \quad (7.3)$$

where $\mathcal{O}$ is the oriented diagram sum divided, configuration by configuration, by the deterministic-cap loop weight. All changes are finite at each mesh. The almost-sure finiteness of vacuum loops, or the finite-cylinder construction of Section 4, justifies this identity even in the plane; unchanged remote loops have ratio one after orientation summation.

For each j , let N_j count closed raw loops surrounding w_j wholly inside a box of a fixed small relative radius at scale r_j . Set

$$U = \prod_j u_+(a_j)^{N_j}, \quad u_+(a) = \frac{\cos((1+a)\lambda)}{\cos \lambda} > 0. \quad (7.4)$$

The patches are disjoint and buffered. For gapped walls a further positive tip factor T_g will be introduced in Section 8; it equals 1 for a complete wall. The normalized paired-wall function is

$$\mathcal{H}_g(z, u) = \frac{\Xi_g(z, u)}{Z_g^{\text{cap}} \mathbb{E}_{\mu_g} [UT_g]}. \quad (7.5)$$

7.2 The ideal exterior observable and its phases

With a complete wall, the capped interior and exterior are independent by the loop weights and domain Markov property. The interior factor is $h_n(z)$ times c_*P_0G and the bounded ratio of its open and deterministic-cap partitions. The ideal exterior object is the same four-change disk-tree observable on $O \cup \{\infty\}$, viewed after inversion about z_* : its boundary order is D, C, B, A , its bases in that order are $(s+1, 1, s, 0)$, its launch is p , and it uses the exterior high designation. Raw boundary arcs form a finite disk tree, since none escapes to infinity almost surely. This definition needs no tiles at infinity and does not change the exterior positive law. Write $H_{VO(u)}^O$ for the tree variable (6.6) with these exterior data, and set

$$J_n(u) = \frac{\mathbb{E}_{\text{open}, O} H_{VO(u)}^O}{P_0G}. \quad (7.6)$$

As for h_n , this is the exterior primitive contraction divided by its open-cap partition and by c_*P_0G ; the tree variable has already had the common source factor c_* removed.

The tree table gives local boundedness and asymptotic equicontinuity of J_n in the probe strip. A relevant nonlocal arc has a macroscopic passage near the bounded wall, and separating nearby probe points requires a boundary-reaching arm. Shielding at the flat marks gives limit zero at p and a value bounded away from zero at d_0 . Consequently every locally uniform subsequential limit J is continuous and nonconstant on this connected strip.

The physical phases agree with the ideal ones for bounded open arcs. First slide the short strings onto the complete wall, avoiding the source path. The new raw profile $H_R - \frac{1}{2} + P_R$ differs from the desired exterior profile by the global constant $-s - \frac{3}{2}$. Since there is one outward source, this contributes only a fixed phase. Let

$$\psi = \arg \left(\left(\frac{1}{z - z_*} \right)' \right) \quad (7.7)$$

with the cut and continuous lift just chosen. The inverted normal is $\theta_O + \psi$ and has total turn 2π : on the clockwise exterior lap, ψ gains 4π before the cut is reset. An oriented arc's image turn is its physical turn plus the endpoint ψ difference, minus 4π times its signed intersections with the outward cut. Boundary normal factors cancel the endpoint terms. For a marked arc, the two departing pieces have additional start terms, canceled exactly by the change of the gauge $e^{i\rho\alpha}$ at the source. Prefix intersections are unchanged topologically. Before the distant strings split, their total charge is -2 , which supplies precisely the remaining image multiplier. Thus the physical and ideal open-arc factors agree up to one common phase whenever those arcs stay before the splitting scales. This is a calculation with turn lifts of smooth disjoint drawn arcs, independent of any invariance of their law.

7.3 A nonzero distant factor

For a closed exterior loop γ , let $A_\gamma = \sum_{w_j \text{ inside } \gamma} a_j$, using its physically bounded inside. Such a loop surrounds p exactly when it surrounds the whole wall. Its ratio to the ordinary weight is

$$\begin{cases} v(A_\gamma) = \cos((A_\gamma - 1)\lambda) / \cos \lambda, & \gamma \text{ surrounds the wall,} \\ u(A_\gamma) = \cos((1 + A_\gamma)\lambda) / \cos \lambda, & \text{otherwise.} \end{cases} \quad (7.8)$$

These formulas follow by counterclockwise intersection with the strings. A closed loop cannot separate the source of a nonzero open-arc term from the connected wall. Thus (7.8) lists all changes to its weight, including when some open arcs go far. Let Y be the product of these ratios; only finitely many are nontrivial.

Lemma 7.1 (Distant normalization). *For sufficiently large fixed L , under the exterior positive tilt $\nu_U \propto U\mu$,*

$$\mathbb{E}_{\nu_U}(Y/U) \geq c_L > 0, \quad \mathbb{E}_{\nu_U}|Y/U|^p \leq C_{p,L} \quad (7.9)$$

for every fixed $p > 0$, uniformly for large R_0 and fine mesh.

Proof. Between consecutive endpoint scales use corridor annuli with radii Kr_j and r_{j+1}/K , for a large fixed geometric constant K . Each winding loop wholly in corridor j has ratio $v(\sum_{i \leq j} a_i) \geq 1$, because the total charge magnitude is 2. Let M_j be the product of these ratios and $M = \prod_j M_j$. Besides the U nests and these corridor loops, all nontrivial loops are counted by a variable X having every fixed exponential moment uniformly under ν_U , with constants independent also of sufficiently large L . Indeed a loop excluding z_* but surrounding an endpoint outside its local cutoff has a passage at that endpoint scale, since it separates the endpoint from z_* . A loop winding about z_* and enclosing some but not all endpoints either lies in its corridor or passes a bordering scale. Fixed box and shell covers, disjoint from other endpoint buffers, and the tilt-edge passage estimates give the claim. Therefore, for a fixed C ,

$$|Y/U| \leq MC^X, \quad Y/U \geq MC^{-X} \quad \text{off } \mathcal{B}, \quad (7.10)$$

where $\mathcal{B}$ is the event that some loop excluding z_* encloses at least two endpoints. Off $\mathcal{B}$, all remaining ratios belong to a fixed finite positive set. Moreover

$$\mathbb{E}_{\nu_U} M^b \leq C_b L^{C_b} \quad (b > 0 \text{ fixed}). \quad (7.11)$$

To prove this, use $O(\log L)$ dyadic passage bands per corridor, group disjoint buffers by colors, and apply Hölder; those buffers avoid the supports of the endpoint tilts.

If a bad loop has smallest enclosed endpoint index j , it crosses corridor j radially, and hence excludes every winding loop counted by M_j . Write $\mathcal{B}_j$ for this event. Successive actual circuit blockers give $\mathbb{P}(\mathcal{B}_j \mid U \text{ patches}, (M_i)_{i \neq j}) \leq CL^{-\alpha}$, uniformly. On $\mathcal{B}_j$ one has $M_j = 1$. First apply this conditional estimate to $\prod_{i \neq j} M_i$, then Hölder with exponent $1 + \delta$ close to one to absorb C^X , interpolating against the polynomial moment bound (7.11). For some $\alpha' > 0$ independent of L ,

$$\mathbb{E}_{\nu_U}[MC^X \mathbf{1}_{\mathcal{B}}] \leq C' L^{-\alpha'} \mathbb{E}_{\nu_U} M. \quad (7.12)$$

The same interpolation, using arbitrary fixed exponential rates for X , gives for every $\varepsilon > 0$

$$\mathbb{E}_{\nu_U}[MC^{-X}] \geq c_\varepsilon L^{-\varepsilon} \mathbb{E}_{\nu_U} M. \quad (7.13)$$

Equivalently, Jensen under the M tilt and the exponential bounds give an arbitrarily small coefficient of $\log L$ in its mean of X . Take $\varepsilon < \alpha'$ and then L large. Equations (7.10)–(7.13), and $M \geq 1$, prove the positive lower bound in (7.9). Its moment bound follows from (7.10), (7.11), and Hölder. $\square$

7.4 Absolute source bounds and factorization

We record the source-path placement used here and in gap closure. Both paths can be taken in regular simple tubes, on their nominal sides of the wall except at the flat launch. For an absolute estimate, temporarily slide the exterior pure-string endpoints at p a small macroscopic distance toward the non-probe, A side of the same solid wall. Keep all gaps and other marks farther away, and include the charge along the traversed bank interval as diagonal port phases. This is an equality of the summed pairing expression: ordered lanes allow a homotopy onto the bank without crossing a source path, and ice conservation gives the port phases. Pure lanes may cross one another. Charges still count from their original

endpoints, so no extra charge is created at a temporary slide-end. The paired factors and the conventions used for gluing do not change.

After this temporary slide each tube has p in the relative interior of a flat entrance face, with a clear portion on both sides; its target is deep inside, and its other sides avoid the wall, strings, and the other tube. Finitely many such choices cover compact probe sets. For each pairing, choose a path that crosses minimally the disjoint raw crosscuts of its tube. A sufficiently small separating crosscut cuts off p locally on the entrance face: the small disk on one side cannot contain the deep target, so must contain the launch. Because the entrance is solid wall, it is an entire launch-straddling raw arc, not a short return of a larger strand. Every other traversed crosscut gives a macroscopic subpassage, even if one full strand visits the tube repeatedly. This construction also avoids entering tiny ordinary boundary-arc disks elsewhere.

The local launch weights are decisive. After the short strings and distant total charge are counted, the effective unpassed raw drop at p is $-s$. Passing the exiting source prefix makes it $-s - 1$, with the Q^{-1} minus sign. Thus small nested launch arches have weight zero while unpassed and unit absolute weight while passed; only the last can mark a nonzero term. The other marked drops are s, t, s , with local unit weights. Ordinary same-bank arcs cancel local caps, and a physical corner without a wire change has zero-drop staggering. All remaining factors cost at most a constant per macroscopic passage. Consequently the source sum, divided by U and the deterministic loop weights, is bounded by

$$|Y/U| C^{1+X_0}, \quad (7.14)$$

where X_0 counts passages near the bounded wall and probe corridors and has uniform fixed exponential moments.

Proposition 7.2 (Complete-wall factorization). *For the complete wall, there are probe-independent complex numbers $a_n(R_0, L)$, bounded above and away from zero in modulus for the fixed large L , such that*

$$\mathcal{H}_{\text{full}}(z, u) = a_n(R_0, L) h_n(z) J_n(u) + o(1) \quad (7.15)$$

uniformly on interior compact probe sets. The error tends to zero with mesh first and then $R_0 \rightarrow \infty$. No limit of a_n is asserted or needed.

Proof. Under ν_U , bulk RSW circuits in the empty scales between the wall and the first endpoints make the probability that a raw open arc reaches the splitting scales tend to zero. On its complement the phase comparison above is exact. Lemma 7.1 and (7.14) make the error negligible even on the exceptional event, by Hölder.

Choose intermediate radii $1 \ll R_1 \ll R_2 \ll R_0$. The ideal tree integrand, including its open-to-deterministic reweighting, is readable inside R_1 when its boundary arcs stay there; set it to zero otherwise. Similarly Y/U is readable outside R_2 unless a raw path joins that scale to the first endpoint bands: new loops at smaller scales have trivial ratios. All product truncation errors vanish under ν_U by the moment estimates and RSW. The inward relative comparison across the intervening bulk annuli then factors their expectation up to $o(1)$. Applied to absolute parts, that comparison also replaces the inner tilted expectation by the unweighted one. The exterior coefficient is the fixed phase times $\mathbb{E}_{\nu_U}(Y/U)$ and the bounded tree/cap normalizations. Interior-exterior independence completes (7.15); Lemma 7.1 bounds the coefficient away from zero. $\square$

8 Closing finitely many wall gaps

Reflection cuts will require vacancies in the wall. The following comparison removes them without estimating a signed observable by an additive error in its partition function.

Choose finitely many separated gap locations away from the marks and source or string corridors. Remove intervals of lengths comparable to h ; at a genuine corner remove such intervals on both adjacent sides. Their endpoints are lattice vertices. At each tip, let N_{tip} count raw arcs joining the two banks of that ray around its tip, wholly inside a sufficiently small cutoff box of scale h . Use disjoint buffered boxes, and set

$$T_g = \prod_{\text{tips}} (\cot \lambda)^{N_{\text{tip}}}. \quad (8.1)$$

All these factors are positive. Other gaps may already be present at fixed sizes and separated locations; include their analogous factors.

Proposition 8.1 (Gap closure). *Keep the distant charges and their locations fixed. Let g denote the gapped wall and f the wall obtained by closing the specified h -gaps, retaining any other gaps and tip factors. There are positive probe-independent numbers b_n , bounded above and below, such that*

$$\mathcal{H}_g(z, u) = b_n \mathcal{H}_f(z, u) + o(1) \quad (8.2)$$

uniformly on interior compact probe sets. The error vanishes in the mesh limit followed by $h \rightarrow 0$. Constants may depend on the fixed remaining features. In particular, a second, smaller collection of gaps may be closed before the first collection.

Proof. Use the same tile pairings in the cut conventions g, f . A port interaction or raw arc is affected if it uses a newly missing port; the gapped diagram joins the filled paths terminating on the two sides there. Write

$$S = UT_f, \quad T = T_g/T_f, \quad \nu_g \propto ST\mu_g, \quad \nu_f \propto S\mu_f. \quad (8.3)$$

The supports of S are away from the closing gaps.

Confinement and the local multiplier. In the common collars at radii $h \ll r_1 \ll 1$, require a primal screen joining the wall rays in the sector carrying primal wire portions, and a dual screen in the other sector, which has no primal through wire. With diverging scale ratio, conditional RSW makes this inner good event tend to probability one under ν_g , even given the remote tilts and deeper tip data. It confines all affected paths to these neighborhoods. Moreover the outer primal connectivity induced by the inner fillings is the same for g, f : every connection using the deeper patch must enter the already connected primal screen, including bypasses through its joined wire portions. Represent the wires by deterministic connections along their designated intervals. The FK Euler formula now makes $\ell_f - \ell_g$ local, equal to a difference of internal component counts plus deterministic constants. This can first be checked with all distant edges fixed.

On this good event there is an exact local multiplier B such that

$$\mathcal{O}_g = B\mathcal{O}_f. \quad (8.4)$$

It is the ratio of the affected spin weights, multiplied by $d^{\ell_f - \ell_g}$. It is positive. To check the latter assertion, view the local patch as a disk with two radial notches ending at the separated tips. A same-bank arc, even one threading the gap, has its usual staggered $d, 1$ weights: simple-arc isotopy fixes its turn with the prescribed normal ends, and its endpoint high statuses are opposite by color parity. Orient an opposite-bank arc from its interior to its exterior port. Its endpoint directions are now the two interior normals; the high statuses agree, since the color to the left is the preceding interior gap at both ends. The matched boundary factors give amplitude

$$\frac{\cos(\rho(W + \theta_i - \theta_j)/2)}{\sin \lambda}. \quad (8.5)$$

For ports on the same ray, turn minus normal difference is $\pm 2\pi$, giving $\cot \lambda$. For different rays it is zero, giving $1/\sin \lambda$. Start with collinear rays, drawing a path around the finite tip or across the slot, respectively. Bending one ray through the intended right angle rotates its endpoint normal by the same amount, so the conclusion persists at a corner gap. All simple crosscuts with the same ends have the same lifted turn by isotopy. New closed internal loops have ordinary weight d , while the filled arcs have homogeneous-interval weights. Every local factor is positive.

We need a comparison uniform as $h \rightarrow 0$:

$$K^{-(1+X_h)}T \leq B \leq K^{1+X_h}T, \quad (8.6)$$

where X_h is a finite sum of passage counts at scale h and has uniform fixed exponential moments under ν_g . The following cap elimination avoids counting every straddle at intermediate scales. In both conventions remove affected boundary arcs of diameter below a small fixed fraction of h , together with their disk descendants. In f these lie on homogeneous continuous intervals, including across the missing positions. In g , a short affected arc with solid ends is near a single tip, cutting off a disk against one continuous bank or across that tip change. Its disk is also short by the straight or single-corner geometry. No such disk spans both separated tips; at a corner both removed side portions have lengths comparable to h . Also remove, in both conventions, any common unaffected descendants needed for either elimination. Such descendants have both ends on a common bank piece and the same neutral disk and intervening arcs; their union is therefore an allowed common removal.

Eliminate innermost arcs first. Zero-change cap increments cancel the $d, 1$ factors. Across a single tip change the two statuses are both high or both low; one unmatched port is transported and no loop is created, leaving exactly the $\cot \lambda$ factor. No whole interval is erased by these local removals. Remaining caps still pair adjacent high-low ports with inherited status, apart from boundedly many end ties. The number of removed tip arcs differs from the tilt count by $O(X_h)$. Every surviving affected boundary arc has a non-short passage at the slot scale, so their number is also $O(X_h)$, using bulk, straight-wall, and tip buffers. For this count, filled paths are disjoint segments of gapped paths. The tilt-edge moment bounds apply also near the cutoff boxes.

Remaining common raw arcs have the same ends. Their residual cap pairings differ at only $O(1 + X_h)$ ports: on every common solid interval, both pair consecutive high-low ports, interrupted only by interval ends and noncommon survivors. Thus their boundary-loop counts differ by at most this number, irrespective of long-range ties. New local closed loops cancel their ordinary spin weights. This proves (8.6) and retains the exact positivity.

Absolute moments, including exceptional configurations. For every fixed power, both $\mathcal{O}_g/(ST)$ and $\mathcal{O}_f/S$ have bounded moments under ν_g , uniformly as the indicated gaps shrink; $\mathcal{O}_f/S$ has the same bound under ν_f . Choose the two simple source corridors as in Section 7, with the ordered lanes and temporary start-clearance slide common to both cut conventions. On each pairing deform the paths to minimal tube separators. Both markers are outward sources and lie on distinct open arcs. Closed loops cannot mark. Apart from entire local launch arches, possible marked crosscut components are counted by macroscopic passages, also for repeated visits of a full strand. Local launch arches are zero while unpassed and have unit magnitude while passed, so only the last nested candidate can mark. This remains true when gaps open elsewhere. Other local flips have unit weights and can be avoided by the paths. Pure strings may meet remote arcs; at a wall mark their effects are exactly their specified bank charges and they do not alter ordinary small disks away from their endpoints.

The short eliminations and the residual cap comparison just used remain valid as count estimates without confinement of long affected paths. Short affected arcs and loops are away from all insertions and are neutral except for their tip factors. Every non-short affected boundary arc costs a bounded factor per X_h . For the other boundary weights, use the residual count of f and eliminate its local disks up to a fixed small macroscopic threshold, chosen relative to retained gaps and feature separations. This includes straddles at the newly filled homogeneous locations. Ordinary common disks cancel caps; flip and launch disks obey the preceding rules; a retained tip gives its designated positive factor, up to cutoff-scale passage errors. Any affected long survivor already costs $O(X_h)$. Thus the residual comparison has no loss per scale between h and macroscopic distance.

Internal closed ratios outside the local bulk tilt patches are trivial unless the loop has a macroscopic passage. For example, a loop enclosing a marker without its start also encloses an end of the marker's boundary arc on a solid of length bounded below, and hence cannot be tiny. The same observation applies to loops cutting off starts on solids. The other strings have only their designated local endpoint nests. Very large loops enclosing all finite data have ratio one. All remote scales, including the distant charges, stay fixed while these gaps close. Consequently the absolute source sums, after their respective positive tilt factors are removed, are bounded by

$$K^{1+X_0+X_h}. \quad (8.7)$$

Here X_0 is a fixed-scale passage count with every fixed exponential moment. Any trajectory of genuinely fixed diameter near a shrinking slot has a fixed-size subpassage outside a small neighborhood of its core, where the ordinary bulk or sector buffers apply. This also covers passages involving the newly opened wall; in f use the disjoint raw segments already noted. Equation (8.7) proves all the stated moment bounds.

Relative decoupling. Choose further scales $r_1 \ll r_2 \ll r_3 \ll 1$, all ratios diverging. There is an exterior-readable approximation F_{out} to $\mathcal{O}_f/S$ from outside the r_2 patches, exact unless raw passages traverse the common collar to r_3 ; set it to zero otherwise. Indeed, on this exterior good event, changing the fills inside r_2 affects only local arcs against homogeneous continuous bank intervals. Eliminate them and their intervening arcs with no net weight and the same remaining caps. To make this elimination common to all fills, remove the union of ports affected, or lying in an intervening interval, for some fill. Stable arcs from such an interval pair inside it and have that same neutral disk with its descendants. The untouched endpoints therefore see precisely the common adjacent-pair reconnection of the deleted blocks. Modified internal loops are ordinary and all remaining source data are unchanged. The moment bounds control its error under both relevant tilted laws.

Let

$$D_{\text{in}} = (B/T)\mathbf{1}_{\{\text{inner good}\}}. \quad (8.8)$$

It is nonnegative, measurable inside r_1 , and has bounded fixed moments and expectation in $[c, C]$, by (8.6) and the high probability of the good event. The outward two-sector case of Lemma 2.4 says that the conditional law of edges outside r_2 , given an inner fill under ν_g , differs by relative $o(1)$ from its law under ν_f . The graph and local wire representatives in the intervening collars and farther out are common; only the inner partition can differ. For several patches, iterate this uniform comparison patch by patch. Conditioning the other interiors adds only local ties, with their common external wires represented along the sides. The factor T is held inside and S is remote, so the relative comparison persists after both tilts. For the remote tilt, the uniform relative density bound changes both its weighted expectation and its normalizing constant by $1 + o(1)$, so it also holds after reweighting and normalization

by S . It follows that

$$\begin{aligned}\mathbb{E}_{\nu_g}[\mathcal{O}_g/(ST)] &= \mathbb{E}_{\nu_g}[D_{\text{in}}F_{\text{out}}] + o(1) \\ &= \mathbb{E}_{\nu_g}D_{\text{in}} \mathbb{E}_{\nu_f}[\mathcal{O}_f/S] + o(1).\end{aligned}\tag{8.9}$$

The errors follow from the preceding rare-event and moment bounds. The first factor is positive, bounded above and below, and independent of the probes. By (7.5) and (8.3), the two outer expectations in (8.9) are exactly $\mathcal{H}_g$ and $\mathcal{H}_f$. This proves the proposition. $\square$

9 Holomorphicity from reflection and extraction

We now supply the holomorphicity premise of Proposition 6.5 by estimating mixed differences of the paired-wall observable. Reflection and angular extraction give the estimate with gaps present; gap closure and complete-wall factorization then transfer it to the disk observable.

Assume now that the coordinates of Ω are rational in the tile frame. Thus its true vertical wall levels are congruent to some X_* modulo a fixed $G_* > 0$. All geometric constructions in this section are fixed before the mesh tends to zero. We first use small compact probe neighborhoods with the following strict margins: the two probe heights are distinct and avoid horizontal walls and feature heights; their abscissas avoid vertical wall levels; and the two probes are not on a common axis diagonal. Distant endpoints are placed at heights outside the probe bands. We remove these temporary exclusions at the end.

9.1 The mixed stencil and the gaps

For $v = z, u$ and a tile-axis unit vector e_k , set

$$\delta_{k,v}^{(2)}F = F(v + 2n^{-1}e_k) - F(v).\tag{9.1}$$

The relevant stencil is

$$(\delta_{1,z}^{(2)} + i\delta_{2,z}^{(2)})\delta_{j,u}^{(2)}, \quad j \in \{1, 2\}.\tag{9.2}$$

Simple two-step extensions with a common arriving prefix express it as a vacuum tensor expectation containing one high and one low local insertion from Section 5. The first is the high combination because the two positive axis directions have relative coefficient i in (9.2). All dependence on the stencil terms is localized in the two small probe boxes; all other paths are common diagonal strings. Fix the starts, homotopies, and lifts as in Section 7, with separate corridors and a final incoming path compatible with both positive extensions. These are valid tensor identities with gaps present.

The *primary gaps* have widths comparable to h . They include all true corners, and openings on every vertical wall in each probe height band. One anchor height $y_{*,i}$ per probe box suffices. Choose $H_i > 0$, with $2H_i$ dividing G_* , sufficiently small that for every probe height y_i in that box,

$$l_i = \frac{y_i + y_{*,i} - H_i}{2}, \quad y_i \in (l_i, l_i + H_i)\tag{9.3}$$

with uniform margins. The two cell intervals are disjoint and lie in clear horizontal feature bands. Put a primary gap at height $y_{*,i}$ on every vertical wall crossing the band. Round its ends to lattice vertices. We will also use *secondary gaps*, of much smaller width σ , at finitely many points of the remaining straight solids. Thus every solid component used in the estimate is a separated straight fragment.

The finite choices and the limiting order are different. First fix the primary width h and choose a finite reflection template; this determines the centers of the secondary gaps. Then fix their width σ sufficiently small for that template. With these choices and the distant

endpoints fixed, take the mesh limit. Only afterward do we close the secondary gaps, close the primary gaps, and send the distant endpoints away:

$$n \longrightarrow \infty, \quad \sigma \longrightarrow 0, \quad h \longrightarrow 0, \quad R_0 \longrightarrow \infty, \quad L \text{ fixed.}$$

No estimate uniform in the size of a shrinking-gap reflection template is required.

9.2 Reflection inequalities and angular extraction

An axis-parallel cut through tile vertices, vacant of solids, endpoints, and stencil boxes, gives a Hilbert-space Cauchy–Schwarz inequality for the plane vacuum. Each child keeps one half and adds its conjugate, arrow-reversed mirror, with exponent $1/2$ in the multiplicative inequality. This is the square-row Gram construction of Section 4, or its finite-cylinder limit. Diagonal strings can cross the cut transversely at its vertices: ice conservation reroutes any runs along the cut within a few vacant rows. Conjugation inverts the reflected charge, while reversing the joined path direction inverts it again. The flux therefore continues across the join, with no new endpoint or phase charge.

For any fixed finite reflection and extraction schedule, take each connected source box to contain all its prospective source edges and their common final incoming segment, with size below the minimum positive cut distance. Cut joins and all other extraction bands then avoid that box; every surviving copy contains an intact reflected image of the same local drawing. When extracting that copy, keep the final incoming segment fixed and reroute only the exterior background. The resulting closed routing difference is supported outside the connected box and has one face coefficient throughout it. Thus the phase is common to every term of the stencil, as required by Proposition 5.3.

Call a stencil copy *favorable* if the slope- $+1$ line through it lies in a clear slab of fixed positive macroscopic thickness, containing no solids, endpoints, other features, or other stencils. Its diagonal-time projection must differ from the others'. Parallel diagonal cuts inside this slab, with many positive-time rows on both sides, allow Proposition 5.3, based on Theorem 4.1, to extract the insertion with cost

$$O(n^{-1-\rho}) \quad \text{for a high insertion,} \quad O(n^{-1+\rho}) \quad \text{for a low insertion.} \quad (9.4)$$

This extraction retains the two external state norms. We spell out which path identities make those norms independent of the choice within the stencil. After an odd number of reflections the local insertion is the L_+, Q^{-1} alternative; its angular parameter is still $\tau = +$, with the reflected high coefficient. Rewrite any backward increment locally as the negative of its forward counterpart. The common start and end and the compensation pairs from Section 5 put both high terms on positive-step paths. The arriving charge belongs to the background even if the original prefix came from the future side. In the vacant slab, funnel that background to common seams and one skirt avoiding the entire marker rectangle. The difference of the original and new routings is a closed chain whose face coefficient is the same at every prospective source edge. Ice conservation therefore gives one common scalar phase, rather than term-dependent phases. These changes stay inside the intact stencil box and vacant band, so all other stencil sums remain in the external states unchanged.

The two state norms are themselves reflected diagrams, now omitting the extracted middle band and using the positive intervening square-layer metric at the two cuts. Their diagonal chains join at shared level vertices with continuous flux. Thus their exponents are again $1/2$. No bound on arbitrary boundary-state norms is being assumed. Other favorable copies to either side, and their mirrors, remain favorable if the bands are sufficiently narrow.

Suppose a node has k favorable copies. Extract one at a median projection. If k_-, k_+ copies are strictly before and after it, its two norm children have $2k_-, 2k_+$ favorable copies.

Each is at most k , and their half-average is $k - 1$. Under the successive half-weights the count reaches zero with expected time at most the initial count. Truncating at a sufficiently large finite depth therefore leaves an arbitrarily small expected residual count. Every extracted insertion is counted once at its node; otherwise total counts are preserved in half-average. All cuts and extraction bands avoid the other intact local data.

9.3 The complete cost of the norm diagrams

Angular extraction has supplied the decay factors while retaining the two exterior norms. We now estimate the reflected configurations defining those norms, so that their wall and charge costs can be compared with the normalization of $\mathcal{H}_g$.

At a final leaf, bound the remaining tensor sum by an absolute winding expansion. Every retained straight segment, its features, and every remote endpoint is an intact reflected copy. In one common + expansion, even-reflection copies use the original local weights, whereas odd-reflection copies use the absolute wrong-expansion weights. Indeed spatial reflection and travel reversal preserve the signed strand turn together, while conjugation reverses its phase. For positive comparison use direct cap wires on even copies and reflected complementary wires on odd copies, completing the disjoint straight slits by any deterministic planar positive caps. The local tables can then be read in their original bank coordinates.

Ordinary local arcs have exactly their high-low cap weights. Each flat mark contributes both banks. In the correct expansion the local magnitudes at all four flips are 1; in the wrong expansion they are zero per local straddle at p, b, d_0 and d at c , on each bank. At p use the effective raw drop $-s$ and the exiting diagonal prefix; at the other marks use s, t, s . Negation of the effective shifts on a flat wall gives the wrong weights. Reflection cuts preserve these outgoing local charge data even if the strings subsequently continue to mirrors. Tip-cross factors on a straight fragment are $\cot \lambda$ in either absolute convention. Endpoints of charge magnitude a have local closed-loop factors $u_+(a)$ on even copies and $u_-(a)$ on odd copies. There are no true corners on the solids.

Choose all local cutoffs smaller than the fixed cut margins. At any unextracted marker, a nonzero term requires its strand to reach a solid or another marker: a closed component cannot change orientation only once. It is therefore macroscopic, and no tiny surrounding loop can isolate the marker. Using vertex instead of source-edge endpoints changes positions by boundedly many mesh steps and costs only bounded passage factors. Every other nonordinary arc or loop has a macroscopic passage. There is no flux to infinity: each diagonal chain has total divergence zero after its source terminations are included, and the mirror joins create no new endpoints. Thus a closed loop enclosing all finite data has ordinary weight; one returning to the data while going far is counted among the macroscopic passages.

Cap elimination therefore bounds the absolute leaf expression, relative to its deterministic caps, by C^{1+X} times the product of the local factors just listed, where X has every fixed exponential moment. A zero local factor means the corresponding absence indicator. Hölder and the positive local-tilt comparisons absorb C^X at arbitrarily small loss in the coefficient of $\log n$. The two banks at a mark count separately: conditional on their exterior data they are two half-sector patches separated by the solid wall, with only outer cap ties. Tip expectations agree up to constants with the original positive tip normalizations. For a complementary copy, reflect across the ray to exchange banks and wire assignments, then use radius and collar comparisons. Surface partition costs of the disjoint straight fragments compare to the products of their original single-fragment costs; the one-step polarity shift supplies the same comparison for complementary copies. These are the separated-patch and surface comparisons of Lemmas 2.3 and 2.8.

Accordingly each retained feature carries its original baseline: the vacuum-relative segment partition, the tip expectation, or $\log F_+(a)$ for a bulk endpoint. The extra logarithmic costs

on wrong copies are

$$2P \text{ per colocated } s\text{-flip pair}, \quad 2D_* \text{ per } t\text{-flip pair}, \quad \log \frac{F_-(a)}{F_+(a)} \text{ per bulk endpoint.} \quad (9.5)$$

The errors are one-sided $o(\log n)$ in the sense of Section 3. No decay from an unextracted insertion is used; its finitely many local terms cost only a constant.

Interpret the exponents $1/2$ on the reflection tree as probability weights. An original segment or other feature has expected copy count one: it is never cut or removed in an extracted middle band. Ensure at least one preliminary split. In each individual child of every split, every retained feature is paired with its opposite-parity mirror. Hence the final expected even and odd counts are both $1/2$ per original feature. The baseline costs average back to

$$\log(Z_g^{\text{cap}} \mathbb{E}_{\mu_g}[UT_g]) + O(1). \quad (9.6)$$

There are three s marks and one t mark, with both banks at each, and two endpoint-charge sets of total magnitude one. Thus the extra averaged cost is at most

$$3P + D_* + \frac{1}{2} \sum_j \log \frac{F_-(a_j)}{F_+(a_j)} \leq 2P + D_* + o(\log n) \leq P + o(\log n). \quad (9.7)$$

The first inequality uses (3.8) twice, and the second follows from (3.6) and $C_i \geq 0$. Since $P \leq -c_0 \log n + O(1)$ for some $c_0 > 0$, this is a strict power saving available to pay for incomplete extraction and all small Hölder losses. Keeping the norm diagrams, the baseline, and the parity costs is essential to this saving.

9.4 Making almost all stencil copies favorable

Lemma 9.1 (Finite reflection templates). *For fixed primary gap widths and every $\varepsilon > 0$, one can choose finitely many reflection and extraction steps and finitely many secondary gaps so that the expected extracted count of each original marker is at least $1 - \varepsilon$. The cuts have strict macroscopic clearances. On any compact set of eligible probe pairs, finitely many such templates suffice, and one may use the union of their secondary gaps. All choices precede the mesh limit.*

Proof. We first work at one probe pair with ideal real cuts, before a small generic perturbation. Divide a bounded vertical interval containing all data, with empty end margins, into finitely many distinctly labeled cells, including the two marker cells (9.3). Horizontal folds act on a word of cell labels. Splitting a word of length m after position j makes the two words

$$WW^{\text{rev}} \quad \text{and} \quad V^{\text{rev}}V, \quad (9.8)$$

where W is the prefix of length j and V the suffix of length $m - j$. Their lengths are $2j$ and $2(m - j)$.

Under the size-biased weights j/m and $1 - j/m$, the vector of label frequencies is a bounded martingale. The expected terminal label frequency multiplied by the initial word length equals the expected terminal label count under the ordinary half-weights. Choose j within a bounded distance of $m/2$ to maximize the conditional frequency variance. These frequencies converge almost surely to a pure label. Here are the details. If the frequency vector π stays away from the vertices of its simplex, then some admissible near-middle prefix has

$$\|\text{counts}_{\leq j} - j\pi\| \geq c > 0. \quad (9.9)$$

Indeed two consecutive choices differ by a single-label vector minus π ; they cannot both have small norm. The mixed length-two case is immediate. The conditional frequency variance is then at least c'/m^2 .

Along the size-biased chain, $\sum m^{-2} = \infty$ almost surely. Lengths change by bounded amounts; the probabilities of its symmetric increments are $1/2 + O(1/m)$. Both the absolute conditional drift and conditional variance of $\log m$ are $O(m^{-2})$. If the sum were finite, localization and martingale convergence would make $\log m$ converge to a finite limit. Then m would be bounded, contradicting the same assumed summability. A bounded frequency martingale cannot accumulate the divergent conditional variance in (9.9) while tending to a mixed limit. Length one, if reached, is already pure. This proves purity and hence arbitrarily small expected impurity at a sufficiently large finite depth.

Outer-end folds retaining the whole word and its reflection can also make its length arbitrarily large; the empty child has zero size weight. In a branch, concentrate on its majority label if that label is a marker. For any prescribed finite run depth, only a proportion bounded by that depth times the minority frequency, plus the end contribution, fails to lie deep in a pure run of the majority. Their expected count can therefore be made arbitrarily small. Adjacent equal-label cells in the folded word alternate orientation. In a pure run of marker cell i , the vertical projection to the original cell is a triangular wave of period $2H_i$. It sends every marker ordinate b' to y_i and $b' + H_i$ to $y_{*,i}$, by (9.3).

We next choose vertical cuts to align the marker abscissas with the primary gaps. Horizontal folds change no abscissa, so the original horizontal support interval bounds every resulting diagram, independently of word depth. Thus the following finite abscissa schedule can be chosen separately for each possible target marker before choosing the horizontal word depth. All copies of a majority marker start from the same original abscissa x_0 . Its farther clear side up to a neighboring vertical wall has width at least a fixed fraction of G_* ; suppose this is the left side, reflecting the following construction otherwise. Take H_i much smaller than this width. Cut just to the right of the current rightmost copy $a_{\max}$ and retain the left child, including all marker columns there. The discarded child's marker count is zero, so this loses no expected majority count. The left tail below x_0 still projects identically. If the cut is c' , the new rightmost copy and right-tail projection are

$$a'_{\max} = 2c' - x_0, \quad s' \mapsto a'_{\max} + x_0 - s'. \quad (9.10)$$

Keep the central projection in the wall-free abscissa interval. The first cut is close to x_0 on the right; subsequent distances $c' - a_{\max} > 0$ can lie in the ample leftward clearance sampled by the old right tail.

Drift the copy abscissas modulo $2H_i$ toward $A_* = X_* - H_i$. Lift their distribution initially to mean x_0 and variance zero. At each step, with lifted mean μ , choose $c' \equiv \mu + v \pmod{H_i}$; reflected lifts are $2(\mu + v) - a$. The new mean is $\mu + v$ and the variance increases by exactly v^2 . Choose Nv to realize the required residue displacement to A_* in the rightward direction, with bounded positive total displacement, and then choose N large. The first cut fits its small right clearance and the later cuts their larger available clearance. Chebyshev's inequality makes all but an arbitrarily small fraction of copies have residue within a small fraction of the primary gap width of A_* . At the last step,

$$a_{\max} + x_0 \equiv 2A_* \pmod{2H_i}. \quad (9.11)$$

For such a successful copy (a, b') , consider its slope-+1 line. Choose the pure-run depth larger than the entire folded horizontal support span divided by the cell width; this finite bound is known from the abscissa schedule before the word depth is chosen. Throughout the horizontal support of all data, the line then projects vertically into the clear marker band, so

meets no horizontal wall or other feature height. In the central portion its abscissas avoid vertical walls. In the left tail, every projected vertical wall lies at X_* modulo $2H_i$, and hence at horizontal offset H_i from a up to the small residue error. The same is true in the right tail by (9.10)–(9.11). The triangular-wave identity therefore sends the ordinate at each such encounter into the primary gap at $y_{*,i}$. The line has a wall-free slab of positive thickness. By designing first the finite abscissa schedule and its required run depths for the fixed h , then the word depth, the expected non-targeted or unsuccessful marker counts are arbitrarily small for each of the two original markers.

Finally perturb every cut generically by a sufficiently small fixed amount. The finite successful inventory and strict clearances persist. Distinct stencil copies acquire distinct time projections: for copies with different reflection histories, their difference depends nontrivially on the last cut where the histories differ; distinct original markers with identical histories were separated by the generic-coordinate assumption. Keep cuts away from endpoints, marked features, tips, and coincident runs along solids. At every transverse solid intersection insert a secondary gap at its preimage on the original wall. There are finitely many such preimages, away from marks and corners after the perturbation. Thus all intended axis cuts become vacant.

Each template works on a neighborhood of its initial probe pair. A finite cover of a compact support gives finitely many templates; take the union of all their secondary gaps, then choose σ sufficiently small. Additional removed portions only improve the clearances. Round cuts to vertex lines. These roundings and the $O(1)$ -mesh shifts of the polygon preserve all fixed margins for fine mesh. Apply the finite median-extraction procedure above, with a further arbitrarily small residual count. This proves the lemma. $\square$

Proposition 9.2 (Mixed derivative saving). *For fixed primary gaps and sufficiently small fixed secondary gaps chosen as in Lemma 9.1, there is $c > 0$ such that*

$$\left| (\delta_{1,z}^{(2)} + i\delta_{2,z}^{(2)})\delta_{j,u}^{(2)}\mathcal{H}_g(z,u) \right| = O(n^{-2-c}) \quad (9.12)$$

uniformly on the eligible compact probe sets. The constants and reflection-tree size may depend on these fixed gaps.

Proof. Choose the expected extraction deficit ε in Lemma 9.1 small relative to c_0 in (9.7). The angular costs contribute at most

$$-(1+\rho)(1-\varepsilon)\log n - (1-\rho)(1-\varepsilon)\log n = (-2+2\varepsilon)\log n. \quad (9.13)$$

The baseline (9.6) cancels the precise normalization in (7.5). The remaining feature cost is at most $-c_0\log n + o(\log n)$ by (9.7). There are finitely many nodes, so all Hölder allowances may be chosen with total smaller than, say, $c_0/4$. Taking $2\varepsilon < c_0/4$ leaves a strictly negative extra exponent. This proves (9.12). It uses finite templates fixed before the mesh, not any uniformity as their gaps shrink. $\square$

9.5 Removing the gaps and the exterior factor

Theorem 9.3 (Holomorphic disk limit). *For every rational orthogonal polygon with the four flat marks and balanced approximations specified above, every locally continuous subsequential limit of h_n is holomorphic. Consequently the boundary identification and crossing formula of Proposition 6.5 hold without its holomorphicity premise for these polygons.*

Proof. Take a simultaneous locally convergent subsequence $h_n \rightarrow h$, $J_n \rightarrow J$. The ideal compactness statements do not use holomorphicity. On a product lattice coset for the step-two stencil, multiply (9.12) by a compactly supported smooth test function $\varphi(z,u)$ in an eligible

product neighborhood and sum with normalized mesh weights. Twice summing by parts and dividing by $(2/n)^2$ gives

$$\int \mathcal{H}_g(z, u)(\partial_1 + i\partial_2)_z \partial_{j,u} \varphi(z, u) d^2 z d^2 u = o(1), \quad (9.14)$$

where the integral denotes the corresponding lattice Riemann sum before passage to a limit. The difference-to-derivative error vanishes because $\mathcal{H}_g$ is locally bounded at fixed gaps by the absolute moment bounds. Every coset gives the same conclusion.

First let the mesh tend to zero. Proposition 8.1, with its positive multiplier bounded above and below, transfers the zero limit from the secondary gaps to the primary-gapped wall as $\sigma \rightarrow 0$. Apply it again as the primary widths $h \rightarrow 0$ to reach the complete wall. Finally use Proposition 7.2, whose scalar is bounded away from zero, and then let $R_0 \rightarrow \infty$. The ideal functions have compact convergence, so their lattice sums are ordinary Riemann sums. We obtain

$$\int h(z) J(u)(\partial_1 + i\partial_2)_z \partial_{j,u} \varphi(z, u) d^2 z d^2 u = 0. \quad (9.15)$$

No convergence of either comparison scalar is required: for each fixed geometric stage its inverse is bounded, and that stage's error tends to zero in the stated order.

Fix an interior z off the finitely many feature projection lines. There is a generic exterior probe box eligible for this z on which some distributional derivative $\partial_j J$ is nonzero. Otherwise both derivatives would vanish off the finitely many excluded lines in the connected strip; continuity would make J constant on the whole strip, contradicting its different limiting values at p, d_0 . Shrink the z neighborhood and this probe box to retain all strict margins. Choose a smooth compactly supported probe test β in this box such that $\langle \partial_j J, \beta \rangle \neq 0$. For every smooth test α supported in the z neighborhood, put $\varphi(z, u) = \alpha(z)\beta(u)$ in (9.15). The resulting identity is

$$\langle (\partial_1 + i\partial_2)h, \alpha \rangle \langle \partial_j J, \beta \rangle = 0.$$

Dividing by this nonzero scalar test pairing proves $(\partial_1 + i\partial_2)h = 0$ distributionally near z . Thus h is holomorphic there. Its continuity extends holomorphicity across the finitely many exceptional straight lines by Morera's theorem, splitting triangles along those lines and passing to the limit. Proposition 6.5 now applies. $\square$

10 Transfer to marked Jordan quadrilaterals

Throughout Sections 10–12, fix $1 \leq q < 4$ and write

$$d = \sqrt{q} = 2 \cos \lambda, \quad \rho = \lambda/\pi, \quad h = 1 - 2\rho \in [1/3, 1).$$

The four-change calculation of Section 6, whose holomorphicity hypothesis is supplied in Section 9, gives the functions

$$f(\chi) = \frac{I_C(\chi)}{I_B(\chi) + I_C(\chi)}, \quad g(\chi) = \frac{f(\chi)}{f(\chi) + d(1 - f(\chi))}, \quad 0 < \chi < 1, \quad (10.1)$$

where

$$I_B(\chi) = \int_{\chi}^1 u^{\rho-1} (u - \chi)^{\rho-1} (1 - u)^{1-3\rho} du, \\ I_C(\chi) = \int_1^{\infty} u^{\rho-1} (u - \chi)^{\rho-1} (u - 1)^{1-3\rho} du.$$

The first function is the probability in the open-cap normalization; the second is the probability of an *actual open path* between the two separately wired intervals. Wiring changes component weights; it does not supply an open path between different intervals.

Proposition 10.1 (Jordan-quad transfer). *Consider the marked, uniformly convergent square lattice polygon approximations $(D_n; a_n, b_n, c_n, d_n)$ to a bounded Jordan quadrilateral $(D; a, b, c, d)$ in the four-mark sense of Subsection 1.3. Keep every edge of the polygon graph random, and separately wire the two closed arcs $a_n b_n$ and $c_n d_n$, with no other primal identifications. If $\psi : D \rightarrow \mathbb{H}$ sends (a, c, d) to $(0, 1, \infty)$ and $\chi = \psi(b)$, the probability of an actual open path between these arcs tends to $g(\chi)$.*

Proof. First consider a fixed rational orthogonal polygon in tile coordinates. The deterministic caps of Section 2 contribute one or two additional loops according to the pairing. Their weight ratio is exactly the factor d in (10.1), so the four-change formula gives g . If its designated color lies on the dual grid, translate the complete tile domain, switches, and caps by one tile step. This preserves its loop law and makes that color the prescribed primal grid, at a geometric cost $O(\delta_n)$. We retain this translated tile graph and its cap law as the test graph; we do not identify it with the full graph of a primal polygon. It is nevertheless on the *same* primal lattice as the target. Each primal edge belongs to one tile, so an absent edge incident to a test vertex can attach only at the geometric tile boundary. Every eligible primal vertex on a designated electrode, including a vertex at a homogeneous corner or on a pendant edge, belongs to that electrode's cap wire. These facts are sufficient for the common-edge comparisons below.

Choose a plane Schoenflies homeomorphism taking $\bar{D}$ to the unit square, with the two electrodes on the left and right sides [52, Theorem 3.1]. Uniform marked boundary convergence and winding number imply the following strict inclusions for all sufficiently large n : D_n contains every closed inset square and is contained in every prescribed neighborhood of the square in these coordinates; its four marked arcs are close to their corresponding sides. Fix $\varepsilon > 0$. In physical coordinates polygonize the pullbacks of

$$[\varepsilon, 1 - \varepsilon] \times [-\varepsilon, 1 + \varepsilon] \quad \text{and} \quad [-\varepsilon, 1 + \varepsilon] \times [\varepsilon, 1 - \varepsilon].$$

Call these the easier and harder test quads, respectively (Figure 4), and put their electrodes on the nearly vertical sides. Take the polygonization error in Schoenflies coordinates smaller than $\varepsilon/10$, with marks in flat side interiors, and retain this margin after all tile translations. In particular, the free sides and the four marked transition neighborhoods retain the strict comparison margins; homogeneous corners along an electrode are allowed. Such simple rational orthogonal polygonizations exist: first use radial shrinking in a Carathéodory disk parametrization to obtain an analytic Jordan approximation, then a fine simple polygon; replace each segment by a rectangular staircase inside its own thin corridor and join adjacent staircases in disjoint small vertex neighborhoods. The marks can be placed in order in side interiors. These constructions give marked uniform convergence as $\varepsilon \downarrow 0$.

For the upper bound, restrict both laws to their common primal edges in the easier quad. Any target edge outside this common graph can attach to it only on its two electrode sides: the easier horizontal sides lie strictly outside the target. Outside pieces cannot join the two electrode sides, even through a target wire block, because the substantial intervening strip lies wholly inside the easier quad and the two target wire blocks lie in separate exterior zones. Thus the target's induced boundary partition on the common graph is weaker than separate wiring of the easier electrodes. Domain Markov and FK monotonicity give a coupling in which the easier common-edge configuration dominates the target's. Every actual target crossing contains a common-edge crossing of the easier electrodes. Indeed, follow it from the initial exterior zone to the opposite zone, taking its last exit from the first and first entry into the second; its intermediate edges must belong to the common graph. There is no escape through the distant horizontal sides.

For the lower bound interchange the roles using the harder quad. Its exterior pieces can attach to the target only on the designated target arcs, and cannot link opposite sides outside

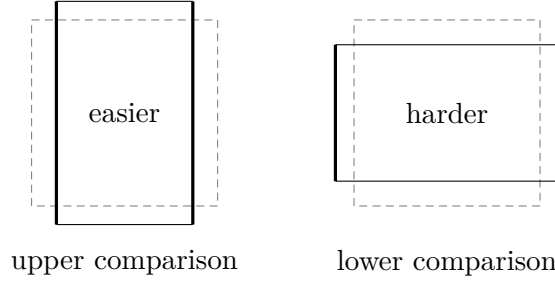


Figure 4: The Jordan-quad comparison in Schoenflies coordinates. The dashed square is the limiting target; thick sides are separately wired electrodes. The easier quad extends beyond the free sides and ends inside the target's electrode sides. The harder quad has the opposite inclusions. Physical test polygons retain strict margins after pullback and lattice approximation.

the common graph. Here it matters that the target contains *all* lattice edges in its closure, and that both graphs use exactly the same primal grid. The target wires therefore dominate the induced harder-quad partition. Every harder crossing contains an actual shared-edge piece joining the two target arcs. This proves the lower comparison. All transitions occur at ordinary graph vertices; shrinking lattice edges have uniformly shrinking diameter in Schoenflies coordinates, so the strict margins justify the discrete versions of the statements.

The conformal cross-ratios of the two test quads tend to χ . Indeed uniformly convergent parametrizations of Jordan curves imply uniform local connectedness of their boundaries: points close in the image have parameters close in the circular order, except for the identified endpoints, because the limiting parametrization is an embedding. Kernel convergence and boundary extension of normalized Riemann maps then give marked conformal convergence [39]; see also the explicit uniform-Jordan and kernel-convergence statements in Camia and Newman [8, Appendix A, Theorems 6–8 and Corollaries A.1–A.2]. Since g is continuous on $(0, 1)$, let first $n \rightarrow \infty$ and then $\varepsilon \downarrow 0$ in the two comparisons. $\square$

Corollary 10.2 (The prescribed bond-percolation crossing). *At $q = 1$, with every edge, including every boundary edge, sampled independently with probability $1/2$ and with no wiring, the probability of the event C_n defined in Subsection 1.3 tends to*

$$\frac{\int_0^\chi [t(1-t)]^{-2/3} dt}{\int_0^1 [t(1-t)]^{-2/3} dt}.$$

Proof. For $q = 1$ the boundary partition does not change the edge law, and $d = 1$ gives $g = f$. The $\rho = 1/3$ specialization of the four-change integral calculation is precisely the displayed expression. The comparison in Proposition 10.1 concerned actual edge paths throughout, so no external wire is counted as a crossing and no interface-event continuity assertion is needed. $\square$

11 Alternating tests after exploration

The Jordan result alone does not apply to a domain cut out by a random exploration. We prove the needed extension directly under the original lattice law. Conditional errors at exceptional histories will be controlled by an *averaged* six-arm estimate.

11.1 Geometry and conditional laws

We reveal a switch in full when the strand first encounters its tile, then follow the prescribed local arc. The revealing filtration records these switch states and the current position on that arc. A stopping time for this filtration may therefore fall partway through an already-revealed tile; it need not be a tile-completion time. Below, a *switch-revealing stop* means any stopping time for this filtration, including such a partial-tile stop.

Lemma 11.1 (A compatible simple drawing). *The prescribed polygon graph and its Dobrushin strand admit a tile and collar representation in a Jordan polygon V_n such that:*

- (i) V_n has the same marked uniform Jordan limit as D_n ;
- (ii) the rounded strand is simple, visits the same ordered local pieces as the prescribed medial strand, and is at curve distance $O(\delta_n)$ from it;
- (iii) at every stop for this revealing filtration, the remaining domain is a simple slit disk. Under deterministic cap closure its unrevealed edges have the original conditional FK law, with known open edges contracted and known closed edges deleted;
- (iv) primal and dual banks of the prefix have physical access to their corresponding known edges and wire vertices within $O(\delta_n)$. Every completion of the unknown edges embeds its actual paths in the slit disk without crossing the past.

The construction can use identical domains and caps away from an interval where two additional boundary changes are inserted.

Proof. Make one tile for each edge of the polygon graph, including boundary edges. Glue the four triangles directed toward each interior square center. The exterior triangle of a boundary edge is a separate tab, with its own outer dual vertex; do not glue different exterior tabs along their radial sides. Each primal boundary vertex retains all its interior incidences. Flatten the exterior tabs toward their bases, to less than half their original height, and add a still thinner Jordan collar. The tabs are then disjoint except at their prescribed vertices. All changes lie within $O(\delta_n)$ of the original boundary, and tracing the tabs and collar along the original parametrization proves (i).

Draw the switches and boundary caps as disjoint arcs. A local wire can be drawn behind its cap in the collar. The unmatched ports continue to the marked outer-boundary points. The finite choices inside each tile and the collar can be made deterministically. At a wired boundary edge, capping its outer dual vertex joins the inside ports in the same way for either switch state. Declaring that edge open for the drawing merely substitutes a cap-sized detour and possibly deletes a microscopic loop. On the free side, the cap around a primal vertex is exactly the turn beside the dual wired arc. Inside the polygon, glued half-segments give the specified medial edges and their quarter-turn pairings. Match corresponding ordered pieces in the two drawings. Each substituted piece lies in boundedly many neighboring tiles, so simultaneous parametrizations have distance $O(\delta_n)$. Zero-length pieces are handled by limits of increasing reparametrizations. This also covers a midpoint visited with both prescribed pairings and proves (ii).

The tile/cap Euler identity gives exactly the conditional FK law asserted in (iii). On a queried primal-open tile the adjacent primal corners are joined by its actual edge; on a queried closed turn the bank stays at one primal corner. Along a cap it stays at one vertex or uses neighbors already belonging to that wire interval. Thus the bank is a known chain attached to the appropriate original interval, and every queried primal-open edge accesses this bank within one tile. The same statement holds for dual edges. Designated outer vertices access their arcs through the collar. These are physical accesses, not additional identifications. At a stop partway through a local arc, the full state of its tile has already been revealed, and following that known arc reveals no additional unknown edge. The conditional law in (iii)

and these bank accesses in (iv) therefore apply at such stops as well. Projecting back to the usual primal or dual lattice moves points by $O(\delta_n)$ and duplicates no edge; separate virtual copies of a vertex may become one physical vertex. $\square$

In a four-open-change experiment, deterministic closure with separate wires of one color differs by a density depending only on the two pairing outcomes. This density and its reciprocal are bounded by constants depending on q . At a switch-revealing stopping time the odds change is still exactly

$$p \mapsto \frac{p}{p + d(1 - p)}. \quad (11.1)$$

One obtains this by conditioning the *full* finite-sample density, not by postulating a Markov property for a limiting curve. Consequently conditional rare-event probabilities in a deterministic-cap law can be averaged under open-cap histories at a bounded multiplicative cost.

Lemma 11.2 (Small-interval behavior). *For f in (10.1),*

$$f(1 - \chi) = 1 - f(\chi), \quad f(\chi) = c_q \chi^h (1 + o(1)) \quad (\chi \downarrow 0), \quad c_q > 0.$$

Proof. The substitutions u^{-1} in I_C and $u = 1 - (1 - \chi)t$ in I_B , followed by the Euler transformation of the beta integral, give

$$I_B(\chi) = \left(\frac{1 - \chi}{\chi} \right)^h I_C(1 - \chi).$$

Both $I_C(0)$ and $I_C(1-)$ are finite and strictly positive: at the possible singularity $u = 1$, the latter integrand has exponent $-2\rho > -1$, while the integrand at infinity has exponent $-1 - \rho < -1$. Substitution into (10.1) proves both claims, with $c_q = I_C(0)/I_C(1-)$. $\square$

Theorem 11.3 (Moving four-change test). *Let a four-change strand in V_n be stopped before completing its strand, revealing only the switches it has encountered. Let U_n be its remaining simple slit disk, with the tip as the new starting mark. Consider histories for which a fixed basepoint has a fixed positive disk in U_n , and the four prime-end marks in a disk chart normalized at that basepoint remain mutually separated by a fixed positive amount. Then, on these histories,*

$$\mathbb{P}_{\text{open}}(E_P \mid \mathcal{F}_n) - f(\chi_{U_n}) \longrightarrow 0 \quad \text{in probability.} \quad (11.2)$$

Here E_P is the primal pairing event, equivalently the actual bridge event after separate primal closure, and χ_{U_n} is its current ordered cross-ratio. The assertion holds for arbitrary mesh-dependent stopping rules satisfying this localization.

For the comparison, first use separate deterministic primal closure and let u_n be the rectangle abscissa in U_n , with the two primal electrodes at 0 and 1. A *possible primal path* is an edge path after all still unknown primal edges have been filled open, without using a wire identification as an edge. We need the comparison boundary to avoid vertices in the central part of $0 < u_n < 1$ that have such a possible path to an electrode: every central vertex of an actual crossing is among them.

We construct its boundary by routing a finite polygonal obstacle near the explored bank. Outside small protection balls and grid cells where the past already supplies four alternating arms, the obstacle has positive clearance from those vertices. The exceptional regions have small conditional attachment cost outside a set of histories of small probability. Rounding the obstacle gives a Jordan domain H , selected from a finite list at each fixed accuracy. There is no inclusion requirement between H and U_n . Instead, matched rectangle fibers align

their abscissas along electrode paths that avoid the exceptional regions. After deleting the small-cost exceptional edges, an original crossing must contain a crossing of a trimmed test quad in H on their common graph, where the test wiring dominates the original induced partition. This gives the upper bound; exchanging colors gives the other bound.

Every annulus used below lies a fixed positive distance inside the original domain before its mesh limit is taken. The inputs are the bulk passage estimate of Section 2, the vanishing full-lattice one-arm bound with wired rims, and the six-arm estimate

$$\mathbb{P}(A_6(r, R)) \leq C(r/R)^{2+c_0}, \quad c_0 > 0, \quad (11.3)$$

where $A_6(r, R)$ is the event of six cyclically alternating actual primal-open and dual-open arms across the annulus. The estimate holds for the critical square-lattice FK law with this fixed $q < 4$ [17, Corollary 6.7]. An outer buffer and the bulk boundary-condition comparison of Lemma 2.3 transfer this estimate from the infinite-plane law to the original finite law, with the outer scale fixed. Bounded open-cap reweightings are harmless. No six-arm estimate conditioned on an arbitrary explored maze is asserted.

Lemma 11.4 (Physical paths and chart oscillation). *Let u_n be the rectangle abscissa just defined. Under the localization in Theorem 11.3, there is a common modulus tending to zero for the oscillation of u_n on a connected path of vanishing physical diameter, including paths with accesses to the prime boundary. Consequently, for $\beta > 0$, a possible primal path starting in $\beta \leq u_n \leq 1 - \beta$ cannot reach a known primal-open edge or a primal electrode inside a sufficiently small physical ball.*

Proof. Away from the normalizing point, a connected set in a ball of radius r is hit before domain exit by Brownian motion from that point with probability tending uniformly to zero as $r \downarrow 0$. Comparison with a fixed enclosing Euclidean annulus gives this bound. In the normalized disk, a connected set of a fixed positive diameter has hitting probability bounded below; this follows, for example, by a crosscut between separated angular sectors and the Beurling estimate. The rectangle maps of the disk form a compact family because the four marks remain separated. Thus a small physical continuum cannot have a macroscopic rectangle-coordinate oscillation. Near the basepoint use interior distortion instead. Endpoint accesses follow by interior approximation.

By Lemma 11.1, every known primal-open edge and fixed primal wire attachment accesses $u_n = 0$ or 1 within $O(\delta_n)$. A short possible connection from the central band to one of them would contradict the just-proved modulus. Small buffers in the band absorb individual tile errors. In particular the two opposite designated intervals have not already met through the revealed bank under the stated nondegeneracy. $\square$

Lemma 11.5 (Deletion and protected-ball costs). *Fix a history and use separate deterministic primal closure. Deleting a set of still random edges changes the actual crossing probability by at most the probability, in the undeleted conditional law, that an endpoint of a deleted edge reaches either wire interval or its revealed primal bank by an actual open path, allowing one initial test step over the deleted edge. Contact at a wire vertex may have length zero. For a central-band set contained in a ball of radius r , this cost is bounded by a quantity tending to zero as $r \downarrow 0$, uniformly in the history and center, provided its required outer radius is fixed and the mesh is sufficiently small. The same holds for any fixed finite number of such balls.*

Proof. The separate wire intervals and their revealed banks are the only nontrivial identification blocks, by Lemma 11.1. Couple the deleted and original laws monotonically. Explore the actual open clusters from all deletion endpoints in the stronger configuration, revealing the corresponding edges in both laws. FK partition monotonicity persists at every step. After

the clusters are enclosed by their closed cuts, the conditional laws outside agree and can be coupled identically. If no explored cluster reaches a wire interval, the crossing decision is unchanged. This proves the first assertion, including contact with a wire along a known bank.

Choose the fixed outer radius small enough for Lemma 11.4. Every component of possible edges inside this ball that meets the central-band set has no accessible wire or known primal-open attachment. Its uncertain boundary connections can occur only at the ball rim. Project to the physical lattice, fill missing edges, and wire that rim. Splitting physical vertex copies only weakens the original induced partition. The full-lattice wired-rim law therefore dominates every relevant actual arm simultaneously. Cropping the ball slightly absorbs tile errors. Its one-arm probability from radius r to the fixed outer radius tends to zero: buffered dual circuits from RSW block such an arm. A finite union bound gives the last assertion. $\square$

11.2 A directed obstacle away from possible connections

Fix a small trimming parameter $a_* > 0$ and then $0 < \beta < a_*/20$. A primal vertex is a *candidate* if its chart coordinate lies in $[\beta, 1 - \beta]$ and it has a possible connection to an electrode. Each candidate supplies a possible arm to a fixed positive physical distance, by Lemma 11.4.

Lemma 11.6 (Flagged cells and their cost). *In a fixed bulk block let π be a simple known primal bank path. Choose fixed scales $\ell \ll R_1$, away from its endpoints and the original boundary. Flag a deterministic grid cell of side ℓ if, in its annulus from 100ℓ to R_1 , the prefix already supplies four cyclically alternating actual arms, its two primal arms being known wired bank paths. If a candidate within $O(\ell)$ of this cell actually connects to an electrode, then six actual alternating arms occur in a cropped annulus from 200ℓ to $R_1/2$. For a bounded region covered by the deterministic cells, the averaged conditional attachment cost of all flagged cells is at most*

$$C\ell^{-2}(\ell/R_1)^{2+c_0} = O_{R_1}(\ell^{c_0}). \quad (11.4)$$

The statement remains valid with one initial test step and with the bounded cap changes in (11.1).

Proof. The extra actual primal arm cannot make a lateral connection to either known primal leg inside the wider annulus, even in the all-unknown-primal-open completion, by Lemma 11.4. On the side containing the extra arm it is bracketed by the two known primal legs. A planar crossing alternative in each intervening strip supplies a dual radial separator. The original dual separator on the other side remains. After cropping at both rims, these three primal and three dual arms are disjoint and alternate.

For precision, thicken actual paths to their tile color territories. An open transversal between two primal territories would give an actual lateral bridge, with an error of at most one tile. Its absence forces a dual territorial radial crossing in the intervening topological rectangle, which shadows a dual edge path in the cropped annulus. If the no-bridge test is made after every unknown primal edge is opened, every dual separator so obtained consists of known closed tiles. Thus the four old legs and the separators used in this argument are actual original edges; no auxiliary boundary condition manufactures them.

Apply (11.3) in the original full-sample law to each cell of the deterministic grid, then sum over $O(\ell^{-2})$ cells. The center may have been selected by the past because the sum was taken over the entire deterministic grid. Bounded cap densities control the averaged conditional probabilities under either history law. Markov's inequality now makes the conditional flagged-cell cost small outside a set of histories of small probability. This is an annealed-to-conditional argument, with R_1 and all comparison constants fixed before $\ell \downarrow 0$. $\square$

Lemma 11.7 (Directed obstacle). *After removing histories of arbitrarily small probability and allowing an arbitrarily small conditional deletion cost, there is a Jordan polygon H containing the basepoint, selected from a finite list at each fixed accuracy, with the following properties. Its boundary follows the original boundary and explored prefix to that accuracy; away from finitely many protected balls and flagged cells it has fixed positive clearance from all central candidates. As the accuracies decrease, H and U_n have the same pointed kernel limits. Artificial marks in H may be chosen with disk angles arbitrarily close to the corresponding angles in U_n .*

Proof. We construct the obstacle in three stages, keeping the scale choices in the order in which they are used.

Retained bank blocks. Retain only prefix excursions reaching depth r from the limiting Jordan boundary, including each complete segment between successive visits to depth $h_b \ll r$ and the last incomplete segment if present. Their number is tight, uniformly in small h_b , because each visits one of a fixed finite family of bulk annuli at depth $r/2$. First bound this number, then choose h_b small. Join their boundary-near endpoints to the exterior of a slightly larger polygon through protected balls of radius $O(h_b)$. Lemma 11.5 controls central attachments in these finitely many balls. No boundary arm exponent for the rough Jordan curve is needed.

Break the retained excursions into blocks of diameter at most $\epsilon \ll h_b$, each contained with margin in a bulk square S of side 10ϵ . Successive displacement thresholds and the bulk passage estimate make the number of blocks tight. After fixing a block-count cutoff, snap its endpoints to a much finer deterministic grid and protect them by balls of radius $e \ll \epsilon$. The actual known primal bank gives a path between the true endpoints in S ; loop erasure gives a simple path π , possibly trivial. Take

$$\ell \ll R_1 \ll e,$$

so all flag annuli are bulk annuli.

A connector avoiding possible attachments. Declare forbidden every closed ℓ -cell within 3ℓ of a candidate, except flagged cells and cells in enlarged endpoint protections.

There is a connector between the protected endpoints avoiding all forbidden cells. Otherwise planar cell separation gives a polygonal circuit through a chain of touching forbidden cells and possibly the exterior of S , separating the two endpoints. It can be chosen away from their protection balls and within a cell-sized error of the cell chain. Choose a candidate witness in each cell. Two successive witnesses can be joined avoiding π inside a disk of radius $O(R_1)$. If they could not, a proper crosscut of the simple path π would separate them in a circle of radius between $4R_1$ and $6R_1$. Its full-path endpoints lie outside because the cell is away from endpoint protections. The crosscut passes within $O(\ell)$ of the witness segment. The two witnesses have possible arms to that circle, on opposite sides of π , and cannot connect locally to π by Lemma 11.4. The two known radial legs of π and the two possible arms consequently interlace. In each intervening strip the optimistic no-bridge test supplies a *known* dual radial separator, as in the proof of Lemma 11.6. The cell is therefore flagged, a contradiction.

Push the separating circuit through the witness points and these avoiding joins. Its displacement is $O(R_1)$, smaller than its distance from the endpoints, so its winding distinction is preserved. Along the exterior of S avoidance is automatic because π has an interior margin. We have obtained a separating circuit disjoint from π , which is impossible. Overlapping endpoint protections or a trivial π are handled by a connector within their union.

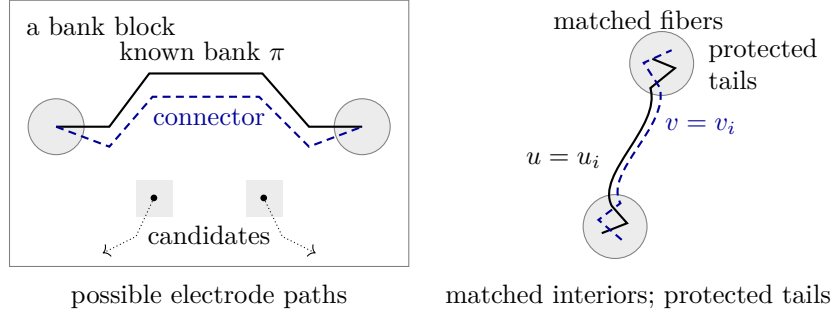


Figure 5: The two comparisons in the moving-domain argument, schematically. Left: a deterministic connector stays near a retained known bank block while avoiding central candidate neighborhoods; the dots have merely possible electrode paths, not known open paths or bridges to the bank. The endpoint balls are charged by the deletion estimate. Flagged cells, where an attachment is instead charged by the six-arm estimate, are not shown. Right: fibers of the rectangle abscissas u in U_n and v in H match on compact interior portions; length–area confines their remaining tails to common small balls. Neither panel asserts that H and U_n are nested or that their slit boundaries converge uniformly.

The exterior-attached obstacle and its kernel. Select one polygonal connector for each allowed finite grid pattern. Their union and the protected exterior attachments form an obstacle K attached to the complement of the larger polygon. The connectors need not form one simple curve. They stay close to the retained prefix, and every omitted prefix piece is boundary-near; conversely the entire prefix is approached by this obstacle or the outer wall. Within distance $\ell/2$ of K , candidates occur only in flagged or protected error regions. The basepoint component of the complement is simply connected, because each obstacle component is attached to the exterior. It is a fixed polygonal domain, possibly with slits and pinches. An interior radial approximation in its Riemann map, followed by a polygonal perturbation, gives a Jordan domain H whose boundary is within $\ell/10$ of the obstacle or outer wall. The outer wall was chosen much farther than ℓ from the original graph. Hence the asserted clearance holds. Every choice is from a finite list at fixed scales and count cutoffs.

Normalize maps of H and U_n at the same basepoint, and transfer marks by finite-grid approximations to their disk angles. On a Hausdorff subsequence of the original boundary and prefix, every compact path in the limiting basepoint component eventually belongs to both constructions, while every limiting boundary point is approached by complements of both. Thus their pointed kernels agree, and their normalized interior charts have the same limit. This is an interior statement; no locally connected boundary for the limiting slit domain is asserted. The fixed basepoint margin and separation of the marks persist throughout. $\square$

The obstacle and the comparison fibers have different jobs: the first keeps the new boundary away from possible crossing attachments, whereas the second aligns the two chart coordinates. Figure 5 illustrates these two geometric steps.

11.3 Alignment of the comparison charts

Proof of Theorem 11.3. It suffices to prove an upper bound for the primal connection under separate deterministic primal closure. Interchanging the colors and using the other deterministic closure gives the complementary upper bound; (11.1) then yields the open-cap limit.

Use Lemma 11.7, and write u and v for the rectangle abscissas in U_n and H . Test on the Jordan subquad of H between $v = a_*$ and $v = 1 - a_*$, with these sides separately wired. At

fixed accuracy there are finitely many such quads; approximate each on the original color grid by the polygon construction of Proposition 10.1. Their limiting crossing probabilities differ from $g(\chi_{U_n})$ by arbitrarily small trimming, chart-angle, and kernel errors, uniformly under the nondegeneracy assumptions.

Matched fibers and their protected tails. To align the charts along crossing paths, take a fine finite grid of abscissa bins in the central part of $(0, 1)$, with width much smaller than both β and a_* . In each bin choose a nearby vertical fiber for each chart. Its central part is a compact internal curve and is matched in the other chart by kernel convergence. Its two remaining tails can be confined to balls of vanishing radius, with matched centers. Here is the required length–area argument. Remove small fractions of rectangle height at top and bottom. For each fixed fraction, inverse maps and derivatives converge on the compact retained rectangles. Let that fraction decrease slowly. For a bin I and a deleted height interval J , Cauchy–Schwarz gives

$$\int_I \text{length}(\Phi(\{x\} \times J)) \, dx \leq (|I||J|)^{1/2} \left(\int_{I \times J} |\Phi'(z)|^2 \, dA(z) \right)^{1/2},$$

where Φ is the inverse rectangle map. The last integral is bounded by the common physical area bound. Choose levels in each bin controlling the tails of both charts. Their initial endpoints match on the retained compact pieces, so matched small endpoint balls contain all four tails. This argument does not require uniform local connectedness of the explored boundaries. Tile errors are absorbed by decreasing the fractions slowly enough. Lemma 11.5 controls the conditional attachment cost of these finitely many endpoint balls.

Deleting edges that obstruct comparison. Delete still unknown edges touching a slightly narrower band, initially $3\beta \leq u \leq 1 - 3\beta$, if their endpoints fail containment in H with a sufficiently small fixed fraction of the clearance ℓ , or if their two abscissas differ by more than $a_*/10$. These deletions have negligible conditional crossing cost. To see this, suppose an endpoint of such an edge reaches an electrode by an actual path, allowing the initial test step. Paths meeting a flagged cell, protection ball, or fiber-endpoint ball while centrally located already have negligible total cost. For every other path, trace it inward from the electrode after its last entry into the broader band $2\beta \leq u \leq 1 - 2\beta$. It crosses a comparison u -fiber on its main internal piece, giving an anchor in H with nearly equal abscissas. It cannot subsequently leave H , since its central vertices are candidates and the wall has clearance from them. Each later crossing of either chart’s comparison fiber is again on its main piece. If v drifted by more than a few bin widths from u , it would cross a v -fiber in a nonmatching u -bin, a contradiction. The endpoint therefore has neither proposed defect. Vertices without a possible electrode connection can be discarded at no crossing cost. Smaller nested bands and clearance constants accommodate complete edge segments and physical-grid projection.

The induced partition on the common graph. Now keep the surviving uncertain edges in $4\beta \leq u \leq 1 - 4\beta$ that also belong to the trimmed test graph. A nonshared original piece can attach to these only on a designated side of the test quad: clearance excludes its free horizontal sides, and the abscissa buffers exclude an attachment from a noncentral piece. No nonshared piece, even with history wires, connects the two designated sides. Such a traversal must pass through the middle shared strip by chart matching, while all identifications lie in the two separate groups near $u = 0, 1$. After matching physical vertices, the induced original partition is thus dominated by the test law’s separate wiring. Every modified original actual crossing contains a shared crossing between the test sides, by taking its traversal from the first

exterior group to the second. Domain Markov and partition monotonicity give the desired test upper bound.

Order of the approximations. The construction has the following quantifiers. If convergence in probability failed with fixed error and probability, choose first a_* small, then β and fiber precision smaller. Choose obstacle accuracies tending to zero, with, at each stage, a deep-excursion cutoff, boundary protection radius h_b , block sizes and count cutoffs, endpoint protections e , the intermediate radius R_1 , and finally the flag scale ℓ . The conditional protection costs and the averaged flag cost (11.4) are made small in that order. For each fixed stage choose the lattice mesh small enough for every geometric margin and each of the finitely many Jordan test limits. Select failing histories outside all exceptional sets, and pass to a common kernel subsequence. The fiber argument and additional deletions along this subsequence have uniform conditional error by Lemma 11.5, and contradict the failure. All six-arm summations took place under the original law before this deterministic selection of histories. This proves (11.2). $\square$

12 Identification of the Dobrushin strand

The moving four-change tests first determine the limiting Loewner driver. We then combine that identification with physical compactness and crossing estimates to identify the entire ordered Dobrushin curve.

12.1 Two small tests determine the driver

Map V_n to $\mathbb{H}$, sending its two Dobrushin ends to 0 and ∞ , with a convergent interior normalization. The inverse maps converge uniformly in disk coordinates, including the boundary, by marked Jordan convergence. Choose the orientation so that the positive real side is primal-free. The rounded simple strand has a continuous Loewner driver $W_t^{(n)}$, with

$$\partial_t g_t^{(n)}(z) = \frac{2}{g_t^{(n)}(z) - W_t^{(n)}}, \quad g_0^{(n)}(z) = z, \quad W_0^{(n)} = 0. \quad (12.1)$$

We suppress the superscript when discussing a single mesh. Fix $T, R < \infty$ and stop at T or the first exit of $|W_t| < R$. All constants in this subsection may depend on these fixed bounds.

Lemma 12.1 (Bounded martingale tests). *For every $x > 2R + 1$, put*

$$M_t(x) = \left(\frac{x g_t'(x)}{g_t(x) - W_t} \right)^h. \quad (12.2)$$

These processes and their reciprocals are bounded on the stopped range, uniformly in the mesh. For arbitrary stopping rules $0 \leq \sigma_n \leq \tau_n$ in that range and uniformly bounded $\mathcal{F}_{\sigma_n}$ -measurable tests F_n ,

$$\mathbb{E}[F_n(M_{\tau_n}(x) - M_{\sigma_n}(x))] \longrightarrow 0 \quad (12.3)$$

under the original single-strand Dobrushin law.

Proof. Insert on the original free side a small separate primal interval whose endpoints in the initial coordinate approximate $x, x + s$. Use the four-open-change experiment. Until the specified stop its cross-ratio is

$$\chi_t = \frac{g_t(x + s) - g_t(x)}{g_t(x + s) - W_t}.$$

The real Loewner solutions remain separated from the driver and from one another. At each fixed $s > 0$ the four marks, including ∞ , therefore stay nondegenerate. A physical point corresponding to iM for sufficiently large fixed M has a fixed unaffected disk; it supplies the basepoint in Theorem 11.3. The trajectory cannot end at either inserted mark within this localization.

Let Z_t be the conditional pairing probability. It is an exact bounded martingale. At any of the allowed stopping rules, Theorem 11.3 gives $Z_t - f(\chi_t) \rightarrow 0$ in probability, and hence in mean, for fixed s . Divide the martingale identity by $f(s/(x+s))$ only at this fixed s . Lemma 11.2 and analytic ODE estimates near x give, uniformly in the stopped range,

$$\frac{f(\chi_t)}{f(s/(x+s))} = \left(\frac{xg'_t(x)}{g_t(x) - W_t} \right)^h + o_s(1), \quad (12.4)$$

after the mesh limit. The expression on the right is bounded above and away from zero independently of small s : $g_t(x) - W_t$ is bounded below, $g_t(x)$ is bounded above, and

$$g'_t(x) = \exp \left(- \int_0^t \frac{2 du}{(g_u(x) - W_u)^2} \right)$$

has the same bounds.

We next compare the laws, after replacing the normalized conditional probabilities by the bounded expressions in (12.4). Separate deterministic four-change closure adds just the small wire interval to the Dobrushin law. Couple monotonically and explore clusters from that interval in the stronger law. The probability that its actual cluster reaches a fixed small outer neighborhood tends to zero as the interval shrinks. Inside the intervening annulus there are no wire attachments away from the rims, since the old primal wire is a positive distance away. Filling to the full-lattice wired-rim law dominates the arm, which is blocked by the buffered RSW circuits. Outside this neighborhood the configurations can consequently be coupled identically. The open-four and deterministic-four laws also have vanishing total variation difference: their density depends only on the pairing, and the exceptional pairing has probability $g(s/(x+s)) \rightarrow 0$. The stopped strands stay outside a fixed neighborhood of the insertion point, by the separated real and nearby complex solutions of (12.1). The drawing and caps were chosen identical off that neighborhood.

Thus bounded stopped-history functions of M transfer with vanishing error, first as the mesh tends to zero and then as $s \downarrow 0$. We never divide a coupling error by $f(s/(x+s))$. The same stopping rules are applied in the coupled experiments, so the conclusion includes mesh-dependent rules and yields (12.3). $\square$

Proposition 12.2 (Driving convergence). *On every compact capacity interval, the Dobrushin driving functions converge in law, uniformly, to $\sqrt{\kappa}B_t$, where B is standard Brownian motion and*

$$\kappa = \frac{8}{1+h} = \frac{4\pi}{\arccos(-\sqrt{q}/2)}. \quad (12.5)$$

Proof. We first work inside the preceding localization. Set

$$X_t(x) = g_t(x), \quad A_t(x) = x^h g'_t(x)^h, \quad M_t(x) = A_t(x)(X_t(x) - W_t)^{-h}.$$

The processes X, A are uniformly bounded and uniformly Lipschitz in time there, with A bounded away from zero. Take $2R+1 < x_1 < x_2$. Their real solutions stay separated: their gap solves

$$\partial_t(X_t(x_2) - X_t(x_1)) = - \frac{2(X_t(x_2) - X_t(x_1))}{(X_t(x_1) - W_t)(X_t(x_2) - W_t)}.$$

It consequently has a positive lower bound on the stopped range.

Here is the two-test convexity argument in detail. At a stopping time σ , freeze X, A and write $F_j(w) = A_\sigma(x_j)(X_\sigma(x_j) - w)^{-h}$. Both derivatives are positive, and

$$\frac{F'_1(w)}{F'_2(w)} = \frac{A_\sigma(x_1)}{A_\sigma(x_2)} \left(\frac{X_\sigma(x_2) - w}{X_\sigma(x_1) - w} \right)^{h+1}$$

has derivative uniformly bounded below by a positive constant on $[-R, R]$. Thus for $a_\sigma = F'_1(W_\sigma)/F'_2(W_\sigma)$, a bounded predictable quantity, integration of the derivative difference on either side of W_σ gives

$$F_1(w) - F_1(W_\sigma) - a_\sigma(F_2(w) - F_2(W_\sigma)) \geq c(w - W_\sigma)^2. \quad (12.6)$$

The change of the frozen X, A between σ and τ costs at most $C(\tau - \sigma)$. Applying (12.3) to these two increments, with the predictable coefficient incorporated into the test, yields

$$\mathbb{E}[(W_\tau - W_\sigma)^2] \leq C\mathbb{E}[\tau - \sigma] + o(1) \quad (12.7)$$

for every sequence of allowed stopping rules. Boundedness and (12.7) give the stopping-time tightness criterion in the Skorokhod J_1 topology. Since every path is continuous, all subsequential limits are continuous, and the convergence may be realized uniformly on compact stopped time intervals.

At inner localization levels chosen to be continuity levels of the hitting operation, ODE continuity and (12.3) make each $M(x)$ a martingale in the limit. Such levels can increase to any outer localization. The identity

$$W_t = X_t(x) - (A_t(x)/M_t(x))^{1/h}$$

and the positive bounds on A, M show that W is a continuous semimartingale. Write its decomposition as a local martingale plus a continuous finite-variation process B^{fv} , and let $[W]$ be its quadratic variation. With $Z = X - W$,

$$dX = 2Z^{-1} dt, \quad d \log A = -2hZ^{-2} dt.$$

Itô's formula gives the finite-variation part of dM/M as

$$hZ^{-1} dB^{\text{fv}} - 4hZ^{-2} dt + \frac{h(h+1)}{2} Z^{-2} d[W].$$

Since M is a martingale, multiplication by Z^2/h yields the identity of signed measures

$$(X - W) dB^{\text{fv}} - 4 dt + \frac{1+h}{2} d[W] = 0. \quad (12.8)$$

Subtract the identities for x_1, x_2 . Their positive separation forces $dB^{\text{fv}} = 0$, and then $d[W] = 8(1+h)^{-1}dt$. Lévy's characterization identifies the localized law as Brownian with the rate in (12.5). The trigonometric identity follows from $\arccos(-d/2) = \pi - \lambda$.

Finally, the probability of exiting a large outer driver bound before T is bounded in the limit by the Brownian probability of hitting inner bounds approaching it. These probabilities tend to zero as the bounds increase. Remove localization to obtain full compact-time driving convergence. Every subsequential law has been identified, so the convergence holds along the full mesh sequence. $\square$

12.2 Physical compactness and avoidable crossings

The use of annular crossings to control random curves was developed by Aizenman and Burchard [2]; conditional avoidable-crossing criteria for compatible curve and Loewner convergence were developed by Kemppainen and Smirnov [24]. We give the physical compactness and traversal arguments needed for the present boundary conventions explicitly below.

Lemma 12.3 (Territorial arm counting). *In a buffered bulk annulus, m disjoint radial medial sublegs force at least m alternating actual primal/dual arms in a further cropped annulus. If only one color is retained, one obtains at least one crossing in a distinct intervening strip per two assignments of that color to successive medial separators.*

Proof. The rounded medial arcs divide tiles into primal and dual territories. A path within a territory shadows a path of its color on the physical lattice: within a tile it passes to the corresponding corner, and passage between opposite corners uses precisely the edge supplied by that switch. The shadowing has an error bounded by one tile. Actual paths stay inside their territories.

Take disjoint simple radial sublegs, perturbing both annular radii within buffers to regular values. Thin parallel crosscuts on their two sides give ordered territorial crossings of opposite colors at each separator. Compress successive runs of equal color in the cyclic list. Each medial separator contributes a color transition, so at least m alternating crossings remain. Shadow them by lattice walks in the central crop. Near the wider rims keep territorial tails until they can be spliced to the walks; thus no positive mesh-width clearance of the thin territorial paths at the rims is needed. Opposite-color paths are disjoint. Same-color members of the alternating list lie in different strips separated by an opposite-color territorial path and cannot meet. Disjoint tails from the inner wider rim preserve order at first hits of the outer cropped rim; taking last passages to those hits gives the claimed actual arms. The same strip ordering, after ignoring the other color's assignments, proves the last assertion. $\square$

Lemma 12.4 (Curve compactness and no triple visits). *The physical Dobrushin strands are tight in the uniform Euclidean curve topology modulo increasing reparametrization. Every subsequential limit has no nonconstant interval on the original boundary, and no interior point is visited three times with nonconstant excursions between consecutive visits.*

Proof. Bulk passage counts in a fixed finite family of buffered annuli are tight by Section 2. We first show that a macroscopic displacement confined to a shrinking neighborhood of the original boundary has vanishing probability. In fixed plane Schoenflies coordinates for D , such a displacement crosses longitudinally one of finitely many boundary subarcs of positive length, after shortening the subpassage to avoid the two change points. On each subarc the boundary type is homogeneous. Its nonwired-color bank is an actual path using that color's edges.

Surround the subarc by a thin Jordan ribbon and place the permitted path in a still thinner band with strict margins. Fill to the full lattice of the bank color and wire the ribbon's outer boundary. This dominates the actual crossing: there is no nearby designated wire of this color, and all exterior attachments to the shared graph occur on its ordinary sides. Longitudinal ribbon crossings become unlikely as the ribbon narrows. One concrete proof chooses any number N of ordered intermediate positions on the subarc, each with its own disjoint fixed neighborhood. At each position two across-ribbon sections can be confined to arbitrarily small physical balls. A longitudinal crossing must then traverse a Euclidean annulus of arbitrarily large logarithmic aspect ratio inside that subribbon; its extremal length tends to infinity. Choose transverse blocker quads with bounded transverse extremal distance and with outer margins beyond the permitted band. The strong quad RSW theorem [17,

Theorem 1.2] gives an actual opposite-color transverse blocker with conditional probability at least $c > 0$, under arbitrary exterior partitions. Approximate the fixed Jordan corridors by polygons with strict margins before taking the mesh limit. Successive conditioning over the disjoint quads bounds the crossing probability by $(1 - c)^N$. Let N grow and then narrow the ribbon. Every geometric scale here was fixed before the mesh limit.

At a fixed displacement accuracy, outside these unlikely boundary travels, every excess collection of successive displacements forces many passages through a finite annulus cover of the compact part away from the ribbon. The passage estimates therefore bound the number of successive displacements at each accuracy. Subdivide curves at these displacement times and apply the usual Arzelà–Ascoli argument modulo reparametrization. This proves tightness and also rules out a nonconstant interval of a limit confined to the original boundary.

Three genuine visits to an interior point, with positive intervening excursions, produce six disjoint medial legs to some fixed positive outer radius. Lemma 12.3 turns them into six alternating actual arms across a cropped annulus. Cover the possible contact points by a deterministic grid of mesh ℓ . The bulk six-arm estimate and the $O(\ell^{-2})$ grid count give a bound $O(\ell^{c_0})$. First take the lattice mesh limit, then $\ell \downarrow 0$, and finally a countable union over interior compact sets and rational excursion radii. This excludes precisely three visits separated by nonconstant excursions; constant waiting intervals are irrelevant to the curve topology. $\square$

Lemma 12.5 (Macroscopic avoidability). *At a switch-revealing stop of the Dobrushin exploration, call a component of $U_n \cap A(z; r, R)$ avoidable if it does not separate the tip access from the target access inside U_n . The conditional probability of a future radial traversal in any avoidable component is bounded by a quantity tending to zero as $r/R \downarrow 0$, uniformly over histories and centers, with r, R fixed before the mesh limit and $\delta_n \ll r$.*

Proof. An avoidable component cannot access both designated primal and dual prime boundary arcs strictly between the rims. A path joining such accesses contains a crosscut separating the tip from the target. The future simple strand traverses one component between rims; choose a color whose designated boundary is absent there. Its bank supplies an actual arm of unqueried edges in a cropped annulus. A known-open edge or wire jump of that color would give a short physical access to its designated boundary there, contrary to the choice of component.

To bound all such components at once, keep their random edges of this color within a cropped grid annulus. Unqueried tiles are uncut; an adjacency of incident edges, or attachment to a known edge or wire vertex, has a corresponding $O(\delta_n)$ path in U_n until it reaches a designated wire. Thus away from the cropped rims the induced partition contains only ordinary physical-vertex attachments. Delete known edges of the opposite state, fill the rest to the full lattice, and wire both rims. FK monotonicity dominates all the relevant actual arms simultaneously by one full-lattice arm event. Buffered dual circuits give its vanishing bound. Sum over the two possible missing colors. The argument uses the conditional FK law at actual revealing stops; an optional last piece of length $O(\delta_n)$ is absorbed by the buffers. $\square$

Corollary 12.6 (No return from a small target cap). *After the strand first enters a shrinking initial conformal cap around its target, the probability of a subsequent displacement of fixed positive physical size tends to zero.*

Proof. At the first entrance the previously untouched cap gives a route from the tip to the target inside a small physical ball. Stop just before entrance, or at the entering piece and use its bank with a one-step buffer. Outside this inner ball every relevant annular component is avoidable. Apply Lemma 12.5. The diameters of the initial conformal caps shrink uniformly because the initial inverse disk maps converge uniformly up to the boundary. $\square$

12.3 The ordered trace, including zero-capacity intervals

Driving convergence and physical compactness leave one traversal issue: a limiting curve could still move during an interval on which its half-plane capacity is constant. The next two lemmas provide the ingredients for excluding this extra motion.

Lemma 12.7 (Interior double visits in every time interval). *For ordinary chordal SLE_κ , $4 < \kappa < 8$, almost surely every nonempty open capacity interval contains both visits to some interior self-double point.*

Proof. The double-point and boundary-hit dimension theorems of Miller–Wu [31, Theorem 1.1 and Corollary 1.7] give, respectively,

$$1 + \frac{\kappa}{8} - \frac{6}{\kappa} \quad \text{and} \quad 2 - \frac{8}{\kappa}.$$

Their difference is $(\kappa-4)^2/(8\kappa) > 0$, so the trace has an interior double point almost surely. Let E_T mean that both visits occur strictly before T . Scaling makes $\mathbb{P}(E_T)$ independent of $T > 0$, while $\bigcup_{m \geq 1} E_m$ has probability one. Thus $\mathbb{P}(E_T) = 1$ for every $T > 0$. At a deterministic time a , the domain Markov property restarts ordinary SLE with additive capacity clock. An interior double point of the restarted trace maps under g_a^{-1} to an interior point of the original remaining domain. Apply this observation to all rational open intervals. $\square$

Lemma 12.8 (Small gates exclude swallowed excursions). *In a joint subsequential limit of the physical strand and its Loewner driving function, the physical curve cannot enter an open component of a Loewner hull that is disjoint from the Loewner trace and the original boundary.*

Proof. Use the continuous Brownian SLE trace [42]; here $4 < \kappa \leq 6$. Suppose that the claimed excursion occurs. By a countable choice, there is a small closed ball J with a larger buffered neighborhood disjoint from the SLE trace and original boundary, approached deeply inside a swallowed component at finite limiting capacity. Fix bounds for this capacity and the driver, and a high basepoint corresponding to iM in the initial half-plane coordinate. Take the first discrete approach to an intermediate buffer around J , before the alleged hit. Along a subsequence of these approach times the prefix keeps a fixed spare distance from J , whereas its kernel viewed from the basepoint excludes J . This follows from driving convergence and equality of the accessible Loewner domains; the whole buffered ball, being disjoint from the trace, lies in one excluded component.

Nevertheless J is still internal in every discrete simple slit disk. Normalize its inverse disk map Φ_n at the basepoint and write $z_n = \Phi_n^{-1}(\text{center } J)$. Kernel convergence forces $|z_n| \rightarrow 1$. Put $\epsilon_n = 1 - |z_n|$, and take circular disk crosscuts centered at $z_n/|z_n|$, with radii between $2\epsilon_n$ and $\sqrt{\epsilon_n}$. If $L_n(t)$ is their image length, Cauchy–Schwarz and the area formula give

$$\int_{2\epsilon_n}^{\sqrt{\epsilon_n}} L_n(t)^2 \frac{dt}{t} \leq 2\pi \text{area}(U_n). \quad (12.9)$$

The right side is uniformly bounded and the logarithmic integration range diverges, so one image crosscut has diameter tending to zero. Its disk cap contains z_n , and its endpoints access the slit boundary. Since that boundary stays a fixed distance from J , the crosscut separates all of J from the basepoint for sufficiently large n .

The target is on the basepoint side. Indeed the driver and capacity bounds give a fixed route high above the half-plane hulls from the basepoint to the target, with positive physical separation from the prefix. A small gate touching the prefix misses this route. A gate with both endpoints on the original outer wall cannot separate two fixed interior balls when its

diameter tends to zero, by uniform Jordan convergence. Adding a boundary-to-interior simple slit disjoint from that gate does not change its separation. Gates with an endpoint at the initial slit attachment are included among gates touching the prefix.

We must use a stopping rule, not a gate selected from the future excursion. Fix a small $\ell > 0$ and stop at the first discrete step at which there exists a crosscut of diameter at most ℓ separating J from both basepoint and target, while the prefix still keeps the chosen spare distance from J and the basepoint is safe. The preceding construction shows that this stop occurs before the offending buffered hit for fine meshes. Choices can be fixed measurably from each finite history. At first occurrence a witnessing gate has an endpoint on the latest microscopic step, including its attachment point. Otherwise it was a crosscut of the previous slit domain with the same separation: adding a disjoint simple slit cannot change that separation. This contradicts first occurrence. The tip therefore reaches the gate and its target-side bank within a ball of radius $\ell + O(\delta_n)$.

Choose a fixed outer radius smaller than the spare distance to J . In the annulus from 4ℓ to this radius about the new gate endpoint, components on the trapped side are avoidable: a tip-to-target route crosses the gate inside the inner ball and stays on its other side. Any later hit of J requires a radial traversal on the trapped side, since every return through the gate is inside the inner rim. Lemma 12.5, applied at this first-gate stop, makes its conditional probability vanish as $\ell \downarrow 0$ after the mesh limit. There is no sum over possible gate locations. This contradicts the alleged excursion. $\square$

Theorem 12.9 (Dobrushin convergence for $1 \leq q < 4$). *For every bounded marked Jordan domain and every marked uniform polygon approximation satisfying Equation (1.3) with the boundary conventions of Subsection 1.1, the Dobrushin strand converges in law, in the uniform Euclidean oriented-curve topology modulo increasing reparametrization, to chordal $\text{SLE}_{\kappa(q)}$.*

Proof. By Proposition 12.2 and Lemma 12.4, take a joint subsequence realized with compact driving convergence and uniform physical curve convergence up to reparametrization. Write $Y(t)$ for the physical Brownian SLE trace, and $\gamma(s)$ for the full limiting physical curve in an arbitrary forward parametrization. Before its first target visit, let $c(s)$ be the half-plane capacity of the hull of its initial segment. Kernel convergence makes c continuous and nondecreasing and identifies it as the limit of the corresponding discrete capacities. These mapped segments are bounded locally away from the target. As the target is approached, $c(s) \rightarrow \infty$: a driver bounded on every finite capacity interval has its finite-time hulls in bounded portions of the initial half-plane.

The domain accessible toward the target after time s is the Loewner remaining domain of Y at time $c(s)$, by kernel convergence and the Loewner ODE. On every interval where c changes, the newly added trace segments of γ and Y have a point in common: a boundary point of the smaller shared remaining domain inside the older domain belongs to both increments. Continuity, with shrinking such intervals, implies

$$\gamma(s) = Y(c(s))$$

at every time outside the interiors of plateaus of c , including their active endpoints. A nonconstant plateau is therefore an extra excursion based at $Y(t)$, with t its constant capacity, contained in the old closed hull or original boundary.

Lemma 12.8 excludes motion inside an open swallowed component. Hence an extra plateau can move only on $Y([0, t])$ or the original boundary. By Lemma 12.4, it has no varying subinterval entirely on that boundary. If the plateau were nonconstant, choose a compact varying subinterval in its interior whose image avoids both the boundary and its basepoint. Every image point has a unique old SLE time: two old visits together with this genuine extra visit would give three interior visits separated by nonconstant excursions, forbidden

by Lemma 12.4. This old-time inverse is continuous. Indeed any sequence of old times has a convergent subsequence in $[0, t]$, and trace continuity and uniqueness identify its limit. Composing with the extra subcurve yields a continuous nonconstant real function, whose image contains a nonempty capacity interval. Lemma 12.7 provides a point in that interval with two old visits, a contradiction. Thus every plateau is constant. The same reasoning at capacity zero excludes an initial extra motion.

Corollary 12.6 excludes further motion after the first limit target visit. The capacity change of time therefore identifies the entire ordered curve with Y , up to the permitted increasing reparametrizations and constant waiting intervals. Finally Lemma 11.1 transfers this conclusion from the rounded simple drawing to the specified medial strand at distance $O(\delta_n)$, including repeated midpoint visits with fixed pairings. All subsequential limits agree, proving the claimed full-sequence convergence. $\square$

13 Peeling along monochromatic walls

Throughout Sections 13–15, fix $1 \leq q < 4$ and write $4 < \kappa = \kappa(q) \leq 6$. We now construct a joint exploration toward interior targets. The boundary conditions in this construction are part of the proof: its initial law is the wired FK law in a square, and its final application will be to the free infinite-volume law of Subsection 1.2.

13.1 Actual walls, caps, and the exact exploration law

Begin with a nearest-neighbor square whose boundary primal edges are deterministically open. Attach to the inward side of each boundary edge the corresponding half-tile. Its two ports are paired around its dual corner; this triangle contains no random switch. More generally, a *homogeneous wall* of color $C \in \{P, D\}$ consists of actual open C -edges and these deterministic triangles on its inward side. Actual paths along the wall are allowed. This distinction from an abstract boundary identification will be used in every crossing argument below.

All strands are oriented with primal on the left. To start an exploration in a primal-walled component, remove one wall triangle. Enter at its earlier port in counterclockwise boundary order; the other port is the force port. On a dual wall reverse the boundary order. The boundary of an active component, in counterclockwise order, is

$$P_W D_W, \quad [D_W, D_O]_D, \quad D_O P_O, \quad [P_O, P_W]_P. \quad (13.1)$$

The first and third pieces are tile sides, while the second and fourth are actual open dual and primal paths with their deterministic inward triangles. Either path may have length zero. Boundary occurrences, rather than just geometric vertices, specify the accesses in this description.

Here is the update at the tip side $P_W D_W$.

- (i) If the next tile is fresh, reveal its switch. In the primal-open case, cut along its open diagonal $P_W P'$, remove the triangle traversed by the strand, and continue from $P' D_W$ on the two-change side. The other half-tile becomes a deterministic wall triangle.
- (ii) If P' is an interior vertex, there is only one active component. If it is a boundary occurrence, the new diagonal is a proper crosscut joining the primal arc to itself. It cuts off a homogeneous primal pocket. Retain the component containing the target. In the pocket, remove the triangle on the *next counterclockwise old-wall edge* starting from the hitting occurrence P' and restart there. For targets on the two-change side, continue the old strand.

- (iii) For a dual-open switch use the same rule with the colors exchanged and clockwise order. If the tip meets a known triangle, remove it and follow its cap to the other leg. This shortens the wall instead of adding an edge.

The rule is understood in the sector seen from the tile. The opposite corner of a newly retained wall triangle lies on its indicated side of the crosscut. A known triangle has two consecutive boundary sides in that sector, so clipping it leaves just the other leg. In particular one never glues the two banks of a zero-width slit. Starting from the square, nonempty completed components are physically Jordan domains: an update either indents the boundary through a new interior corner, cuts it by a proper crosscut, or clips consecutive boundary edges. The same observations apply when a homogeneous wall is opened. The only degenerate terminal configurations lie within a bounded number of tiles of the target. They cannot occur on a bounded radial horizon from a macroscopic starting domain.

Lemma 13.1 (Exact pocket law). *Conditionally on any finite exploration history, the unqueried switches in its remaining components have the positive cap laws described by their boundaries. Distinct components are conditionally independent. Before two targets split into distinct components their explorations are identical. A strand between successive pocket switches is a portion of one original closed loop.*

Proof. At a cut, every matching connection is either the consumed switch pair or a local cap on one of its sides. Consequently the factors $q^{\ell/2}$ for unstarted loops, and the factor for eventually closing the current strand, have no remaining dependence involving both sides of the cut. Their product therefore factors over the components. A two-change component has one strand between its ports; closing these ports outside gives its marginal cap law. The deterministic triangles implement exactly those identifications. No random switch is hidden in such a triangle.

Until a switch into a homogeneous pocket, every traversed pair belongs to the loop through the originally opened cap. At a pocket switch that loop remains on the other side of the actual open diagonal. The new cap is on a loop not yet traced in the retained pocket. This also proves the last assertion. For finitely many targets keep the cut-off components, including their boundary states, and process active branches in a fixed priority order. The factorization gives the claimed conditional independence before taking any limit. $\square$

13.2 Radial clocks and forcepoint resets

For an interior target z let

$$t = \log \frac{\text{crad}(U_0, z)}{\text{crad}(U_t, z)}, \quad \Phi_t : U_t \longrightarrow \mathbb{D}, \quad \Phi_t(z) = 0,$$

where the argument of $\Phi'_t(z)$ is fixed. On $t \leq T$, Koebe's theorem supplies a ball about z of radius bounded below in terms of T and the initial conformal radius. The chart modulus of Section 11 therefore applies uniformly to connected short paths and to their boundary accesses.

Let e^{iW_t} be the tip and e^{iO_t} the force access. Choose lifts so that

$$\theta_t = O_t - W_t \in [0, 2\pi] \tag{13.2}$$

is the angle of the counterclockwise dual boundary arc. The force access is at the wall-color endpoint of the force side chosen at the last opening: after a primal opening this is P_O , approached beside the interior of the side.

We use the following interpolation, which fixes the behavior at collisions. Move along the small tip side to the near endpoint of the new diagonal, grow the diagonal as a slit, and, on

the retained two-change side, grow the other leg of the traversed triangle to clip it. Move along the resulting boundary leg to the new port. For a known triangle grow its other leg across to clip it. Boundary motion itself has zero growing time. When the slit makes a pocket crosscut, the force access collapses to the tip: $\theta = 0$ in a primal pocket and $\theta = 2\pi$ in a dual pocket. At a primal restart, move O counterclockwise *ahead*, to the far endpoint of the next edge, and remove that edge's triangle by growing its two legs in wall order. Then continue with W at the new start. Reverse both directions at a dual restart. Triangle removals on separated sides of a common crosscut may subsequently be processed separately.

These updates have disjoint growing-time interiors and produce exactly the designated target components. They differ from the medial strand, including its reset connectors, only inside a bounded number of adjacent tiles in the accessed sectors. The chart modulus implies that all boundary-angle jumps vanish uniformly on bounded horizons. The capacities of individual updates also vanish: their cuts are chart-small, and kernel convergence preserves the basepoint component. The total angular oscillation within one update vanishes as well. Indeed its cut is in a vanishing sector of its starting chart; the normalized new maps converge to the old map on compact interior sets and, by reflection, on closed boundary arcs away from that sector. The new tip is in that sector, while a force access is either regular or belongs to the same collapsing sector.

Between the designated resets, the radial boundary equation is

$$dO_t = \cot(\theta_t/2) dt. \quad (13.3)$$

If its endpoint is hit but the force side is retained, continue using the access beside that side. Convergence in the retained sector, which can be checked by truncating the access with short circular gates, gives the same one-sided equation on the next slit. Its improper integral at a collision is finite: near the collision the motion has one sign and the force access has a finite angular limit. No additional jump of O is introduced. These conventions also apply when the end of the newly opened triangle is reached. Stops at completed updates suffice in what follows; allowing the switch of a just-started update changes nothing in the limit.

13.3 Crossings in the moving charts

We need actual paths reaching the explored wall, including where that wall is irregular. The following form specifies both the geometry and the information retained after a crossing is found.

Lemma 13.2 (Wall crossings and screening). *Fix a buffered corridor of bounded aspect in a normalized disk chart. Its terminal wall intervals lie strictly inside designated homogeneous arcs; one terminal interval may instead be an interior bar. After the mesh limit at this fixed scale, an actual crossing of either prescribed color X has conditional probability bounded below, uniformly in the exploration history and its boundary partitions. At an X -wall it reaches an actual wall vertex. At an opposite-color wall it reaches the X -corner of an inward wall triangle. In the latter case a segment inside that triangle, ending at the midpoint of its base, completes the crossing to a screen and meets an actual X -path only at its starting corner.*

The two-sided bounded-density collar comparison of Section 2 holds for such buffered chart collars on a homogeneous wall, with a bounded number of through-wire blocks.

Proof. We explain the reduction to a simple square-lattice quad, so that the up-to-boundary crossing theorem applies with its exact hypotheses. For the color X , retain the edge-connected component around the basepoint of square faces whose centers have ℓ^∞ distance more than $k\delta$ from the planar complement; a fixed $k = 20$ suffices. The failing faces are edge-connected, as one sees by rasterizing a thickened path in the connected complement, including slit banks.

Other components of passing faces are assigned to the exterior. Both face families are then edge-connected. Alternating interior and exterior faces at a self-touch vertex would separate one of these families, so the retained polygon has a simple boundary. It lies in unqueried territory and eventually contains every compact chart core.

Every exterior edge-neighbor of the retained passing component is originally failing: a passing neighbor would belong to that component. Thus discarded passing islands do not affect the following access argument. An incident failing face on its perimeter can be followed, through $O(k)$ adjacent failing faces, to the first wall contact. Thus every perimeter point has a short exterior access to the actual wall. The chart modulus implies that the perimeter approaches the unit circle uniformly and that its separated accesses have the correct cyclic order. The winding about the basepoint fixes that order globally.

Approximate the two longitudinal sides of the desired corridor by simple X -lattice crosscuts inside this polygon. First approximate their compact interior portions with positive clearance; then append terminal paths in disjoint chart neighborhoods, stopping at the first perimeter contact. Grid shadowing and loop erasure preserve the chart margins, since each short connected error is chart-small. Together with the appropriate perimeter intervals these crosscuts bound a simple lattice quad. The transverse margins and angular order bound its extremal distance by a constant depending only on the buffered corridor.

We use continuum extremal distance here, as required by Theorem 1.2 of Duminil-Copin et al. [17]. In the normalized disk choose a narrower, positive-width smooth family of longitudinal reference curves through the buffered corridor. Its terminal subarcs stay away from the four side-endpoint neighborhoods. Construct the two lattice crosscuts, including their terminal tails, in disjoint outer side neighborhoods, leaving this entire narrower family clear. Its compact central portions lie in the retained polygon for all sufficiently fine meshes. Crop each reference curve around that central portion at its first polygon-perimeter hits in the two directions. If one terminal side is an interior bar, crop at its lattice approximation at that end, retaining fixed margins from the bar endpoints. Since it avoids both longitudinal crosscuts, the cropped curve lies in the quad and joins its two designated terminal arcs. The fixed terminal margins keep these contacts away from the four marked corners; these are nondegenerate arcs, not single-point electrodes.

For completeness, transfer an arbitrary nonnegative square-integrable density on the quad to the disk chart and extend it by zero to the fixed reference corridor. The reference curves may be parametrized by a smooth rectangle with bounded metric distortion. Fubini and Cauchy–Schwarz give one reference curve whose squared weighted length is at most a constant times the squared-density area integral. Cropping only decreases that length. Conformal invariance of the length-area quotient therefore bounds the continuum extremal distance of the lattice quad by a constant depending only on the buffered corridor. No comparison with a vertex-weight modulus is needed.

Theorem 1.2 of Duminil-Copin et al. [17] now gives a positive crossing probability for the simple quad at p_q , for $1 \leq q < 4$, with arbitrary boundary partition. Equivalently, use the free law as the lower comparison. Do not query edges running along its terminal perimeter sides. From a reached vertex, an outward failing-face chain gives $O(k)$ available X -edges toward the intended wall. The extension uses only those unqueried terminal-side edges and edges outside the quad. Finite energy gives a uniform positive conditional cost. Stop at the first wall contact; if an opposite-color wall blocks the next step, the current vertex is already the X -corner of its triangle. This proves the crossing assertion, including its endpoint convention. Circuits and interior connections with strict margins follow by ordinary box-crossing and gluing.

For the collar statement, find the same-color screen nearest the unknown-facing side of a deterministic buffered subcollar. Paths may be restricted to the strip between the prescribed

wall intervals, pruned at terminal returns, and augmented by ordered spokes to common endpoints. Minimizing the region on the unknown side selects a simple outermost crossing. Any competing path entering strictly nearer that side has an open excursion that can replace part of the selected path and decrease this region. Hence the selection tests only on and on that side of the screen. Its other side retains an FK law with a boundary partition.

Actual paths of the screen's color must attach to it in order to cross. An opposite-color path cannot pass it except through the bounded through-wire blocks; the added segment in a wall triangle creates no other bypass. Splitting or merging those blocks changes a weight by a bounded factor. The primal and dual screening searches used in the collar proof of Section 2 thus apply without change to arbitrary-event densities. All chart margins are fixed before the mesh limit, as required. $\square$

13.4 A bound on generations

Lemma 13.3 (Disjoint passages and reference walls). *The number of edge-disjoint actual monochromatic cycles surrounding a fixed macroscopic ball in the initial square, and using no deterministic initial wall edge, is tight as the mesh tends to zero. Along finitely many target branches with bounded radial horizons, the number of reference-wall generations defined below is jointly tight.*

Proof. Such cycles provide edge-disjoint vertical passages in a fixed-height middle horizontal band. By bounded degree and Menger's theorem, their number is bounded by a fixed multiple of the maximal number of vertex-disjoint passages. For primal paths one may add random switches on the initial wired boundary edges, which does not change the interior marginal under consideration. For dual paths use the rectangle of interior faces. Thus it suffices to work in an ordinary fixed-aspect color-lattice rectangle, with its induced positive FK law.

Prune bottom-to-top paths so they do not return to their terminal rows. Select the leftmost open path, then the leftmost path to its right that is vertex-disjoint from it, and repeat. To justify the information claim, attach noncrossing spokes just below and above the terminal rows to common endpoints, pad the rectangle sides, and minimize left area. Another eligible path entering strictly left gives an excursion with distinct endpoints on the augmented path and permits replacement in either traversal direction, decreasing left area. End excursions through a spoke use only terminal contacts. The selected path is therefore determined by its own edges and the edges on its left. If m vertex-disjoint passages exist, at least $m - 1$ remain to its right, and the argument iterates.

After a selection p , there is a uniformly positive conditional probability of blocking another disjoint passage. If p comes within ten mesh steps of the right wall on some horizontal half-row, force an opposite-color connection along that row from outside the right wall to first adjacency to p ; its finite-energy cost is bounded. If p already blocks, no forcing is necessary. Otherwise follow p on its right through opposite-lattice face centers, joining consecutive ones inside the right wedges at its vertices. Erase loops and trim terminal returns to obtain a simple path p_* between the extreme interior half-rows. Those rows are single intervals on the right side because p was pruned. With them and the rightmost interior column, p_* bounds a simple opposite-lattice quad of fresh edges.

Horizontal rows give a bounded-extremal-distance family across this quad: there are order-height disjoint rows, each of length at most order-width. Lemma 13.2, or directly Duminil-Copin et al. [17, Theorem 1.2], supplies an opposite-open crossing with uniformly positive probability. Extend it one step beyond the right boundary at finite-energy cost. At its other endpoint, append a segment in the incident square to a vertex of p . This screen blocks all actual vertical paths on the right avoiding vertices of p . Its edges are fresh under the leftmost-path conditioning. Consequently the successive selections have a geometric tail;

blocker tests need not themselves be exposed when continuing the search. In particular, if N is their number, then for constants depending only on q and the fixed rectangle shape,

$$\mathbb{P}(N \geq m) \leq (1 - c)^{m-1}, \quad \sup_{\delta} \mathbb{E}[e^{aN}] < \infty \quad \text{for } 0 < a < -\log(1 - c). \quad (13.4)$$

This proves the first assertion, and also gives the corresponding bound for a strictly bulk rectangle.

A generation begins in a homogeneous C -pocket, whose present wall is its *reference wall*. Read the C -arc from O to W in counterclockwise order for primal and clockwise order for dual. It consists of a prefix of the remaining reference interval followed by a suffix of actual open edges revealed during this generation. A triangle step shortens this path; a fresh diagonal extends it or returns to an earlier vertex. At a same-color switch retaining part of the prefix, the next gap is precisely its first retained reference edge. The description is thus preserved, even when the prefix becomes empty. These switches take place on the reference wall.

Start a new generation at the first homogeneous pocket whose retained arc has no reference edge; every opposite-color switch has this property. Its wall consists entirely of actual diagonals discovered since the previous start. Consumed edges are not reintroduced by known triangles. The successive new reference cycles are consequently edge-disjoint, separately for each color. They surround the basepoint ball present up to the fixed horizon. If a wall walk has repeated vertices, its winding extracts a surrounding cycle. The first assertion bounds the number of generations. For finitely many branches sum the bounds, allowing an ancestral generation to be counted along each of its descendants. $\square$

13.5 Local chordal tests in a peeled domain

Put $h = 1 - 2\rho$, with ρ as in Section 3. At a causal completed state with θ in a compact subinterval of $(0, 2\pi)$, map the active domain to $\mathbb{H}$ so that the tip is 0, the force access is ∞ , and the target has imaginary part one. The positive real arc is dual-designated. Stop a short subsequent segment at fixed driver and chordal-capacity bounds, chosen so the target remains safely inside and connected to the force access. There are no resets on this segment. Let g_t and $W_t^{\mathbb{H}}$ be its chordal map and driver.

Proposition 13.4 (Tests away from coincidence). *For $x > 0$ beyond these localization bounds, the quantities*

$$M_t(x) = \left(\frac{xg'_t(x)}{g_t(x) - W_t^{\mathbb{H}}} \right)^h \quad (13.5)$$

satisfy the approximate martingale identities between arbitrary causal stops on the segment. Tests may include the entire earlier exploration history of finitely many targets. The errors tend to zero after the mesh limit. The assertion concerns the averaged identities for this history law, not a uniform assertion over all deterministic mazes.

Proof. We supply the modifications needed to apply the moving-domain comparison of Section 11.

The test notch. Fix first a small chart scale $s > 0$. Cut a notch around x , of size between s and $2s$, with two transverse ends on the old dual wall. Put a new separate primal wire only on the middle portion of its new wall, a positive chart distance from those ends, and keep dual designations on both flanks. Approximate a regular semicircular or polygonal notch by openable dual diagonals near its ends and openable primal diagonals in its strictly interior middle, with transition tile sides. The artificial diagonals are deterministic in the modified experiment. The result has four changes with the same triangular-cap convention as before.

This construction is possible in unqueried territory. Compact interior parts have ordinary positive-clearance approximations; end parts shadow dual-grid paths in the prescribed accessed corridors up to first actual dual-wall contact. Loop erasure preserves their order by the chart margins. We also choose the notch to have controlled physical terminal tails. Length-area, applied to a positive-width family of chart curves, bounds the integral of the inverse-curve tail lengths below height fraction ϵ by $O(\sqrt{\epsilon})$, since the physical area is bounded. Thus a curve can be chosen with tails vanishing at shrinking height cutoffs. One may make the height cutoff decrease slowly with the mesh. Its retained interior is approximated with positive one-sided clearance, and the new primal interval stays at positive physical distance from all previous and locally stopped trajectory pieces for fixed s .

Exact identities and comparison of the histories. Sample the notched law conditionally at the start of the segment. Follow its strand until the localized stop, measuring the stop in the unnotched geometry. The strand cannot reach or swallow the notch data: real and complex Loewner ODE bounds keep its growth away from that chart neighborhood. Every locally discarded pocket is homogeneous and contains neither notch nor force data.

The pairing probability with the two primal wires separate is an exact martingale. Identifying the two primal intervals gives the complementary dual comparison; its law is a bounded tilt depending only on the pairing. These identities are taken at completed states. For additional targets keep reporting geometry and scheduling from the unnotched cuts until the local stop or an active horizon, without changing a target assignment near the notch. Still-shared targets see this same growth, and separated components have their ordinary independent laws by Lemma 13.1. If a target clock freezes, truncate the identity there.

Lemma 13.2 compares the exterior history laws, for fixed s , by bounded density constants. To obtain a coupling error tending to zero as $s \downarrow 0$, bracket the exterior primal partitions at the notch by the extremal partitions, expose actual attachment clusters in the stronger law, and block them by dual wall-to-wall paths in successive dyadic chart half-collars. These lie away from the old primal wire, so there is no primal wire bypass. Conditional success probabilities at each fixed-shape scale are uniform after the mesh limit. Only the separate-primal-wire law is needed in this last coupling, not its pairing tilt.

Rough-domain obstacles. For a later stopped notched domain, its four-change rectangle is nondegenerate for fixed s . The central possible-attachment vertices and their optimistic color paths are defined exactly as in the moving-domain argument. Accessible identifications and deterministic open adjacencies have short accesses to their assigned wires. Away from those wires a central attachment begins with an actual uncertain-edge path to macroscopic distance. Protected-ball deletion still has only rim attachments in the relevant components; splitting a physical vertex weakens these connections. No notch cap supplies a shortcut of a color away from that color's electrodes.

Use the initial square, the relevant ancestral and current exploration pieces, and the notch as obstacles. First truncate the numbers of branches and generations, using Lemma 13.3. Within generation j , retain excursions reaching depth r_j from the initial boundary and the earlier-generation obstacles, starting and ending at much smaller depth h_j , or at an active endpoint. The earlier obstacles contain its reference wall up to a step-sized error. Away from them, each retained leg is jump-free and has actual open paths of both colors beside one loop portion. Bulk passage bounds, on a finite grid of annuli, make the number of these legs and of their short internal blocks tight. A trajectory does not retrace a loop portion: at a switch the old loop remains in the other component.

Choose accuracies backwards through the generations, approximating each earlier obstacle much better than the descendant depth h_j . Excursion endpoints then attach to earlier routes

in protected small balls, and omitted shallow pieces contribute only r_j to the Hausdorff error. Each retained block has a known actual path π of the requested color in a short bulk patch. The routing and flag construction of Section 11 therefore applies on a finer deterministic grid. Route the notch on its removed side, with clearance on compact interior portions, and attach its short terminal tails in protected balls. Specifically, fix the two protection radii, choose still smaller physical tails as above, then approximate the retained interior by a polygon with one-sided clearance. At fixed height cutoffs, normal-family bounds allow all perturbations and snappings to be chosen from finite lists. Make earlier obstacle accuracies smaller afterward, and take mesh errors last. The resulting rounded Jordan comparison domain contains every required internal compact path and has all the limiting complement contacts needed for kernel convergence; its marked prime data agree in interior charts.

Why the flag cost is the original six-arm cost. Use original past switches and actual wire edges for the known legs. In testing the absence of a short optimistic bridge to π , allow even the original unnotched ambient completion. An actual path leaving the old domain first hits its designated wall; opposite-color wall diagonals block it. The same is true on first contact with a previously queried edge of its color. Near the current old wire portions, the chart modulus excludes a shortcut from a central vertex in a sufficiently small physical disk.

There are two different locality checks near the notch. A primal central path that locally crosses it stays near the old positive dual-wall patch in the starting chart, far from any old primal contact or locally grown trajectory; it cannot reach a known primal history edge there. For a dual central vertex near a flagged historical detail, the entire notch is far away in the old chart. Indeed a central dual abscissa near the notch must avoid both dual flanks and their ends, and hence lie in a compact interior chart patch beside the new primal interval. That patch has positive physical distance from the old history and the locally stopped pieces, first by the regular notch margins and then by the ODE bounds. Kernel convergence and analytic continuation away from the local hull give the same assertion on every shape subsequence.

All these arguments follow connected paths in the old domain until their first old-wall contact. Boundary-sector identifications cannot undo the actual diagonal barriers. The sector-duality argument thus produces the original known flags from the two witness arms, and an actual extra arm produces six alternating arms in the original ambient switch lattice.

For dual extra arms near flags, the required narrow arm pieces lie outside the notch and its narrower protecting collar. The arbitrary-event collar density comparison transfers their probability, including the local history, to the original law; the pairing tilt costs another bounded factor. For primal extra arms, first compare the stopped history through the outer collar. Conditionally on that history the extra-arm event is increasing in primal edges. Restore the deleted notch and wire only the new middle primal vertices locally. This dominates the modified law: the dual flanks had imposed no primal wiring, and their known dual edges had only removed primal choices. The flagged local arm pieces are physically separated from that new interior wiring by a vacant bulk collar, containing no earlier history. The bulk density comparison then transfers their probability to the original conditional law.

For either color, summing over fine-grid flagged cells consequently costs a bounded multiple of the original annealed six-arm sum. Its decay follows from Duminil-Copin et al. [17, Corollary 6.7], after the collar transfer of Section 2. Take the grid scale small after all collar margins, trims, and block cutoffs. Protected-ball errors, in contrast, are estimated directly conditionally in the modified law. The chart and notch may depend on the segment-start history: at fixed scales their margins and comparison constants are uniform under the stated normalizations.

The alternating formula and the shrinking notch. These checks are the additional inputs required by the upper barrier comparison of Section 11. Its directed routes have clearance from central candidates except for the controlled deletions. The level fibers of the remaining kernel and comparison template agree outside protected end balls, giving the same shared-graph partition domination for the abscissa-trimmed quad. Fix generation cutoffs, block cutoffs, grids, and all intermediate scales before the mesh limit. The preceding averaged bounds make adverse conditional costs negligible in history probability. The two color upper bounds and the pairing tilt therefore identify the stopped conditional pairing probability with the separate-wire value $g(\chi)$.

Let $Z_t^{(s)}$ be this exact conditional pairing probability and $\chi_t^{(s)}$ the cross-ratio of its stopped notched domain. At fixed s , the preceding identification gives $Z_t^{(s)} - g(\chi_t^{(s)}) \rightarrow 0$ in mean at every allowed stop. The initial value $g(\chi_0^{(s)})$ is measurable at the segment start and bounded below for fixed s . We may therefore divide the exact martingale identity by it and remove the mesh error before shrinking the notch. The remaining ratio satisfies

$$\frac{g(\chi_t^{(s)})}{g(\chi_0^{(s)})} = M_t(x) + o_s(1)$$

locally uniformly on the stopped range. To see this conformal normalization, remove the same small boundary hull near x in the starting and later chordal charts. The map between them is analytic near that hull, with derivative $g'_t(x)$ bounded above and below and bounded second derivative. Rescaling the notch gives uniform Jordan shape convergence, so the flattened prepoint interval widths have ratio $g'_t(x)(1 + o(1))$, while their base positions have ratio $(g_t(x) - W_t^{\mathbb{H}})/x$. The asymptotic $g(\chi) \sim c\chi^h$ gives (13.5), whose value is bounded above and away from zero on the stopped range. Only after this replacement do we use the vanishing separate-wire coupling error to return the bounded martingale tests to the original law as $s \downarrow 0$. That coupling error is never divided by the vanishing initial pairing value. This order of limits proves the proposition. $\square$

14 Identification of the joint angular tree

The local chordal tests determine the angular dynamics while the tip and force accesses are distinct. We must also determine the continuation when they coincide and show that branches have independent continuation noises after their target domains separate.

14.1 Interior evolution and an oriented boundary estimate

All assertions in this section are first made for finitely many fixed deterministic interior targets and bounded radial horizons. The horizons may subsequently increase. The small-update estimates of Subsection 13.2 absorb port ambiguities and the difference between an interpolated stop and a completed-state stop.

Lemma 14.1 (Interior characteristics). *Away from $\theta \in \{0, 2\pi\}$, every limiting branch has the local characteristics*

$$\begin{aligned} dW_t &= \sqrt{\kappa} dB_t + \frac{6 - \kappa}{2} \cot(\theta_t/2) dt, \\ dO_t &= \cot(\theta_t/2) dt, \quad \theta_t = O_t - W_t. \end{aligned} \tag{14.1}$$

Here t is that branch's radial clock; the martingale identities hold relative to the full causal exploration history.

Proof. Use two ordered test points x in Proposition 13.4. The relative-convexity inequality (12.6) and the inversion argument of Proposition 12.2 identify the stopped chordal driver and its bracket. They also give stopped second-moment estimates. On a sufficiently short segment started with $\theta \in [e, 2\pi - e]$, the chordal and radial clock rates and coordinate changes are bounded above and below. Thus crossing a fixed interior angular interval in vanishing radial time has probability tending to zero, uniformly at causal starting stops. A macroscopic angular increment started near an endpoint must cross such an interior interval. Aldous' criterion and the vanishing within-update oscillations give continuous tightness of θ ; in particular its numbers of fixed-size alternating swings are tight.

After the tightness of W, O proved below, localize on rational start ranges and inner continuity levels to pass the re-centered identities to a joint subsequential limit, retaining tests of the past. The ordinary chordal-to-radial target change, before any loss of the force access, gives (14.1). This is precisely the stopped coordinate change in Sheffield [46, Proposition 3.12]; no continuation through collisions is inferred from that proposition. $\square$

The missing tightness, and later the absence of an endpoint push, come from an estimate whose direction depends on the restart rule illustrated in Figure 6.

Lemma 14.2 (Fast motion near coincidence). *There are positive constants c, C with the following property. Start at a causal stop with $\theta < r/16$, where $r > 0$ is small. For $0 < u \ll r^2$, the probability that W advances counterclockwise by r within additional radial time u , while $\theta < r/16$ until that advance, is at most*

$$C \exp(-cr/\sqrt{u}) + o(1). \quad (14.2)$$

The error vanishes in the mesh limit for fixed r, u . The constants in the limiting estimate are uniform in these parameters. Near 2π reverse the direction of motion and exchange the colors.

Proof. Work in the disk chart at the starting stop. The radial Loewner ODE confines the updates during time u to a strip of depth $C\sqrt{u}$ from its boundary. A driver advance of r forces a positive angular traversal through a buffered part of the corridor from angle $W_0 + r/5$ to $W_0 + r/2$.

Here is the boundary-coordinate justification of this implication. Stop at the first $|W - W_0| = r$, so the lift cannot wrap around. If the advancing growth never passes, for example, $W_0 + 2r/3$ in the starting chart, a neighborhood of radius comparable to r at $W_0 + 5r/6$ remains untouched. Its prepoint image stays $O(\sqrt{u})$ from its original angle. First check the ODE at depth a large constant times $\sqrt{u}$. Near the untouched arc the map reflects analytically; radial monotonicity bounds the log-radius harmonic part by the absolute original log radius. In the stated neighborhood this is $O(r)$, so the interior gradient estimate bounds the angular derivative. The boundary estimate follows by reflection. The untouched prime access is distinct from the tip, and the small boundary relocations also miss it. The driver therefore cannot pass its lifted image, contrary to the stipulated advance. This proves the required physical traversal in the starting chart. All within-step and strand-surrogate errors vanish for fixed r, u .

Place order $r/\sqrt{u}$ disjoint buffered bars across that angular corridor. Each has width comparable to $\sqrt{u}$ and extends to a larger fixed multiple of the allowed depth. By Lemma 13.2, each has a uniformly positive conditional chance of an actual dual-open path from the dual corner of a triangle on the primal wall to its inner end. Complete this path across that triangle to the midpoint of its primal base.

One successful bar prevents the required advance before a dual reset. For completeness, follow the actual traversed switch pairs and add a microscopic connector at each primal restart.

At that restart the connector passes from the encountered pair through the neighborhood of the open primal diagonal and hitting vertex to the new cap's *earlier* port, approaching along its near base endpoint. It stays in the retained sectors and avoids all actual dual paths. A positive angular passage within the shallow strip would have to cross the completed bar positively. It cannot cross its actual dual part. Traversal of the cap pair in the fixed triangle crosses the added segment only negatively, because primal is on the left. If this triangle itself is selected for the restart, the connector reaches the earlier port without crossing the segment. Every other reset connector stays beside its own primal vertices and avoids that fixed segment. An excluded triangle is never used again. Thus the positive crossing is impossible. This argument is in the prime closure of the stop domain, where all short accessed moves are uniformly chart-small.

While $\theta < r/16$ there is no dual reset. Multiplying the conditional failure bounds for the disjoint bars proves (14.2). Also, a negative motion of order r is impossible in this zone: O is nondecreasing, including its primal resets, and $W = O - \theta$. The reflected argument proves the other endpoint statement. $\square$

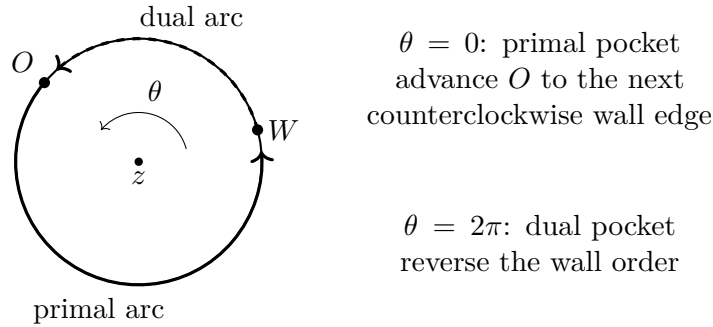


Figure 6: Normalized chart of a two-change component. The angle $\theta = O - W$ is measured counterclockwise along the dual arc. The arrows on the circle indicate boundary order, not strand orientation. The forcepoint reset uses the next old-wall edge on the retained pocket. A full turn changes the endpoint of the reflected angular interval. The drawing is schematic; microscopic tip and force sides are suppressed.

Away from coincidences use Lemma 14.1. In an endpoint zone, a fast exit toward the interior is controlled by the tightness of θ , and fast motion remaining in that zone is controlled by Lemma 14.2 and the sign of O . Thus W and O are continuously tight, also at variable stops. Moreover, on a sufficiently small fixed endpoint zone, both the continuous part of O and every reset have the same sign. Its variation on each visit is bounded by its angular oscillation there. Count visits by alternations between nested endpoint zones, whose number is tight; outside these zones the cotangent rate is bounded. We obtain tightness of

$$\text{Var}_{[0,T]}(O) \quad \text{and} \quad \int_0^T |\cot(\theta_t/2)| dt. \quad (14.3)$$

In any limit the endpoint sets have zero radial Lebesgue measure, O has locally finite variation, and (13.3) holds off those sets. A remaining singular part could only be nonnegative at 0 and nonpositive at 2π . We next rule it out.

14.2 Instantaneous reflection and removal of a singular push

Proposition 14.3 (The endpoint convention is determined). *In its own radial clock, the limiting gap is the unique instantaneous reflected diffusion on $[0, 2\pi]$ with interior generator*

$$\mathcal{A}f(\theta) = \frac{\kappa}{2}f''(\theta) + \frac{\kappa - 4}{2}\cot(\theta/2)f'(\theta). \quad (14.4)$$

The forcepoint has no singular push:

$$O_t = O_0 + \int_0^t \cot(\theta_s/2) ds. \quad (14.5)$$

Equation (14.1) consequently holds globally, including collisions, with locally integrable drift.

Proof. For a smooth function constant near both endpoints, the localized interior identities glue to the martingale problem for $\mathcal{A}$. To admit a smooth function with $f'(0) = f'(2\pi) = 0$, approximate it by such cutoffs with uniformly bounded second derivatives. Its cotangent term is bounded near the ends, since $|f'(\theta)| = O(\min(\theta, 2\pi - \theta))$. The endpoint time is zero, so dominated convergence gives the limiting martingale identities and, by applying the same argument to f^2 , their brackets.

In particular $X = \cos(\theta/2)$ solves

$$dX_t = \frac{\sqrt{\kappa}}{2}\sqrt{1 - X_t^2} dB_t - \frac{3\kappa - 8}{8}X_t dt, \quad -1 \leq X_t \leq 1. \quad (14.6)$$

The diffusion coefficient is one-half Hölder and the drift is Lipschitz; the one-dimensional square-root uniqueness criterion gives pathwise uniqueness. Hence this equation fixes the instantaneous reflected law of θ ; a sticky alternative is also excluded by zero endpoint time.

Near either endpoint the scaled distance R has Bessel dimension

$$\mathfrak{d} = 3 - \frac{8}{\kappa} \in (1, 2). \quad (14.7)$$

Its drift differs from $(\mathfrak{d} - 1)/(2R)$ by a bounded smooth term of order R . Stopped Girsanov, equivalently in the squared coordinate where the excess drift divided by the diffusion coefficient extends boundedly, transfers the usual Bessel path properties. Thus θ is locally Hölder of every exponent below $1/2$, and each endpoint set has Hausdorff dimension at most

$$d_0 = 1 - \mathfrak{d}/2 < 1/2. \quad (14.8)$$

The needed Bessel Hölder statement is, for example, Hutzenthaler et al. [22, Theorem 1.2 and Corollary 2.14]. The zero-set statement follows from the inverse-local-time description in Bertoin [6, Section 2.2, p. 21] and the stable-range dimension in Bertoin [6, Corollary 5.3, p. 42]. Only a bounded stopped perturbation at one endpoint is being used here.

Write the possible remaining term as the signed measure

$$d\nu = dO - \cot(\theta/2) dt.$$

It is supported on the two endpoint sets and has the sign specified after (14.3). We give the time-block argument explicitly, so that a Hausdorff-dimension bound, rather than an unstated Minkowski bound, suffices. Choose $\eta > 0$ so small that

$$p = \frac{1}{2} - \eta > d_0, \quad u_m = 2^{-m}, \quad r_m = u_m^p.$$

By the Hölder property, every sufficiently fine dyadic block of length u_m meeting an endpoint set has its end-gap at most $r_m/32$ throughout. Indeed choose a Hölder exponent strictly

between p and $1/2$. Apply Lemma 14.2 with harmless fixed buffer changes to the finitely many blocks at level m , taking the mesh limit first. The probability that some level- m block stays in the stated endpoint zone and has one-signed O -variation exceeding Cr_m is at most

$$C(1 + T/u_m) \exp(-cu_m^{-\eta}). \quad (14.9)$$

This is summable in m . The estimate for O follows from that for W because their difference is the end-gap, of size at most $r_m/32$ on such a block. Borel–Cantelli gives, almost surely, the variation bound Cu_m^p on every sufficiently fine endpoint-meeting block, separately for the two endpoints and on each fixed horizon.

On an endpoint neighborhood, dO is one-signed, and so is the cotangent time measure. Its restriction to the endpoint set is zero, whereas the restriction of dO there is precisely $d\nu$. Consequently $|\nu|(I) \leq \text{Var}_I(O)$ for every small interval in that neighborhood. Cover the endpoint set by intervals of variable lengths whose sum of p th powers is arbitrarily small, using (14.8). Replace each by a bounded number of comparable dyadic intervals that meet the set. The preceding variation bound yields arbitrarily small $|\nu|$ -mass. Thus $\nu = 0$, proving (14.5).

Finally, a Bessel process of dimension greater than one is a semimartingale with integrable inverse-distance drift and no local-time term in its ordinary coordinate [22, Lemma 3.2]. Undoing (14.6) in the endpoint neighborhoods, and using the interior identities elsewhere, gives $d\theta = -\sqrt{\kappa} dB + ((\kappa - 4)/2) \cot(\theta/2) dt$. Together with (14.5) this is (14.1) globally. There is therefore no unspecified boundary pushing rate. $\square$

14.3 Common clocks and independence after separation

The single-branch conclusion does not by itself identify the joint tree. In particular a narrow discrete corridor may survive after its two limiting target kernels have separated. We address this before using any continuum tree theorem.

Proposition 14.4 (Joint target tree). *For every finite family of deterministic interior targets, the peeling converges in its joint radial driving data to the target-invariant branching radial $\text{SLE}_\kappa(\kappa - 6)$ tree. Branches agree until their target kernels separate and then have independent continuation noises. At successive full turns the force side is reversed, equivalently the angular gap remains instantaneously reflected on $[0, 2\pi]$.*

Proof. Use the sum of currently accrued target clocks as a common parameter v , freezing each target at its prescribed horizon. Every clock $t_i(v)$ is nondecreasing and 1-Lipschitz. Several run together while their targets share an active component; separate components are processed in serial priority order. A flat clock has a constant state. All tightness bounds and the interior identities hold between adapted common-clock stops, including tests of the whole past. Work in other components during an interrupted segment is conditionally independent of its unqueried switches by Lemma 13.1; neither the schedule nor the stop gives advance information.

Extract limits jointly with the clocks. The preceding endpoint arguments can be applied in each marginal's own radial clock, using tightness and inverse-clock extraction across flat stretches. Thus the martingale part M_i of W_i is a continuous local martingale relative to the common past, with

$$d\langle M_i \rangle = \kappa dt_i. \quad (14.10)$$

The Loewner equations determine its target kernels and normalized maps.

While targets z_i, z_j remain in one component, put $\xi = \Phi_i(z_j)$. Radial Loewner evolution and the conformal-radius change give

$$dt_j = P^2 dt_i, \quad d \log |\xi| = P dt_i, \quad P = \frac{1 - |\xi|^2}{|e^{iW_i} - \xi|^2}. \quad (14.11)$$

Their marked boundary angles correspond by the normalized disk Möbius map. On a compact interval with $|\xi| < 1$ these formulas are regular and pass to the limit. The targets are still in the same kernel exactly while they retain an interior connecting path: kernel convergence preserves paths with positive clearance. Thus they have the same growth up to kernel separation.

Suppose now their limiting kernels have separated but their discrete components have not yet split. Along such a portion $|\xi| \rightarrow 1$ by monotonicity. On a subinterval where both limiting angle gaps stay in the interior, the two marked prepoints remain separated in both charts. To see the resulting clock separation explicitly, set $w = e^{iW_i}$, $o = e^{iO_i}$ and $F_\xi(u) = (u - \xi)/(1 - \bar{\xi}u)$. The disk Möbius identity is

$$|F_\xi(w) - F_\xi(o)| = \frac{(1 - |\xi|^2)|w - o|}{|w - \xi||o - \xi|}.$$

If both chord gaps are bounded below, the product in the denominator is at most $C(1 - |\xi|)$, while $|w - o|$ bounds the larger factor below. Thus one of w, o is within $C(1 - |\xi|)$ of ξ . The choice cannot change on a connected interior-angle subinterval when $|\xi|$ is sufficiently close to one. If it is the tip, P is of order $(1 - |\xi|)^{-1}$; if it is the forcepoint, P is of order $1 - |\xi|$. Equation (14.11) then forces one limiting clock to be flat. This remains true along subsequences of exact discrete split times. After an exact split, serial growth gives disjoint time interiors for the two clocks. The fixed-priority schedule has only finitely many target-group split and freeze stages. On each stage, exhaust the open set where both angle gaps are interior by countably many compact subintervals; the preceding clock-separation argument applies on each of them. Endpoint angle sets have zero mass for the corresponding own clock by Proposition 14.3. The covariance inequality for continuous local martingales and (14.10) therefore show that

$$d\langle M_i, M_j \rangle = 0 \quad \text{after their kernel separation.} \quad (14.12)$$

Frozen targets are omitted from this comparison.

We spell out why these characteristics determine independent continuations, without asserting that the clocks are independent. In each still-accessible active group choose its first indexed target as representative. Until a separation or freeze, the representative's motion determines all other states and clocks by (14.11). Separation and freeze times are intrinsic stopping stages: mutual accessibility is read from the radial ODE up to first approach to unit modulus. While two other members remain regular with the representative, they cannot lose their shared access before either is lost from that representative. A nonrepresentative's horizon is likewise read by its clock correspondence. At a multiple separation, take the continuous state limits in all own clocks; the subgroups are exactly those retaining internal mutual paths in their kernels. Serial speeds of unrelated groups have no effect on these stages.

For each target accumulate its martingale coordinate only over the suffix during which it is an active representative, keeping it flat otherwise. More precisely, with the predictable choice at the stage endpoints, set

$$N_i(v) = \int_0^v \mathbf{1}_{\{i \text{ is an active representative at } s\}} dM_i(s).$$

Once it becomes representative it remains representative of its descendant group until it freezes. These representative martingales N_i are pairwise orthogonal by (14.12). Their terminal brackets can be finite, so make the completion explicit. Let T_i be the prescribed horizons and take the deterministic common terminal parameter $V = \sum_i T_i$, extending all stopped states constantly if necessary. Divide each N_i by $\sqrt{\kappa}$, and enlarge the common

filtration after V with independent Brownian motions $\widehat{B}_i$, independent also of the entire original terminal sigma-field. Set

$$\widehat{N}_i(V + s) = N_i(V)/\sqrt{\kappa} + \widehat{B}_i(s), \quad s \geq 0,$$

and use $\widehat{N}_i(v) = N_i(v)/\sqrt{\kappa}$ for $v \leq V$. The extended processes are continuous local martingales in this enlarged common filtration, remain pairwise orthogonal, and have infinite terminal brackets. Knight's orthogonal-martingale time-change theorem now gives independent standard Brownian motions in their respective bracket clocks [28, Theorem 2, p. 191]. Restricting to each original terminal bracket supplies precisely its original representative martingale. The auxiliary continuation changes neither the past filtration nor any stage of the exploration.

For the conditional assertion, let $\mathcal{G}_v$ be the common causal filtration and let S be a separation or freeze stage. Restart the active representative martingales at S , stop at the next common stage, and pad constantly to a deterministic horizon. After localizing to bounded brackets, these restarted processes remain orthogonal martingales under $\mathbb{P}(\cdot \mid E)$ for every $E \in \mathcal{G}_S$ of positive probability, with the same bracket identities. Completing them as above and applying Knight's theorem under this measure gives the same product Wiener law for every E . Thus the continuation noises are independent of $\mathcal{G}_S$. A continuing representative uses its original martingale increments, and a newly selected one starts its representative suffix. No conditioning on terminal bracket values or future separation data is involved.

The pathwise uniqueness of (14.6), followed by (14.5), determines the angles and maps through the next intrinsic stage. A newly assigned representative uses its assigned independent noise; a continuing one uses the continuation of the same Brownian solution. A change caused solely by a representative's horizon gives a fresh continuation at that stopping stage. Induction over the finite target groups identifies the usual common-branch, independent-after-separation construction.

The resulting equations and angular convention are the target-invariant radial tree of Sheffield [46, Propositions 3.13–3.14 and Section 4.2]: a one-sided force weight $\kappa - 6$, with the side reversed in an enclosed component after a full turn. Our sign convention is $\theta = O - W$; exchanging the sign of its Brownian motion translates to the opposite angular convention. Increasing the finite horizons proves the assertion at all finite clocks. $\square$

15 Recovery of the nested loop collection

We now identify the full loop curves encoded by the limiting exploration tree, including their traversal order and multiplicity as collection elements. The argument first compares loops in a fixed wired square and then transfers this identification to the free plane law. Within the square, we first match loops by the dense-set targets they surround. Winding information then identifies their traces, and a separate argument rules out additional traversal portions. These are distinct steps: agreement of surrounding-loop labels alone does not identify either the complete trace or its order of traversal.

Continue first in a fixed axis-parallel square with the primal-open wall and wired law of Subsection 13.1. Draw the two pairs at each tile contact disjointly in a midpoint disk of radius less than $\delta/100$, taking each turn on its prescribed side. The resulting actual loops are simple, pairwise disjoint, and avoid actual open diagonals of both colors. Use the analogous drawing for wall triangles and orient every loop primal-left. Matching successive traversal pieces inside those disks changes each loop by at most $2\delta/100$ in Euclidean uniform curve distance, and by a constant multiple of this in spherical distance, simultaneously for all loops. Thus the drawing convention has no effect on the desired topology.

15.1 Compactness and two sides of every limiting loop

Lemma 15.1 (Positive-law loop geometry). *The closed loop collections in the fixed square are tight for matching uniform curves modulo reparametrization and orientation. Constant limit curves may be removed. Their positive-diameter subsequential limits have the following properties almost surely.*

- (i) *Each deterministic interior point lies on no loop. Its absolute winding number about any loop is 0 or 1, and is stable along a converging loop sequence.*
- (ii) *No interior point has three distinct visits in total among the loop elements. Visits include multiple traversals after constant pauses are removed and also multiplicity of different elements.*
- (iii) *At every interior point of a positive loop, every neighborhood contains open disks of both absolute winding statuses for that loop.*

Proof. Passage counts and compactness. Consider a vertical band of fixed positive width, with its vertical sides interior to the square and its height equal to that of the square. Each disjoint medial passage between its sides has an actual dual path beside it: consecutive adjacent dual vertices are identical or joined by an open dual diagonal, including at wall caps. Crop to a slightly narrower band. The medial separators continue beyond both cropped sides and are disjoint from dual-open edges, so they divide the band into strips. Each accompanying dual path lies in the strip on its separator's dual side, with at most two assignments to a strip. Paths in different such strips cannot communicate in the actual dual graph there. Hence many passages force many distinct actual dual components crossing the cropped band.

The graph of internal dual faces is a full lattice rectangle with arbitrary induced boundary data; it has no dual crossing of the initial deterministic primal wall. The disjoint-passage proof of Lemma 13.3 bounds its crossing-component count. Even with all boundary vertices wired, an *actual* left-right crossing is bounded away from probability one by an up-to-boundary primal blocker. End that blocker just beyond the extreme dual rows by bounded finite-energy extensions to neighboring internal primal vertices. The rectangle's dual boundary edges themselves are random, so the crossing theorem applies. The same argument holds for horizontal bands.

A finite family of trimmed bands at each fixed scale bounds both the number of loops with diameter above that scale and the number of their disjoint subarcs with a given positive displacement. These bounds yield compactness modulo reparametrization. To make the extraction explicit, subdivide each positive loop by successive first-displacement times into arcs of diameter at most $2^{-j+O(1)}$, retaining marks from previous subdivisions. The number of marks at each fixed j is tight. Pass to subsequences on which their positions and cyclic orders converge. Assign nested parameter intervals with maximal length tending to zero and positive lengths at every fixed subdivision, splitting an interval further in two when necessary. The corresponding polygonal curves approximate uniformly. Diagonalize over diameter cutoffs tending to zero, retaining multiplicities of matched elements. This proves collection tightness.

Avoidance and the visit bound. For a deterministic interior point, RSW circuits in successive concentric annuli obstruct an actual arm to a fixed positive distance. The probability that a macroscopic loop approaches that point tends to zero with the inner radius. A countable union over diameter cutoffs gives (i). Winding is stable under uniform convergence away from the point, and each simple discrete loop has absolute winding 0 or 1.

Three distinct positive visits near an interior point would yield six medial passages through arbitrarily small annuli out to a fixed positive scale. This includes visits belonging to different

collection elements. In a trimmed intermediate annulus, order those radial separators and assign their actual color arms to the intervening strips. Arms in distinct strips stay separated. Where a strip receives two assignments retain both if their colors differ, and one if their colors agree. Each separator marks a color transition, so the resulting cyclic sequence has an alternating subfamily of at least six arms. Radial buffers justify all croppings. The six-arm bound of Duminil-Copin et al. [17, Corollary 6.7], transferred to this positive square law through a bulk collar, has exponent $2 + c$ for some $c > 0$. A grid union over possible centers therefore vanishes like a positive power of the inner scale. Taking mesh limits first proves (ii).

A circuit attached to each macroscopic bank. Fix a bulk core rectangle, with buffer space around it, and a color X . We claim that, with probability tending to one as $m \rightarrow \infty$ after the mesh limit, the actual cluster of every bottom-to-top core crossing contains an X -open circuit in the buffered patch enclosing a disk of radius c/m .

In a first subrectangle ending at a middle row, label the actual connected components joining its two terminal rows. Their number is tight by the passage bound. Reveal the past up to that middle row. Above it use m fresh transverse slabs of height $h \asymp 1/m$, with all levels rounded on the X -grid. For a given label, its sources are its actually connected bottom vertices. At each new row ask whether these sources connect to that row in the cumulative core rectangle. A full crossing survives all these tests under one of the initial labels.

When a label survives to a row, choose the leftmost simple open path from its sources to that row. The edges strictly to its right are fresh with domain Markov data. Indeed attach auxiliary spokes from one point below to the sources and from the new top row to a point above, in a disk with these auxiliary endpoints on its boundary. Simple paths use spokes only at their ends. Minimizing left area selects a path determined by its own edges and those on its left: any path entering strictly left supplies an open excursion between two distinct selected-path vertices that decreases the left region. The ordinary lattice part p is actually connected to the label.

At the slab's middle height, approach horizontally from the right exterior of the core and choose x on the right at distance $h/16$ from p . The ball $B(x, h/3)$ lies in the wider fresh patch, far from the past, spokes, and slab ends. The fresh graph on the right of p inside a slightly buffered ball dominates the restriction of a free comparison on the full ball graph. All attachments from omitted left-side or path edges go through vertices of p , which are wired by actual connectivity in the conditioned law. This argument applies to either color, using the dual FK law for dual X .

By interior RSW and FKG in the free ball comparison, with probability at least a fixed $c' > 0$ there is an open circuit between radii $h/64, h/24$, connected there to one between radii $h/8, h/4$. The outer circuit meets the deterministic geometric path p , since p comes within $h/16$ of x and exits the ball. Both use the same color graph. The inner circuit is wholly on the right of p , as is its connection up to first contact with p . Therefore this event supplies an actual attached disk-circuit using only fresh right-side edges in the slab. Conditionally on survival, failure to acquire such an attachment costs a factor at most $1 - c'$ at each stage. Truncate the initial number of labels and iterate. This proves the claim.

Visibility of both winding statuses. Use countably many core tests with rational positions and buffered sizes. A varying portion of a limiting loop in an interior ball crosses two horizontal or vertical levels of one such test with clearance. In its discrete approximants the two actual color banks supply crossings. Their clusters lie on opposite sides of the simple loop and do not cross it. Choose the buffered test much smaller than the loop's diameter, so neither of the attached disk-circuits can contain the entire loop. The disks strictly inside the two circuits then have the two corresponding winding statuses. Shrink them to retain

positive clearance; subsequential limits preserve disks of positive radius in the prescribed ball. The preceding estimate makes failure arbitrarily unlikely. Every neighborhood of a point on a nonconstant loop contains a varying portion, so the countable tests prove (iii) simultaneously. $\square$

15.2 Canonical loops and their continuation in the tree

Take joint subsequential couplings of the loop collections and of the angular trees for a countable dense set $\mathcal{Z}$ of deterministic interior targets, with arbitrarily large finite clock horizons. Lemma 15.1(i) ensures that all these targets avoid all positive limiting loops almost surely. Incidental lattice-edge positions do not require an extra convention: each active finite horizon retains a ball around its target.

We specify the established continuum input before making the lattice identification. For $4 < \kappa < 8$, the target-invariant tree of Sheffield [46, Sections 4.2–4.3] constructs the surrounding loops by full-turn angular excursions and completes each partially traced loop through the other branches. At the time of that paper, its stated continuity and invariance conclusion in Proposition 5.1 was conditional. Here the needed hypotheses follow from Miller and Sheffield [29, Theorem 1.3] and Miller and Sheffield [30, Theorem 1.2 and the ensuing CLE discussion]. For our single one-sided force weight, $\kappa - 6 > -2$, so force collisions do not reach the continuation threshold. For reversal, the two one-sided weights $\kappa - 6$ and 0 are both at least $\kappa/2 - 4$. The starting square is a Jordan domain, and all specified targets are distinct interior points off the loops. These results supply continuous non-simple CLE loops and their conformal and anticonformal invariance under exactly the parameters used here. They assert no lattice convergence or traversal identification.

The tree in Proposition 14.4 therefore has a coupled canonical nested CLE_κ collection in the square. Starting from a primal wall, let $b_0 = 0$, and let $b_1, b_2, \dots$ be the successive hits of the alternating opposite endpoints: first 2π , then 0 , and so on. Let a_k be the last visit to the starting endpoint before b_k . The canonical surrounding loop is denoted $\tilde{L}_z^k$. It follows the branch toward z on (a_k, b_k) and is then completed to its starting point by the tree's continuation of the current strand. These are the successive loops with nonzero winding about z , and every canonical loop is obtained from a target in $\mathcal{Z}$.

The endpoint reversal amounts to using reflected exploration chirality in the new enclosed component. Nesting supplies independent CLEs in those components, and anticonformal invariance preserves their unoriented laws. Thus the alternating endpoint convention gives the same nested ensemble. The scalar full-turn description is also the one in Schramm et al. [45, Proposition 2]; that result identifies the conformal-radius decrement, while the joint branch coupling and loop completion used here come from the exploration-tree construction. Endpoint accessibility of the diffusion with $1 < \mathfrak{d} < 2$ makes every b_k finite, and $b_k \rightarrow \infty$: otherwise its continuous angle would make infinitely many full 2π swings on a compact clock interval.

We will need the following concrete interpretation of the completion.

Lemma 15.2 (Default continuation and local double visits). *At an interior-angle time $s \in (a_k, b_k)$, the remaining traversal of $\tilde{L}_z^k$ is the default strand from W_s to O_s in the current domain. In a chart sending those accesses to $0, \infty$, every finite portion of this continuation is ordinary chordal SLE_κ and is recorded by branches to targets from $\mathcal{Z}$ that remain accessible high above that portion. Every open nonconstant traversal interval of $\tilde{L}_z^k$ contains two visits to one square-interior point.*

Proof. The stopped chordal equivalence is only valid while its force access is available. To continue after the original target is cut off, use the branch that retains this access, as

prescribed by the tree construction. A bounded finite chordal-capacity portion has bounded hull and driver. An open set of targets mapped to sufficiently large heights remains accessible above it, with tip and force angles strictly separated on that portion. The countable dense set supplies such a target. Its branch records the default continuation even if it has already separated from z . Target invariance gives consistency when the target or the finite portion is enlarged. This realizes all finite portions and hence the complete strand; it does not extend Proposition 3.12 of Sheffield [46] beyond its stopping condition.

For a single branch stopped at a finite radial time, its continuous trace together with the starting square boundary is a locally connected planar continuum. The current target domain is a complementary component of this continuum. Torhorst's theorem says that every complementary domain of such a continuum has locally connected boundary. The Riemann map from the disk onto that domain therefore extends continuously to the closed disk [40, historical comments and Corollary 3.2]. Thus the prime-access conformal images used here are continuous physical curves.

For ordinary chordal SLE, Miller and Wu [31, Theorem 1.1 and Corollary 1.7] give double points in the interior when $4 < \kappa < 8$. Indeed their double-point dimension exceeds the boundary-hit dimension by $(\kappa - 4)^2/(8\kappa) > 0$. Both visits occur at finite times. Scaling makes the probability of such a double visit before time t independent of $t > 0$; since its union over finite times has probability one, this probability is one for every $t > 0$. The domain Markov property then gives an interior double visit in every rational open capacity interval. Conformal maps preserve an interior double visit. Arrange a single probability-one event over all targets w in the fixed countable dense set, all rational w -clock times r , and a countable exhaustion of strict angle and chart localizations. The first time that the w -clock reaches r is causal. Conditional stopped chordal equivalence and the simultaneous rational-interval double-visit property therefore apply until its positive localization exit. Every open interval of a full-turn piece contains such a rational start.

For an interval in the default completion, choose a rational time s in the original successful excursion and a finite chordal portion containing a compact subinterval of the desired interval. As above, a target w from the same countable set at sufficiently large chart height records that portion with interior angle, even after the original target is cut off. Its radial and chordal clocks are strictly increasing there, so the subinterval contains a rational w -clock start. The universal probability-one event permits this realization-dependent choice of w and r . Applying the local double-visit property before its positive exit time proves the assertion. No Markov property is applied at a non-stopping excursion start or at an arbitrary postselected time. $\square$

15.3 Episodes and full-turn labels

A discrete *episode* starts when a cap is opened and ends at the next switch retaining a homogeneous pocket. Its color is the homogeneous wall color just before the opening. Call it successful when its loop winds around the target.

Lemma 15.3 (Discrete winding order). *As long as the target is macroscopically separated from a consumed tile or star, an episode is successful exactly when its terminal switch changes wall polarity. These polarity changes enumerate all loops surrounding the target, in their order from outside inward. A positive loop with positive clearance from the target is reached before a finite radial horizon depending on that clearance. Conversely a successful switch by a bounded horizon has macroscopic clearance from the target.*

Proof. The loop through the newly opened cap lies inside its homogeneous starting pocket. At the cap, the side toward the base, of the starting color, is exterior to this loop; the other

side is interior. During the episode the actual color banks preserve connections to these sides. If a bank hits its same-color wall, the retained homogeneous target pocket is connected to the cap along that bank. The traversed loop is outside the pocket: its past pairs have been removed, the new open diagonal is a barrier, and the continuation to the old force side stays in the main component. The target can reach that pocket's wall color without crossing the loop. Comparing this color with the starting color proves that the target is inside the loop precisely at a polarity change. The same argument works with separate boundary occurrences.

Every other unstarted candidate surrounding the target stays intact in the retained component. Actual diagonals and walls cannot cross it, and a loop wholly inside a simply connected pocket has its interior there too. Its deterministic pairs in any wall triangles are also intact, because cuts by other loops never traverse them. If the loop entered at a new cap surrounds the target, it is outermost among the remaining candidates: its access to the homogeneous wall at that cap crosses none of them. Thus successful episodes enumerate the surrounding loops, with possible unsuccessful same-color episodes between them.

Before such a loop's episode, growth remains outside its enclosed region; during the episode it follows that loop up to step-sized errors. A ball centered at the target inside its region therefore survives until the switch. If its radius is $r > 0$, then $\text{crad}(U_t, z) \geq r$ up to that time, bounding the radial clock by $\log(\text{crad}(U_0, z)/r)$. Conversely, Koebe's bound on a fixed horizon gives a positive-radius ball in the newly retained pocket, and its surrounding successful loop lies outside that ball, up to vanishing step errors. $\square$

An endpoint approach could otherwise be mistaken for a reset in the limit. The next estimate excludes this possibility.

Lemma 15.4 (Endpoint resets cannot be skipped). *At a bounded-clock completed state with θ approaching 0, the probability of returning to a fixed positive angle before a primal-pocket reset tends to zero, with the mesh limit first and the endpoint tolerance second. The analogous assertion near 2π uses a dual-pocket reset. Consequently the true successive discrete polarity-change times b_k^n converge in probability to b_k on sufficiently large finite horizons.*

Moreover, in an episode that succeeds by a bounded horizon, after a fixed interior-angle advance the probability of retreating arbitrarily near its starting endpoint before its successful switch tends to zero in the same order of limits.

Proof. Near 0 the entire dual wall arc and both end sides are in a tiny boundary neighborhood of the stopping chart. Until a primal reset, the growing dual bank remains actually connected to that arc; stop also before a fixed interior-angle threshold, so no opposite-end switch has yet occurred. In successive buffered dyadic chart half-annuli around this neighborhood, place actual primal transversal paths from primal wall to primal wall. Lemma 13.2 gives a uniformly positive conditional success chance at each separated scale, after the mesh limit at the chosen cutoffs. A successful path confines the connected dual bank, and hence the slit pieces, inside its small wall sector.

A hull confined to such a sector has a small new angular footprint: its basepoint-side maps converge to the identity at the origin and, by reflection, on each closed boundary arc away from that sector. The trapped dual arc remains in its short footprint. Therefore it cannot produce a fixed positive angular advance. Letting the number of half-annuli grow proves the first assertion. Exchange colors and reverse directions at 2π .

A true polarity change approaches the specified opposite endpoint, up to vanishing update errors. Uniform angle convergence and induction therefore preclude a premature macroscopic limit for b_k^n . Conversely, after the limiting first endpoint hit, instantaneous reflection produces a positive excursion within any prescribed additional time almost surely. Truncate the clocks and choose an exit size so that failure has arbitrarily small probability. Start at the first

completed-state approach within a small tolerance after the previous true polarity change. The just-proved estimate excludes an excursion of this size before the corresponding reset. Let the tolerance tend to zero. An earlier matching reset already suffices. Thus no endpoint hit is skipped, and $b_k^n \rightarrow b_k$.

For the final assertion, stop at successive approaches to the starting endpoint after a fixed-size advance. A successful episode that makes such an approach must then advance macroscopically without the corresponding same-color reset, since that reset would end the episode. The first assertion bounds each such event. The number of fixed-size swing tests is tight by angular tightness, so truncation and iteration give the stated retreat estimate. $\square$

It follows in particular that on every compact subinterval of (a_k, b_k) , the prelimit target branch is, with probability tending to one, in the same episode as at its k th successful switch. The angle stays away from the starting endpoint on that compact interval, and the preceding polarity change occurs before it.

Include the true winding-loop labels $L_z^{n,k}$ in the coupling and extract their limits L_z^k , increasing clock and diameter cutoffs as needed. They have positive size at bounded b_k^n by Lemma 15.3. They exhaust the positive limiting loops winding about z : any such loop has positive clearance from z and is reached before a bounded horizon, while the number of full swings there is tight. Winding tests are stable. For finite lists of labels stabilize also the indicators of equality as collection elements and then diagonalize. Distinct elements in this bookkeeping have distinct discrete representatives, even if their geometric limits might initially appear identical.

15.4 Ordered containment and equality of the labels

Lemma 15.5 (The canonical traversal is contained in order). *For each $z \in \mathcal{Z}$ and $k \geq 1$, the entire traversal of $\tilde{L}_z^k$ occurs in cyclic order in L_z^k , possibly with extra traversal portions inserted. Its order occupies at most one turn of L_z^k . If two canonical labels give the same element, their lattice-limit labels give the same collection element.*

Proof. Fix a rational $s \in (a_k, b_k)$. In any open subinterval of (a_k, s) choose $t < t'$ strictly inside it. The target kernel strictly shrinks between these times. There is a new boundary point inside the earlier open kernel; it belongs to the continuous canonical trace over $[t, t']$. Every small ball about this point with closure in the earlier kernel must encounter new prelimit growth between the corresponding times. Otherwise the ball stays clear of new boundary. It also meets a compact internal patch of the later kernel. Connectedness then puts that whole ball in the retained prelimit component, extending the limiting kernel through its alleged boundary point, a contradiction. These prelimit boundary pieces are slit updates tracing the successful episode, up to vanishing physical step errors.

Apply this test simultaneously to any finite ordered list of such intervals. Completed-update stops can be moved slightly inside them because capacities and angular within-step errors vanish. We obtain points of the discrete winding-loop traversal, in chronological order, converging to canonical trace points in those intervals.

Now take any finite list of finite chordal intervals on the default continuation after s . Choose, realization by realization, a target $w \in \mathcal{Z}$ in the current kernel whose chart image is safely above the entire bounded portion in question. It was still accessible with z up to s . Their time correspondence there is regular and passes to the mesh; they cannot have split earlier because the limiting kernel contains internal connecting paths with clearance. The branch toward w follows the default continuation with tip and force angles separated by a positive margin on the selected portion, by Lemma 15.2. Therefore its discrete branch has no reset there and continues the same original winding loop, in order, after the common prefix.

The new-boundary test applies to these intervals in the w -clock. The target w need not have a successful episode for this loop. Its post hoc choice costs no uniformity assumption: all targets, clocks, and regular sharing correspondences from the countable set have already been included in the coupling.

By continuity, shrinking the testing intervals and increasing the finite continuation range shows that every finite increasing list of canonical traversal points has representatives in order on L_z^k . On a countable dense set of canonical parameter values, take a single nondecreasing cyclic choice of representative parameters by compactness and a diagonal subsequence. Omitted endpoint values are included by continuity. The order uses at most one turn: the discrete episode and its no-reset continuation traverse a simple closed loop from its released cap without passing that cap a second time. Two separated canonical visits cannot collapse to one representative parameter, because every intervening open traversal interval has actual variation.

If $\tilde{L}_z^k = \tilde{L}_w^j$ but the two lattice-limit elements were distinct, each would contain the same canonical interior double visit from Lemma 15.2. This would give more than two total limiting visits at that point, contrary to Lemma 15.1(ii). Thus equal canonical labels have equal lattice-limit labels, with multiplicity understood as stipulated above. $\square$

The converse is essential: ordered containment alone would permit several canonical labels inside one discrete limiting element.

Lemma 15.6 (The converse label identification). *With equality understood as equality of collection elements, almost surely and simultaneously for all $z, w \in \mathcal{Z}$ and $k, j \geq 1$,*

$$L_z^k = L_w^j \iff \tilde{L}_z^k = \tilde{L}_w^j. \quad (15.1)$$

Proof. Only the forward implication remains. On the event that the discrete labels are shared, the two branches cannot split before entering this common loop: before its episode the loop and its interior remain intact across every cut by other loops. Choose clock horizons large enough to include both successful switches except on an arbitrarily small event.

While they share a domain, write $\xi = \Phi_z(w)$. For a fixed small $e > 0$, stop at first approach to $|\xi| \geq 1 - e$, if it occurs before the relevant winding switches. For fine mesh this can be a completed state also satisfying $|\xi| \leq 1 - e/2$. An exact split cannot precede this threshold inside the comparison range: the ODE and clock change are regular for $|\xi| \leq 1 - e/2$, and elementary capacities vanish there.

We first show that an *unstarted* loop surrounding both targets at this stop has probability tending to zero as $e \downarrow 0$. Place separated dyadic chart half-annuli around the boundary direction of ξ , between radius a multiple of e and a fixed small radius. Choose their buffered wall intervals to avoid the two marked accesses and sides; deleting the bounded number of affected scales leaves a number tending to infinity. Each bar separates a sector containing the radial access from w to the wall from the sector containing z and a remote wall portion.

Use an actual-color transversal path in each bar, choosing the color to match one wall end. At that end it reaches an actual wall vertex. At the other it does so as well if the wall has the same color. If that wall has the other color, complete the path from its specified-color triangle corner to the base by the segment of Lemma 13.2. After trimming at first wall contacts this is a simple crosscut in the prime closure. Its added segment meets only the one local cap pair, once.

An unstarted simple loop inside the domain surrounding both targets must cross this screen at least twice: it must meet both of the separated sectors, and its interior contains both targets. It cannot cross the actual open path, and cannot traverse that one cap pair twice. Thus one successful screen precludes the event, even with mixed wall colors. The conditional

success probability of each fixed-aspect buffered bar is bounded below by Lemma 13.2. Iteration, with the mesh limit first, proves the asserted small probability.

We next obtain a shared interior precursor for a shared successful loop. If the common loop is processed before the near-separation threshold is needed, its episode visibly passes through interior angles, comparable in the two charts while $|\xi| \leq 1 - e/2$. If the loop is still current at the threshold, its z -angle cannot be arbitrarily close to that episode's starting endpoint and later succeed without a same-color reset, by Lemma 15.4. Otherwise, its progression from the start to the current state has already passed a fixed interior margin. The w -angle at that shared state has a comparable interior margin, with constants allowed to depend on e . The unstarted case has just been made unlikely. Thus, except on an arbitrarily small event, the two successful episodes have a common state before kernel separation, with both angles interior by fixed positive margins.

From this precursor to the respective winding switches, neither angle returns arbitrarily near its episode's starting endpoint, by the successful-episode retreat estimate in Lemma 15.4. Choose in order the clock cutoff, the near-separation threshold, the interior margins, and the retreat tolerance, making their error probabilities successively small. With strict buffers a subsequential precursor remains in a common kernel with $|\xi| \leq 1 - e/2$, lies after both previous polarity changes, and lies in the interiors of both limiting successful excursions. Their endpoint histories and starting excursion time agree along the common branch by the regular chart correspondence. They share the successful front to this interior state and use the same default continuation to the same force access for its rest. Therefore their canonical loop elements coincide. Let all the preceding errors tend to zero. Countability of the labels proves the simultaneous assertion. $\square$

15.5 Trace and traversal identification

We have matched the lattice-limit and canonical labels and placed each canonical traversal in its partner's cyclic order. Away from the square wall, it remains to rule out additional trace points and additional traversal portions in that partner.

Proposition 15.7 (Identification away from the square wall). *Every positive lattice-limit loop with an interior point has a canonical label. Matched elements have the same winding indicators on $\mathcal{Z}$. If either matched element stays away from the square wall, the two elements are equal as curves modulo reparametrization and orientation. Without that restriction, every canonical element is still contained in order in its lattice-limit partner.*

Proof. The two-winding-disk property of Lemma 15.1(iii) provides a disk of winding one for every positive limiting loop with an interior point. It contains a target from $\mathcal{Z}$, so the loop is one of its exhausted winding labels. For a labeled loop, Lemma 15.6 makes its dense-target winding indicators identical to those of its canonical partner: winding one at a target is equivalent to equality with one of that target's successive surrounding-loop labels, in both collections.

Suppose an interior trace point of the lattice-limit loop were outside its canonical partner trace. A small ball about it avoids that canonical trace and has constant canonical winding. The two-winding-disk property supplies disks of both lattice winding statuses in the same ball, each containing a dense target. This contradicts the indicator equality. Hence the entire interior lattice trace lies on its partner. The reverse trace inclusion follows from ordered containment. If the canonical partner is internal, the connected lattice trace containing it cannot reach the wall without adding an interior point outside that partner. If the lattice trace is internal, so is the contained partner. Thus in either wall-away case the full traces are equal.

It remains to exclude extra traversal of this trace. The cyclic representative selection in Lemma 15.5 is strictly increasing: collapsing a canonical parameter interval to one representative would make that entire traversal interval constant. Its generalized inverse is therefore continuous, nondecreasing, and of degree one. Jumps of the representative selection become plateau intervals of this inverse. Outside their interiors, the lattice traversal agrees with the canonical traversal at the assigned parameter; a plateau can only insert a portion anchored at one canonical point.

Assume such a portion varies. By trace equality it lies on the canonical trace. At a visited value different from its anchor there can be only one canonical parameter: two canonical visits plus this inserted visit would violate the total-visit bound of Lemma 15.1(ii). On a varying subarc away from the anchor, the assignment to that unique canonical parameter is continuous, by compactness. It therefore sweeps a nontrivial canonical traversal interval all of whose values have unique canonical visits. This is impossible by the interior-double-visit property of Lemma 15.2. Hence every inserted portion is constant. Equality modulo reparametrization follows, including the preserved multiplicity of collection elements. $\square$

15.6 From squares to the free plane law and the sphere

Theorem 15.8 (Nested full-plane convergence for $q < 4$). *For every fixed $1 \leq q < 4$, the loop collection of the free infinite-volume critical square-lattice FK law converges, through the full mesh sequence, to the nested whole-plane $\text{CLE}_{\kappa(q)}$ in the spherical loop-matching topology of Subsection 1.2. The limit is Möbius invariant and includes every nested generation.*

Proof. We first give the uniform comparison estimates that allow an upgrade from a local limit. For a fixed compact disk K , the probability that any discrete loop visits K and reaches a distant surrounding circle tends to zero as that circle tends to infinity, uniformly for sufficiently small meshes and for larger initial wired squares. Actual monochromatic circuits in buffered annuli block such a loop; conditional RSW gives a fixed success chance at each separated scale. Use a slightly enlarged K and strict radial margins when passing to limits.

The free infinite-volume law and the homogeneous-square law can be coupled identically on a growing intermediate disk with probability tending to one as the square grows. Bracket the exterior partition by the minimal and maximal partitions, use monotone FK coupling, and expose actual boundary-attachment clusters in the maximal sample. Dual RSW screens make their probability of reaching the deep interior tend to zero. Once exterior influence is separated, the remaining conditional laws agree and one uses identical switches. Combining this with the preceding escape estimate makes the *entire* loops visiting K coincide, since those loops stay in the intermediate disk. This argument uses the specified free plane law and FK domain Markov coupling; it assumes no interface limit.

In a fixed-square subsequential coupling, Proposition 15.7 identifies all wall-away macroscopic loop elements, with their traversals. Its remaining possible discrepancies involve the wall. The escape estimate makes it unlikely that a lattice-limit loop both hits K and reaches that wall. The same holds for a canonical CLE loop, because that loop is contained in its lattice-limit partner. Thus, when the square is large, the square CLE and the free plane lattice law have arbitrarily close matching configurations on loops visiting K , in the successive mesh and square limits.

The large-domain local limit of nested CLE is established in Miller et al. [32, Appendix A, Theorem A.1]. Its comparison uses conformal identifications of the relevant random-region loop configurations, with maps approaching the identity on compact sets; it does not assert equality at unchanged Euclidean coordinates. For $4 < \kappa < 8$ this construction agrees with the whole-plane branching construction by Gwynne et al. [21, Remark 2.6], and its law is invariant under inversion, hence under Möbius maps, by their Theorem 1.1. Centered squares

and disks give the same local limit: normalize their conformal radii, and the comparison maps tend to the identity on compact subsets. All these parameter restrictions hold for our $4 < \kappa \leq 6$.

To pass from this local law to the required spherical topology, fix $\epsilon > 0$ and choose K so large that its exterior has spherical diameter less than ϵ . Loops entirely outside K are then irrelevant at that matching accuracy. The uniform escape bounds control the loops that visit K but leave a much larger disk. They hold for square CLEs uniformly as the squares grow, by their subsequential square-law couplings and the containment argument above. Consequently the laws of the square CLEs are Cauchy for spherical matchings: compare two large squares on a large fixed intermediate disk to the same free-plane lattice law, let the mesh tend to zero, and use the escape tails and the exterior diameter bound.

Any resulting spherical limit has the established local whole-plane CLE law. That local law, together with these tails, determines the full spherical collection: at each accuracy all relevant loops are seen in a bounded window except on an arbitrarily small event. Conversely, the same comparisons force every subsequence of the free-plane lattice laws to this limit. Hence convergence holds for the full mesh sequence. The target labels ranged over all $k \geq 1$ and the clock cutoffs increased without bound, so no nested generations have been discarded. This proves Theorem 1.1(ii) for $1 \leq q < 4$. $\square$

16 Square-bond exponent consequences

We prove Corollary 1.3 by extracting two kinds of arm information from the complete unoriented loop limit. With fixed buffers at the annular rims, the absence of a surrounding loop detects a radial arm of either color; self-duality then compares this with one primal-open arm. Even alternating arms are instead detected by the number of loop traversals between the annular rims. These two descriptions give bounds at each fixed annulus ratio. Quasi-multiplicativity converts those bounds into lattice arm exponents, after which the near-critical scaling relations give the thermodynamic exponents.

The square model in this section has $q = 1$; all critical arm probabilities are at $p = 1/2$. On mesh δ , let $\pi_{1,\delta}^\square(a, b)$ be the probability of an actual primal-open path from $\overline{B}_a$ to $\mathbb{C} \setminus B_b$. For even m , let $A_{m,\delta}^X(a, b)$ be the event of m disjoint cyclically alternating arms between these sets, and write $\pi_{m,\delta}^X(a, b)$ for its probability. Here $X = \square$ means primal-open/dual-open square bonds, and $X = \triangle$ means the two colors of critical triangular-site percolation. Write Γ_δ^X for the corresponding complete interface-loop collection. The annuli in this notation are circular.

16.1 One arm from surrounding loops

Let $B_\delta(a, b)$ be the square-bond event of a radial arm of *either* color. For a complete loop collection Γ , let $H_o(\Gamma; a, b)$ be the event that some loop is contained in $\{a < |z| < b\}$ and has nonzero absolute winding about 0; define $H_c(\Gamma; a, b)$ using $\{a \leq |z| \leq b\}$. Put $Z(\Gamma; a, b) = H_o(\Gamma; a, b)^c$, and use subscripts δ and CLE for the square-bond and whole-plane CLE_6 collections. Absolute winding is independent of loop orientation.

For fixed $0 < a < b/4$ and all sufficiently small δ , the rounded disjoint tile loops used in Section 15 give

$$B_\delta(a/2, 2b) \subset Z_\delta(a, b) \subset B_\delta(2a, b/2). \quad (16.1)$$

Indeed, a complete surrounding loop separates the two rims, and actual open paths of either color stay in the complementary territories, so neither can cross that loop. Conversely, if neither color crosses the cropped annulus, planar cutset duality supplies a primal-open circuit and a dual-open circuit surrounding 0 between its rims, up to $O(\delta)$ displacement. They

are disjoint and nested. The cluster of the inner circuit's color is confined by the outer opposite-color circuit, so its outer medial boundary is a complete surrounding loop between them. The fixed collars absorb the displacement.

On the locally finite matching space, $H_o(\cdot; a, b)$ is open: a witnessing loop has positive rim clearance and positive spherical diameter, and a sufficiently close matched loop preserves its absolute winding. The event $H_c(\cdot; a, b)$ is closed. A closed-annulus witness has spherical diameter bounded below in terms of a, b ; local finiteness therefore lets a subsequence of witnesses be matched to one fixed limit loop, which retains closed-annulus containment and nonzero winding. Since $H_o(a, b) \subset H_c(a, b) \subset H_o(a/2, 2b)$, Portmanteau and Theorem 1.1(ii) give

$$\mathbb{P}(Z_{\text{CLE}}(a/2, 2b)) \leq \liminf_{\delta \downarrow 0} \mathbb{P}(Z_\delta(a, b)) \leq \limsup_{\delta \downarrow 0} \mathbb{P}(Z_\delta(a, b)) \leq \mathbb{P}(Z_{\text{CLE}}(a, b)). \quad (16.2)$$

This uses the open complement of H_c and the closed event Z ; it does not require continuity of an event indicator. Combining (16.1)–(16.2) at rescaled radii yields, for $16a < b$,

$$\begin{aligned} \mathbb{P}(Z_{\text{CLE}}(a/4, 4b)) &\leq \liminf_{\delta \downarrow 0} \mathbb{P}(B_\delta(a, b)) \\ &\leq \limsup_{\delta \downarrow 0} \mathbb{P}(B_\delta(a, b)) \leq \mathbb{P}(Z_{\text{CLE}}(2a, b/2)). \end{aligned} \quad (16.3)$$

By scale invariance put $p_0(t) = \mathbb{P}(Z_{\text{CLE}}(t, 1))$. The Camia–Newman identification of the no-surrounding-loop event with the triangular-site either-color annular crossing limit, together with the one-color estimate of Lawler–Schramm–Werner, gives

$$p_0(t) \asymp t^{5/48} \quad (t \downarrow 0). \quad (16.4)$$

Here either-color probability lies between one-color probability and twice that probability; fixed rim dilations accommodate either open or closed rim conventions, as in Camia and Newman [9, Theorem 2 and Section 1.1] and Lawler et al. [27, Theorem 1.2 and Section 3]. At square-bond $p = 1/2$, the dual law translated by $(-\delta/2, -\delta/2)$ is the primal law. The shift is absorbed by the collars, so $\mathbb{P}(B_\delta(a/2, 2b))/2 \leq \pi_{1,\delta}^\square(a, b) \leq \mathbb{P}(B_\delta(a, b))$ for small δ . Consequently, for fixed $t = a/b < 1/16$,

$$\frac{1}{2}p_0(t/64) \leq \liminf_{\delta \downarrow 0} \pi_{1,\delta}^\square(a, b) \leq \limsup_{\delta \downarrow 0} \pi_{1,\delta}^\square(a, b) \leq p_0(4t). \quad (16.5)$$

16.2 Even alternating arms from loop traversals

For even m , let $T_m(\Gamma; a, b)$ mean that there are m parameter intervals on members of Γ , with disjoint interiors on each individual loop, each visiting both $\{|z| < a\}$ and $\{|z| > b\}$. Intervals on distinct loops are counted separately. This definition uses the retained loop traversals, not merely their traces, and is independent of roots and orientations.

For either lattice, fixed $0 < a < b$, and all sufficiently small meshes,

$$A_{m,\delta}^X(a/2, 2b) \subset T_m(\Gamma_\delta^X; a, b), \quad T_m(\Gamma_\delta^X; a/2, 2b) \subset A_{m,\delta}^X(a, b). \quad (16.6)$$

For the first inclusion, take simple last-inner-to-first-outer subpaths of the alternating arms within the two collars. They cut the annulus into m strips with opposite-colored sides. Planar separation in each strip supplies an interface crosscut, belonging to an interval of a complete loop. The crosscuts are disjoint because the open arms separate the strips. Conversely, crop the loop intervals to disjoint radial crosscuts. The territorial arm counting of Lemma 12.3 gives at least m cyclically alternating actual arms in the central crop. The same two-colored

strip argument applies to hexagonal cluster boundaries. The collars absorb the square drawing's $O(\delta)$ displacement and separate any shared interval endpoints before cropping.

A witnessing loop has spherical diameter bounded below by the positive spherical distance between the two rims. A sufficiently close one-to-one matching therefore carries its witnessing intervals, and their order and disjoint interiors, to matched loop circles. It follows that $T_m(\cdot; a, b)$ is open and, when $4a < b$,

$$\overline{T_m(\cdot; a, b)} \subset T_m(\cdot; 2a, b/2). \quad (16.7)$$

For the closure assertion, match the witnessing loops to the finitely many relevant limit loops and pass to subsequences of the interval endpoints on their circles. Cyclic order is preserved; the separated rim visits and uniform continuity prevent a witnessing interval from collapsing. The enlarged rims make its limiting visits strict.

We also need local matching for triangular sites. A monochromatic circuit between a fixed window and a remote boundary confines every loop meeting the window. RSW makes the failure probability tend to zero as the boundary recedes, uniformly in the mesh, so it suffices first to work in a fixed disc. In Camia and Newman [9, Theorem 5], $K_\delta(\varepsilon)$ is a tight bound on the steps needed to find every loop of diameter $> \varepsilon$, and any fixed number of tracked exploration paths and domain boundaries are jointly tight as ordered curves. Each produced loop is a bounded concatenation of subcurves of those tracked curves. Compactness of subcurve endpoints gives tightness of the ordered loop curves, while at most one loop per step gives tightness of their number. Diagonalizing over buffered diameter cutoffs therefore gives subsequential tightness in one-to-one matching.

For any such matching limit, forgetting multiplicities and then roots and orientations at positive diameter cutoffs gives the Hausdorff-set limit of Camia and Newman [9, Theorem 1], which is the usual CLE_6 law [46, Section 1.1]. There can be no duplicate loop elements in the matching limit. Every positive CLE_6 loop has an interior double visit with nonconstant excursions, by Lemma 15.2. Two copies would therefore give at least three visits in total. Matching convergence would produce six disjoint radial medial sublegs around that point, hence six alternating triangular arms. The uniform six-arm bound of Camia and Newman [9, Lemma 6.1] excludes any such point as the inner radius tends to zero, exactly as in Lemma 15.1(ii). Thus the local matching limit is CLE_6 with each loop counted once. This argument uses the ordered-curve metric for individual loops and does not infer one-to-one matching from Hausdorff-set convergence alone.

Apply Portmanteau to the open event T_m and its closure, using (16.6)–(16.7) and the common CLE_6 limit. For $64r < R$ this gives the exact buffered comparison

$$\limsup_{\delta \downarrow 0} \pi_{m,\delta}^\square(r, R) \leq \liminf_{\delta \downarrow 0} \pi_{m,\delta}^\triangle(8r, R/8), \quad (16.8)$$

and the same inequality with the lattices interchanged. The intermediate events for (16.8) are $T_m^\square(2r, R/2)$, its closure inside $T_m^{\text{CLE}}(4r, R/4)$, and the open event $T_m^\triangle(4r, R/4)$. Thus rim tangencies need not have probability zero.

Let $g_m(t)$ be the triangular fixed-annulus limit for the alternating word in the parallelogram annuli of Nolin [36, Theorem 20]. That theorem and its following fixed-annulus statement, based on the triangular-site arm calculation of Smirnov and Werner [50], give $g_m(t) = t^{(m^2-1)/12+o(1)}$ as $t \downarrow 0$. Deterministic inclusions between circular and parallelogram annuli change each rim by a fixed factor. Hence, for one fixed $K \geq 64$ and each fixed $0 < t < 1/K$,

$$g_m(t/K) \leq \liminf_{\delta \downarrow 0} \pi_{m,\delta}^\square(t, 1) \leq \limsup_{\delta \downarrow 0} \pi_{m,\delta}^\square(t, 1) \leq g_m(Kt). \quad (16.9)$$

This is an envelope for fixed annuli, not an assertion that the unbuffered square-bond annular probability has a limit.

16.3 From fixed annuli to lattice exponents

Completion of the proof of Corollary 1.3. At $q = 1$, the arm separation, localization, and quasi-multiplicativity results of Duminil-Copin et al. [17, Propositions 6.2, 6.3, and 6.5] apply to each fixed square-bond color word. Fixed-shape comparison transfers them from their square annuli to the circular annuli used here. In particular, for $j = 1$ or fixed even j there are constants $0 < c_j \leq C_j < \infty$ such that

$$c_j \pi_j(r, R) \leq \pi_j(r, \rho) \pi_j(\rho, R) \leq C_j \pi_j(r, R) \quad (16.10)$$

at separated radii, where $\pi_j(r, R)$ denotes the corresponding unit-mesh annular probability. Equations (16.4), (16.5), and (16.9) say that, after taking the mesh limit at each fixed ratio M , both envelopes have logarithmic cost $\alpha_j \log M + o(\log M)$, with $\alpha_1 = 5/48$ and $\alpha_j = (j^2 - 1)/12$ for even j . Choose a large fixed M , then take all sufficiently large inner radii so that their ratio- M probabilities lie between the two envelopes with an arbitrarily small logarithmic error. Iterating (16.10) over ratio- M annuli adds at most $\max\{|\log c_j|, |\log C_j|\}$ per annulus. First let the number of annuli tend to infinity, then divide by $\log M$ and let $M \rightarrow \infty$. Monotonicity handles the last partial annulus, and quasi-multiplicativity handles any fixed admissible inner radius. A fixed finite inner connection compares the annular one-arm event with connection from 0. This proves (1.6). The argument uses only fixed-ratio limits followed by quasi-multiplicativity, not a uniform rate of convergence in the annulus ratio.

Near-critical consequences. Let $L(p)$ be a finite-size crossing length with a fixed sufficiently small RSW-admissible threshold, using the subcritical color on the appropriate side of $p = 1/2$. Kesten's square-bond scaling relations [26], in the arm-event formulation discussed by Nolin [36, Sections 7–8.1], give

$$\begin{aligned} |p - \tfrac{1}{2}| L(p)^2 \pi_4(r_2, L(p)) &\asymp 1, & \theta(p) &\asymp \pi_1(L(p)) \quad (p > \tfrac{1}{2}), \\ \chi(p) &\asymp L(p)^2 \pi_1(L(p))^2, & \xi(p) &\asymp L(p). \end{aligned} \quad (16.11)$$

Here r_2 is any fixed admissible inner radius for four arms; changing that radius costs only a constant. The second-moment length defined with the Euclidean norm is comparable to the version with the ℓ^∞ norm. Substituting $\pi_4(r_2, R) = R^{-5/4+o(1)}$ first gives $L(p) = |p - \frac{1}{2}|^{-4/3+o(1)}$. Substituting $\pi_1(R) = R^{-5/48+o(1)}$ in the remaining relations gives $\beta = (4/3)(5/48) = 5/36$ and $\gamma = (4/3)(2 - 10/48) = 43/18$, proving (1.7). $\square$

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